18. Troubleshooting Guide

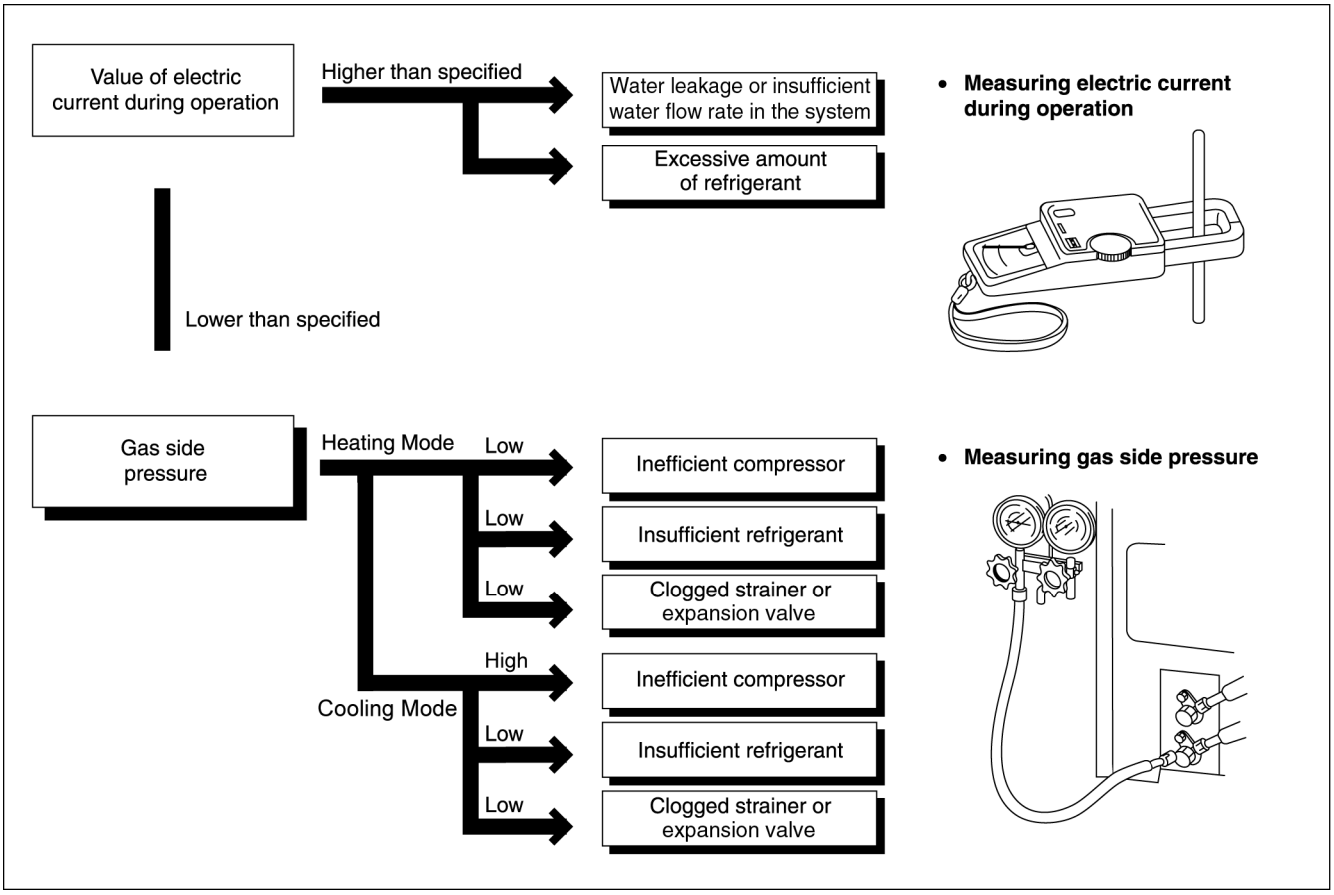
18.1 Refrigeration Cycle System

In order to diagnose malfunctions, make sure that there are no electrical problems before inspecting the refrigeration cycle.
Such problems include insufficient insulation, problem with the power source, malfunction of a compressor and a fan.
The normal pressure of the refrigeration cycle depends on various conditions, the standard values for them are shown in the table on the right.

Normal Pressure (Standard)

	Gas pressure MPa (kg/cm^{2}G)
Heating Mode	2.3 ~ 2.9 (23 ~ 29)
Cooling Mode	0.9 ~ 1.2 (9 ~ 12)

★Condition: ● Outdoor temperature 7°C at heating mode and 35°C at cooling mode.
● Compressor operates at rated frequency.



18.2 Relationship between the Condition of the Air-to-Water Heatpump Indoor and Outdoor Units and Pressure and Electric Current

Condition of the Air-to-Water Heatpump indoor and outdoor units	Heating Mode			Cooling Mode		
	Low Pressure	High Pressure	Electric current during operation	Low Pressure	High Pressure	Electric current during operation
Water leakage or insufficient water flow rate in the system						
Excessive amount of refrigerant						
Inefficient compression						
Insufficient refrigerant (gas leakage)						
Outdoor heat exchange deficiency						
Clogged expansion valve or Strainer						

- Carry out the measurements of pressure, electric current, and temperature fifteen minutes after an operation is started.

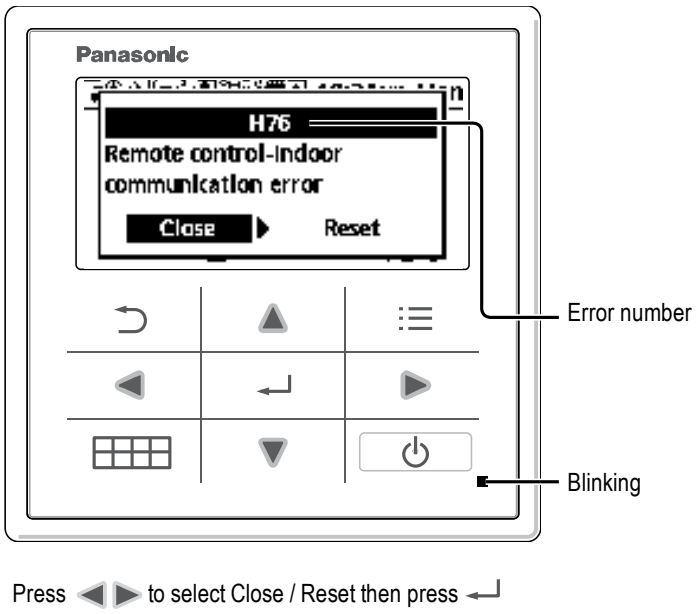
18.3 Breakdown Self Diagnosis Function

18.3.1 Self Diagnosis Function (Three Digits Alphanumeric Code)

- When abnormality occur during operation, the system will stop operation, and OFF/ON control panel LED will blink and error code will display on the control panel.
- Even error code is reset by turning OFF power supply or by selecting ERROR RESET, if the system abnormality is still unrepaired, system will again stop operation, and OFF/ON control panel LED will again blinks and error code will be display.
- The error code will store in IC memory.

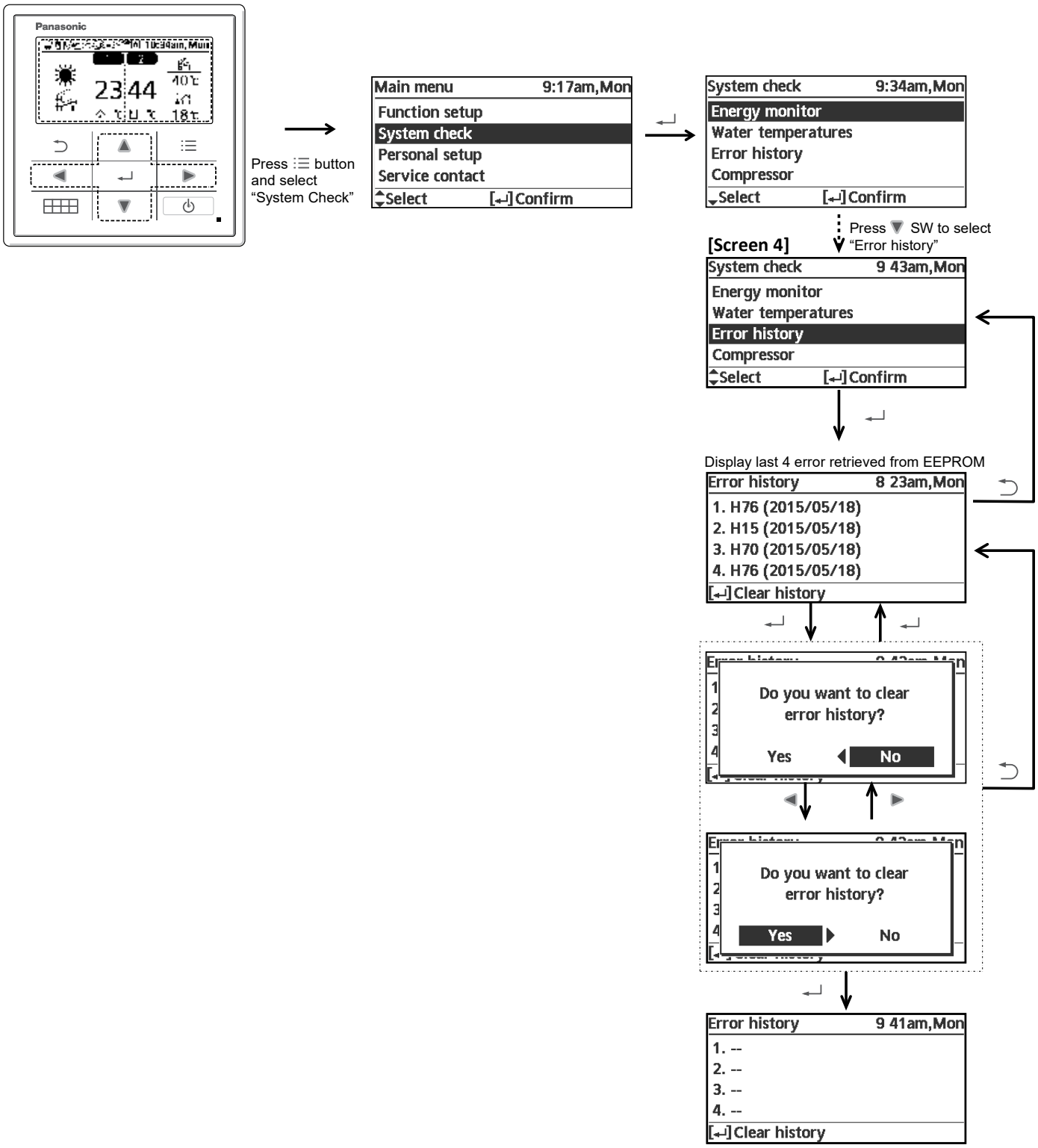
- **To check the error code**
  - When an abnormality occurs, system will stop operation and OFF/ON control panel LED will blink.
  - Error code of the abnormality will be display on the control panel.
  - To determine the abnormality description, the error code table needs to be referred.

eg:



- **To display past/last error code**
  - Turn ON power supply.
  - Refer below procedure to retrieve the error code history.

- **To permanently delete error code from IC memory**
  - Turn ON power supply.
  - Refer below procedure to clear error history.


Panasonic
Press ☰ button and select "System Check"
Main menu 9:17am,Mon
Function setup
System check
Personal setup
Service contact
Select [↵]Confirm
System check 9:34am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Press ▼ SW to select "Error history"
[Screen 4]
System check 9 43am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Display last 4 error retrieved from EEPROM
Error history 8 23am,Mon
1. H76 (2015/05/18)
2. H15 (2015/05/18)
3. H70 (2015/05/18)
4. H76 (2015/05/18)
[↵]Clear history
Do you want to clear error history?
Yes No
Do you want to clear error history?
Yes No
Error history 9 41am,Mon
1. --
2. --
3. --
4. --
[↵]Clear history

18.4 Error Codes Table

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
H00	No abnormality detected	—	—
H12	Indoor/Outdoor capacity unmatched	90s after power supply	• Indoor/outdoor connection wire • Indoor/outdoor PCB • Specification and combination table in catalogue
H15	Outdoor compressor temperature sensor abnormality	Continue for 5 sec.	• Compressor temperature sensor (defective or disconnected)
H20	Water pump abnormality	Continue for 10 sec.	• Indoor PCB • Water pump (malfunction)
H23	Indoor refrigerant liquid temperature sensor abnormality	Continue for 5 sec.	• Refrigerant liquid temperature sensor (defective or disconnected)
H27	Service valve error	Continue for 5 minutes	• High pressure sensor (defective or disconnected)
H28	Abnormal solar sensor	Continue for 5 sec.	• Solar temperature sensor (defective or disconnected)
H31	Abnormal swimming pool sensor	Continue for 5 sec.	• Pool temperature sensor (defective or disconnected)
H36	Abnormal buffer tank sensor	Continue for 5 sec.	• Buffer tank sensor (defective or disconnected)
H38	Brand code not match	When indoor and outdoor brand code not same	—
H42	Compressor low pressure abnormality	—	• Outdoor pipe temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
H43	Abnormal Zone 1 sensor	Continue for 5 sec.	• Water temperature Zone 1 sensor
H44	Abnormal Zone 2 sensor	Continue for 5 sec.	• Water temperature Zone 2 sensor
H62	Water flow switch abnormality	Continue for 1 min.	• Water flow switch
H63	Abnormal low pressure sensor	4 times in 20 minutes	• Low pressure sensor (defective or disconnect)
H64	Refrigerant high pressure abnormality	Continue for 5 sec.	• Outdoor high pressure sensor (defective or disconnected)
H65	Abnormal deice water circulation	water flow > 7 l/min continuously for 20 seconds during anti freeze deice	• Water pump
H67	Abnormal External Thermistor 1	Continue for 5 sec.	• Room temperature Zone 1 sensor
H68	Abnormal External Thermistor 2	Continue for 5 sec.	• Room temperature Zone 2 sensor
H70	Back-up heater OLP abnormality	Continue for 60 sec.	• Back-up heater OLP (Disconnection or activated)
H72	Tank sensor abnormal	Continue for 5 sec.	• Tank sensor
H74	PCB communication error	Communication or transfer error	• Indoor main PCB and Sub PCB
H75	Low water temperature control	Room heater disable and deice request to operate under low water temperature	• Heater operation must enable to increase water temperature
H76	Indoor - control panel communication abnormality	—	• Indoor - control panel (defective or disconnected)
H90	Indoor/outdoor abnormal communication	> 1 min after starting operation	• Internal/external cable connections • Indoor/Outdoor PCB
H91	Tank heater OLP abnormality	Continue for 60 sec.	• Tank heater OLP (Disconnection or activated)
H95	Indoor/Outdoor wrong connection	—	• Indoor/Outdoor supply voltage
H98 / F95	Outdoor high pressure overload protection Cooling high pressure overload protection	—	• Outdoor high pressure sensor • Water pump or water leakage • Clogged expansion valve or strainer • Excess refrigerant • Outdoor PCB
H99	Indoor heat exchanger freeze prevention	—	• Indoor heat exchanger • Refrigerant shortage
F12	Pressure switch activate	4 times occurrence within 30 minutes	• Pressure switch
F14	Outdoor compressor abnormal revolution	4 times occurrence within 20 minutes	• Outdoor compressor
F15	Outdoor fan motor lock abnormality	2 times occurrence within 30 minutes	• Outdoor PCB • Outdoor fan motor
F16	Total running current protection	3 times occurrence within 20 minutes	• Excess refrigerant • Outdoor PCB

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
F20	Outdoor compressor overheating protection	4 times occurrence within 30 minutes	• Compressor tank temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
F22	IPM (power transistor) overheating protection	3 times occurrence within 30 minutes	• Improper heat exchange • IPM (Power transistor)
F23	Outdoor Direct Current (DC) peak detection	7 times occurrence continuously	• Outdoor PCB • Compressor
F24	Refrigeration cycle abnormality	2 times occurrence within 30 minutes	• Insufficient refrigerant • Outdoor PCB • Compressor low compression
F25	Cooling/Heating cycle changeover abnormality	4 times occurrence within 30 minutes	• 4-way valve • V-coil
F27	Pressure switch abnormality	Continue for 1 min.	• Pressure switch
F30	Water outlet sensor 2 abnormality	Continue for 5 sec.	• Water outlet sensor 2 (defective or disconnected)
F32	Abnormal Internal Thermostat	Continue for 5 sec.	• Control panel PCB thermostat
F36	Outdoor air temperature sensor abnormality	Continue for 5 sec.	• Outdoor air temperature sensor (defective or disconnected)
F37	Indoor water inlet temperature sensor abnormality	Continue for 5 sec.	• Water inlet temperature sensor (defective or disconnected)
F40	Outdoor discharge pipe temperature sensor abnormality	Continue for 5 sec.	• Outdoor discharge pipe temperature sensor (defective or disconnected)
F41	PFC control	4 times occurrence within 10 minutes	• Voltage at PFC
F42	Outdoor heat exchanger temperature sensor abnormality	Continue for 5 sec.	• Outdoor heat exchanger temperature sensor (defective or disconnected)
F43	Outdoor defrost sensor abnormality	Continue for 5 sec.	• Outdoor defrost sensor (defective or disconnected)
F45	Indoor water outlet temperature sensor abnormality	Continue for 5 sec.	• Water outlet temperature sensor (defective or disconnected)
F46	Outdoor Current Transformer open circuit	—	• Insufficient refrigerant • Outdoor PCB • Compressor low
F48	Outdoor EVA outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor EVA outlet temperature sensor (defective or disconnected)
F49	Outdoor bypass outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor bypass outlet temperature sensor (defective or disconnected)

18.5 Self-diagnosis Method

18.5.1 Connection Capability Rank Abnormality (H12)

Malfunction Decision Conditions:

During startup operation of cooling and heating, the capability rank of indoor checked by the outdoor is used to determine connection capability rank abnormality.

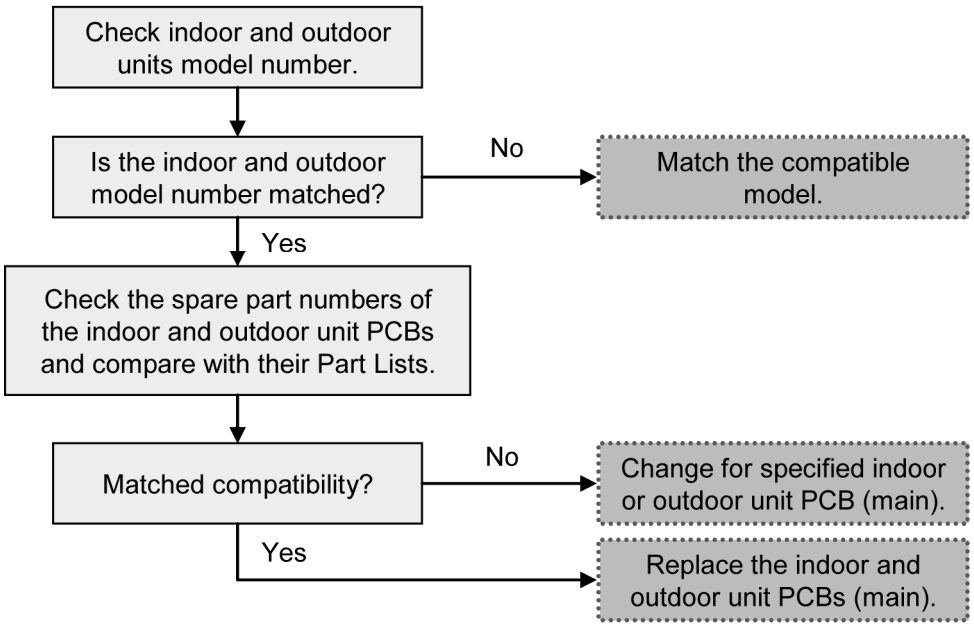
Malfunction Caused:

1 Wrong model interconnected.
2 Wrong indoor unit or outdoor unit PCB (main) used.
3 Faulty indoor unit or outdoor unit PCB (main).

Abnormality Judgment:

Continue for 90 seconds.

Troubleshooting:



18.5.2 Compressor Tank Temperature Sensor Abnormality (H15)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the compressor tank temperature sensor are used to determine sensor error.

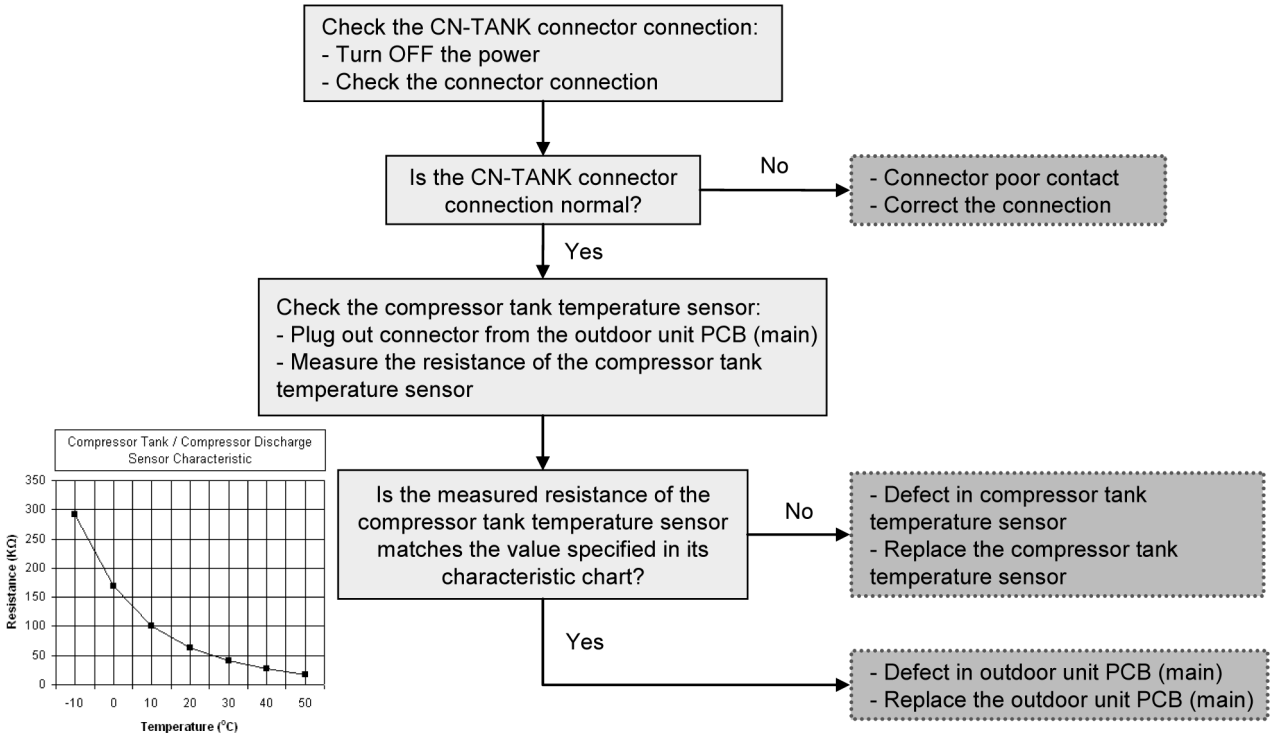
Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:



18.5.3 Water Pump Abnormality (H20)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the rotation speed detected by the IPM of water pump motor during water pump operation is used to determine abnormal water pump (feedback of rotation > 6,000rpm or < 700rpm).

Malfunction Caused:

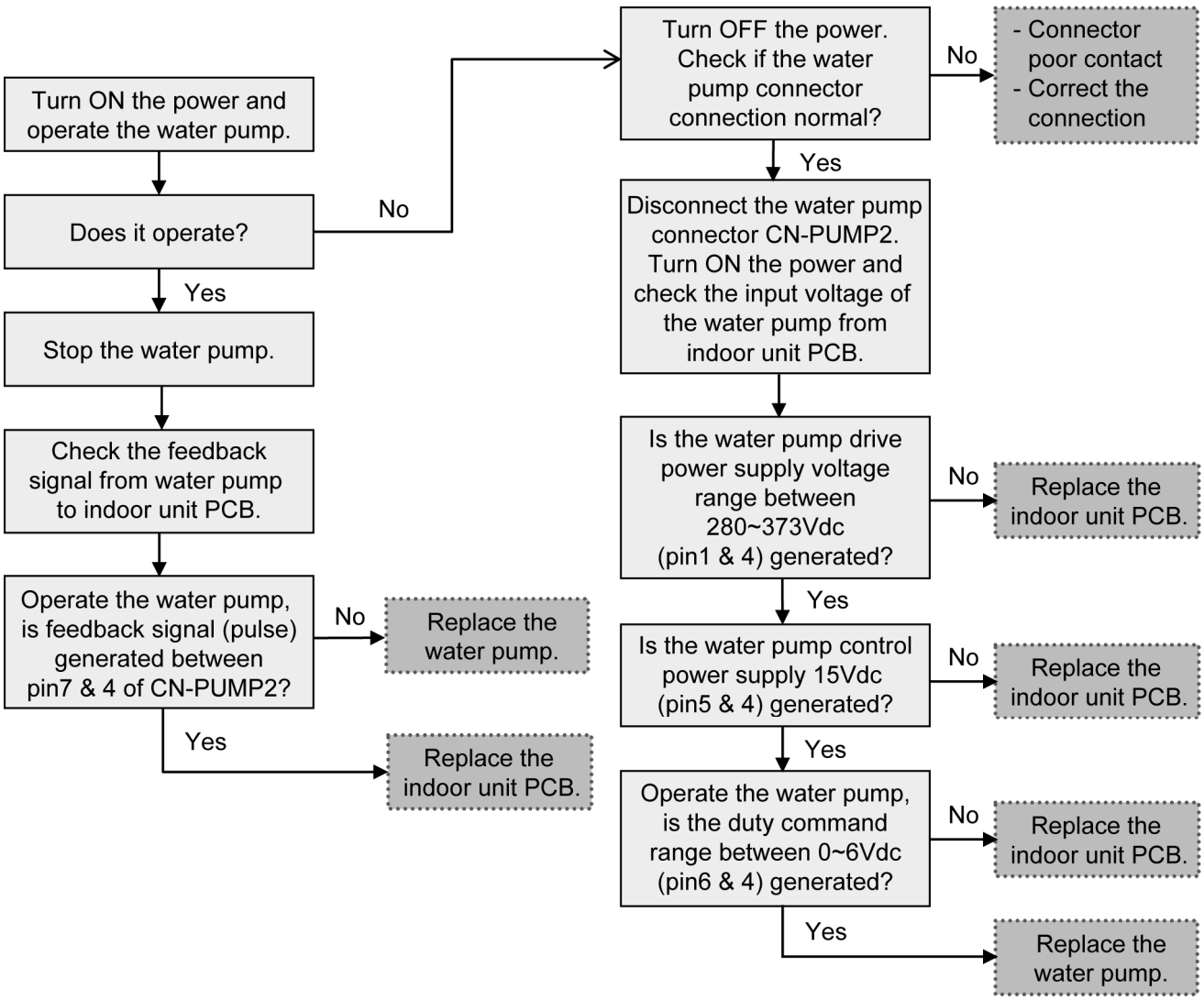
1 Operation stop due to short circuit inside the water pump motor winding.
2 Operation stop due to breaking of wire inside the water pump motor.
3 Operation stop due to breaking of water pump lead wires.
4 Operation stop due to water pump motor IPM malfunction.
5 Operation error due to faulty indoor unit PCB.

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.4 Indoor Refrigerant Liquid Temperature Sensor Abnormality (H23)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor refrigerant liquid temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

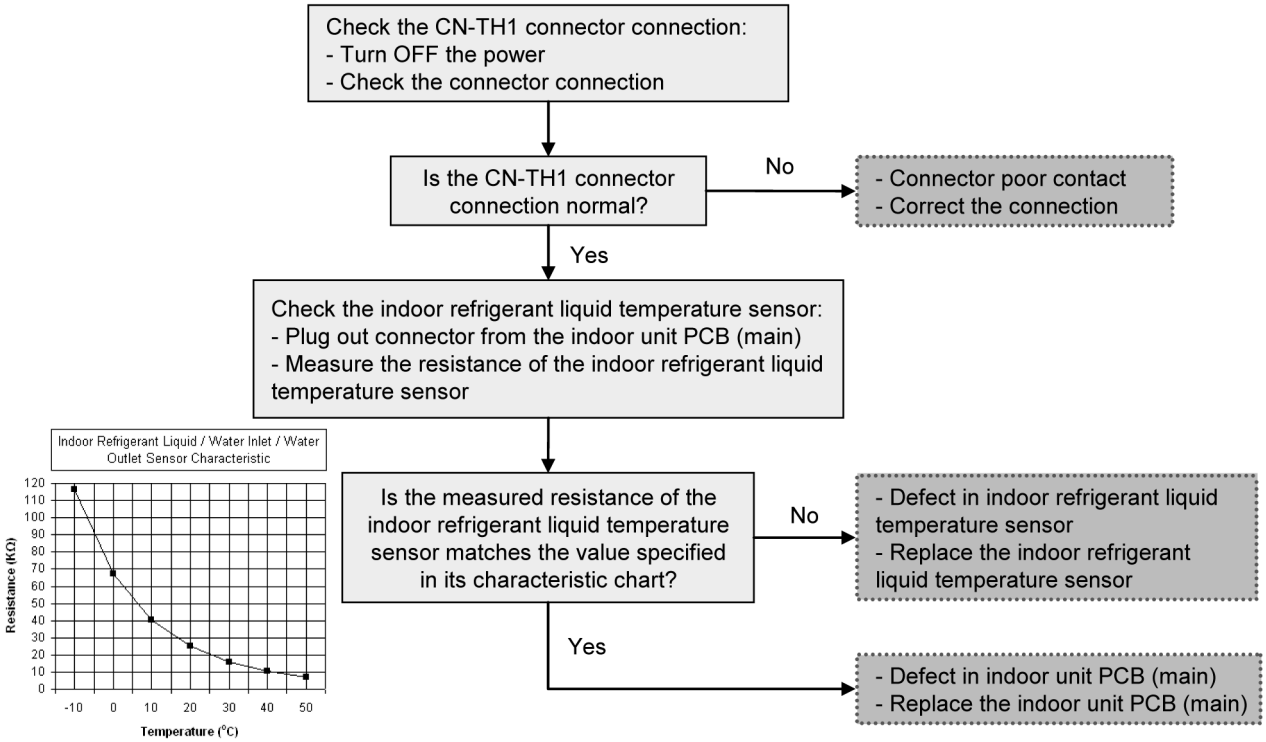
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.5 Service Valve Error (H27)

Malfunction Decision Conditions:

During cooling operation, when:-
[a] Indoor refrigerant pipe temperature at compressor startup - present indoor refrigerant pipe temperature < 3°C
[b] Present high pressure - high pressure at compressor startup < 3kg/cm^2
**Judgment only for first time cooling operation and not during pump down operation.

Malfunction Caused:

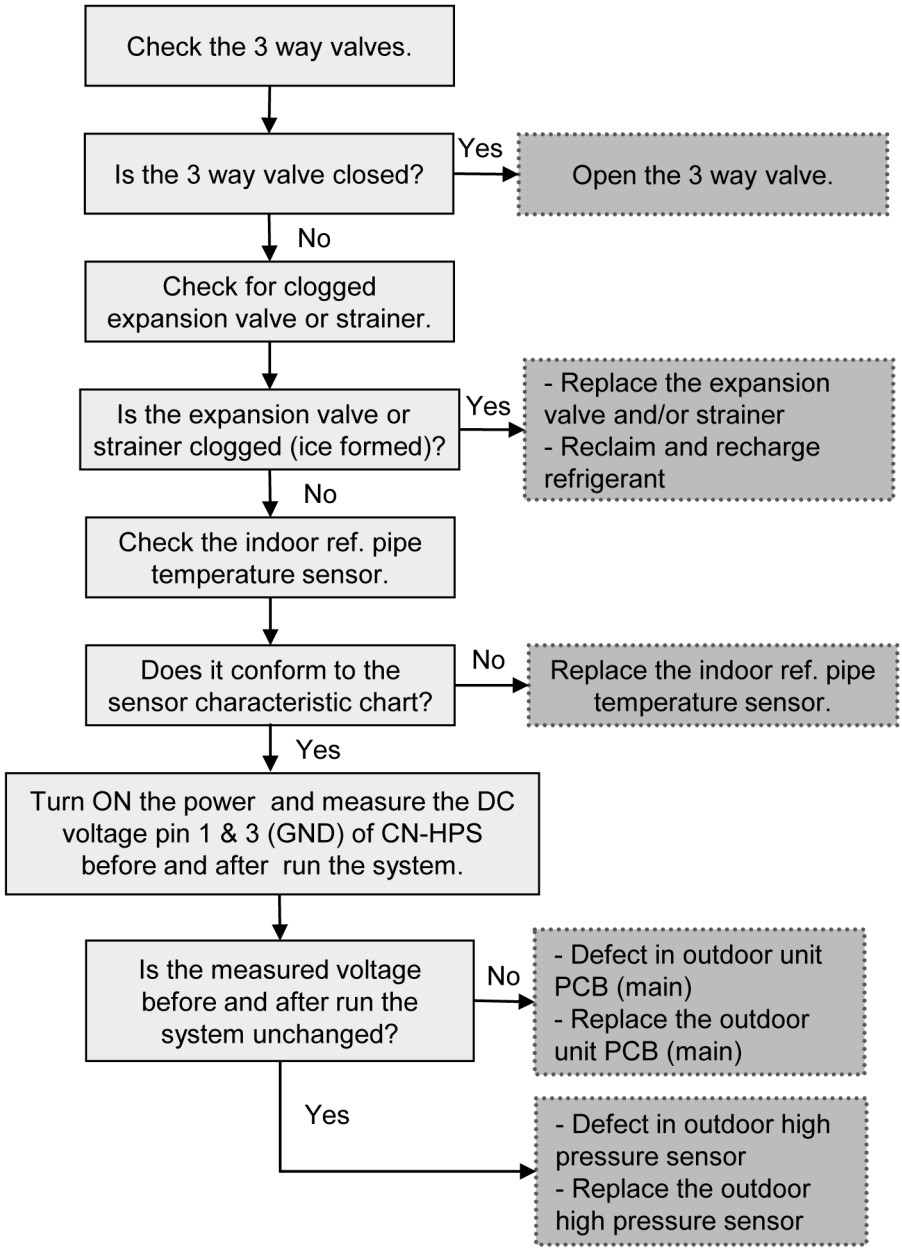
1 3 way valves closed.
2 Faulty high pressure sensor.
3 Faulty indoor refrigerant pipe temperature sensor
4 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 minutes.

Troubleshooting: Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



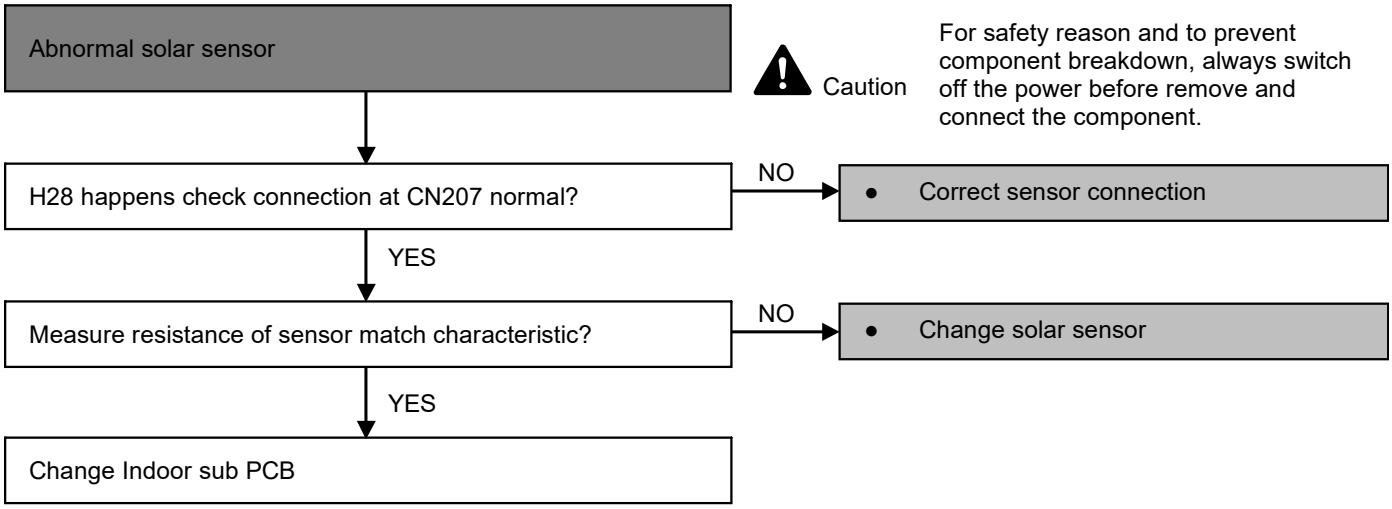
18.5.6 Abnormal Solar Sensor (H28)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty solar sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



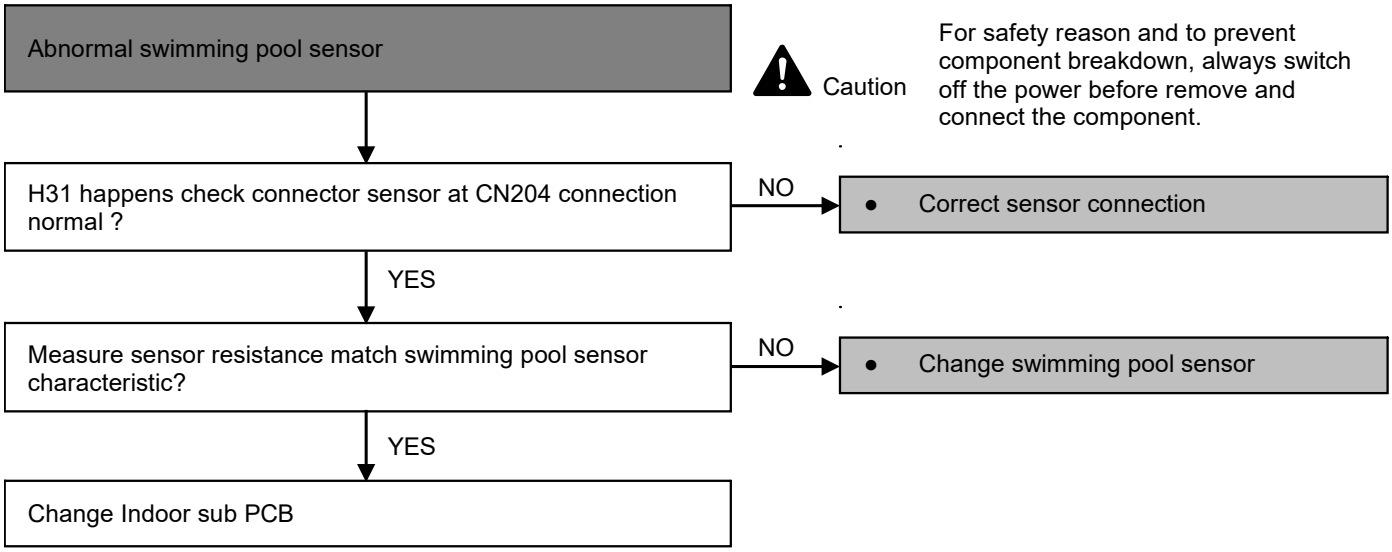
18.5.7 Abnormal Swimming Pool Sensor (H31)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty swimming pool sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



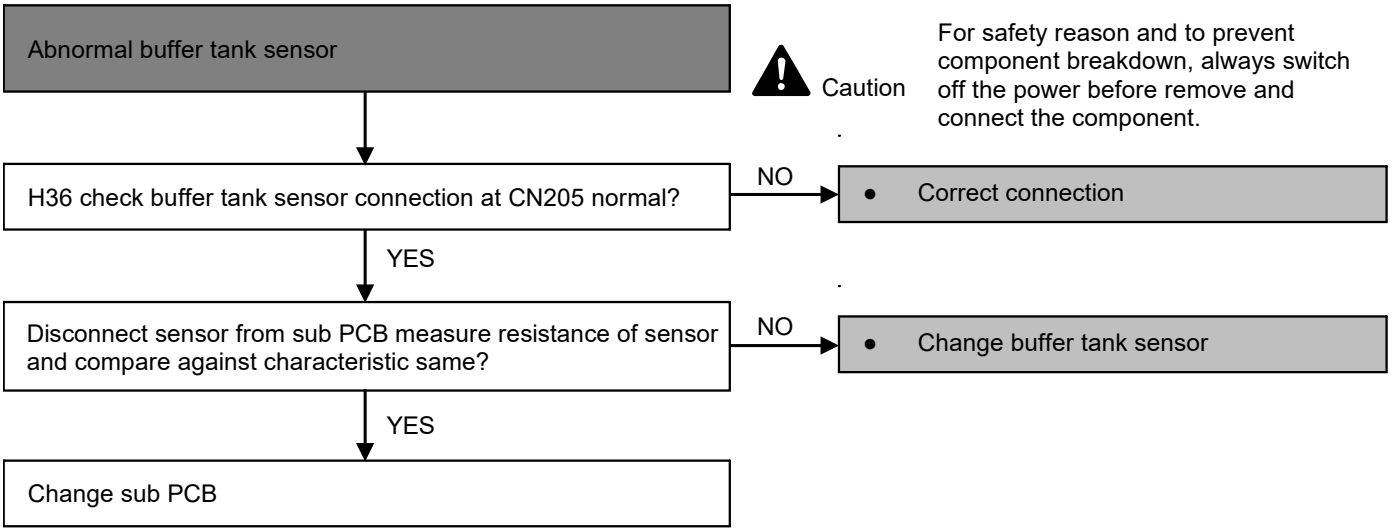
18.5.8 Abnormal Buffer Tank Sensor (H36)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

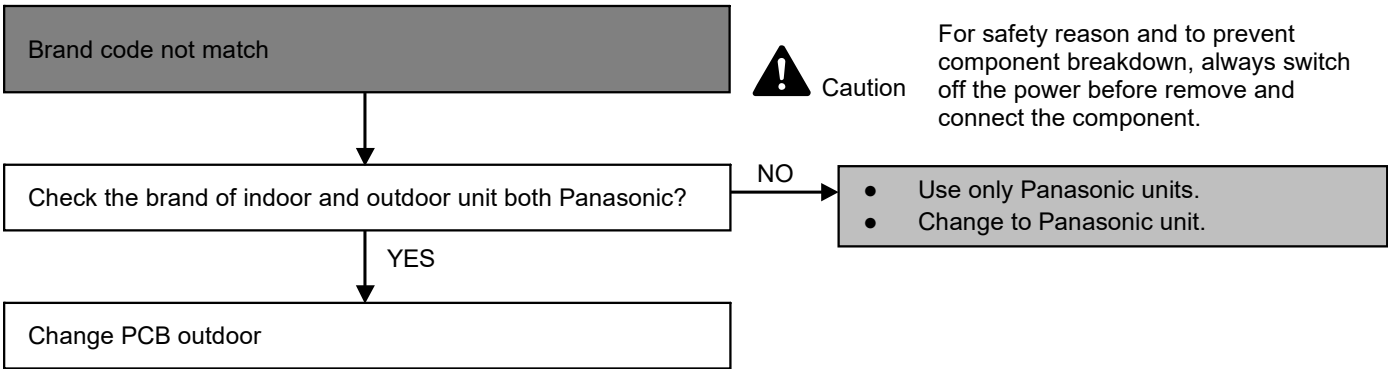
Continue for 5 seconds.



18.5.9 Brand Code Not Matching (H38)

Malfunction Caused:

1 Indoor and outdoor brand code not match.



18.5.10 Compressor Low Pressure Protection (H42)

Malfunction Decision Conditions:

During operation of heating and after 5 minutes compressor ON, when outdoor pipe temperature below -37°C or above 26°C is detected by the outdoor pipe temperature sensor.

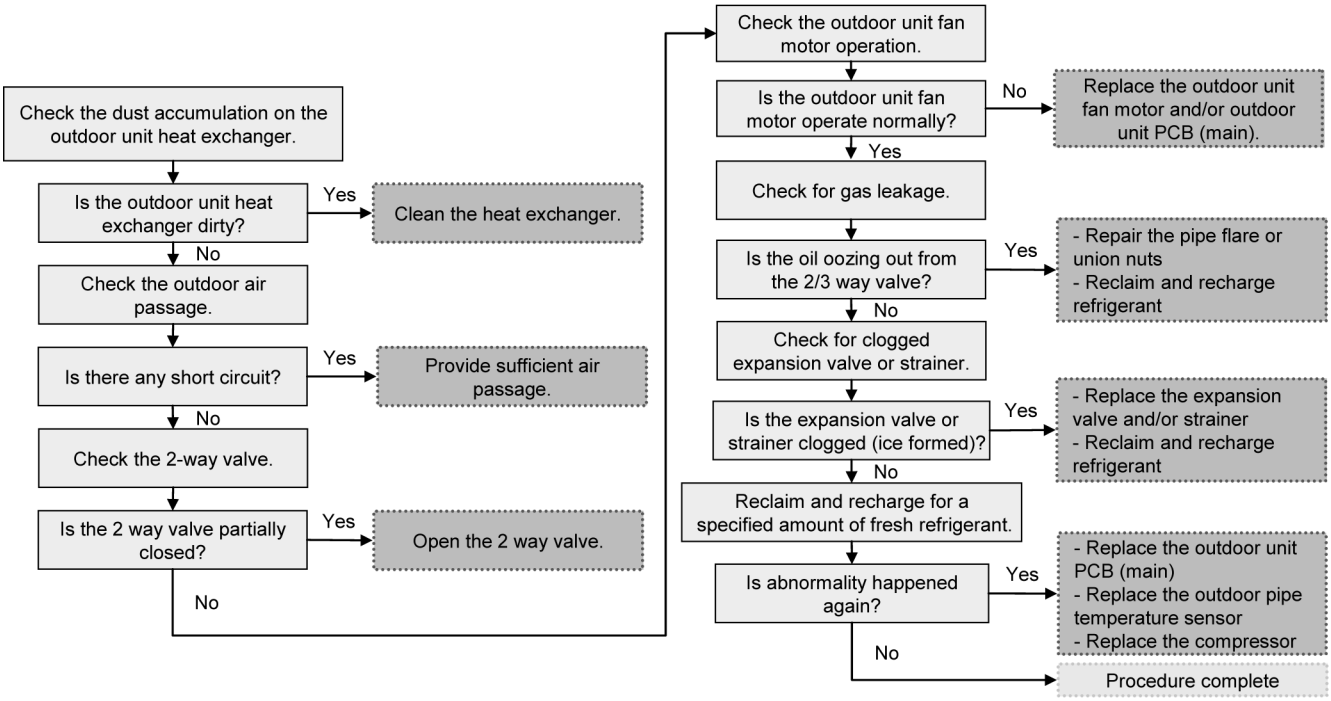
Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 2 way valve partially closed.
4 Faulty outdoor unit fan motor.
5 Refrigerant shortage (refrigerant leakage).
6 Clogged expansion valve or strainer.
7 Faulty outdoor pipe temperature sensor.
8 Faulty outdoor unit main PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



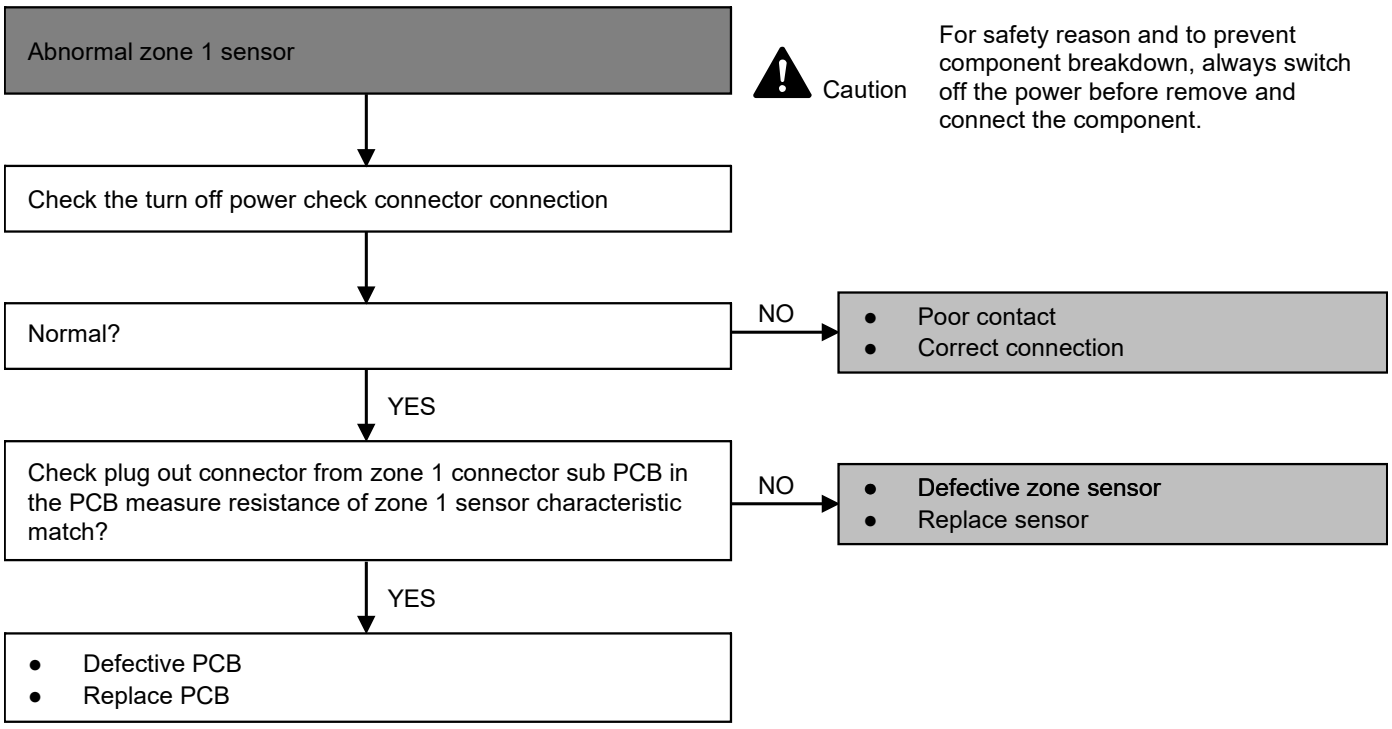
18.5.11 Abnormal Zone 1 Sensor (H43)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



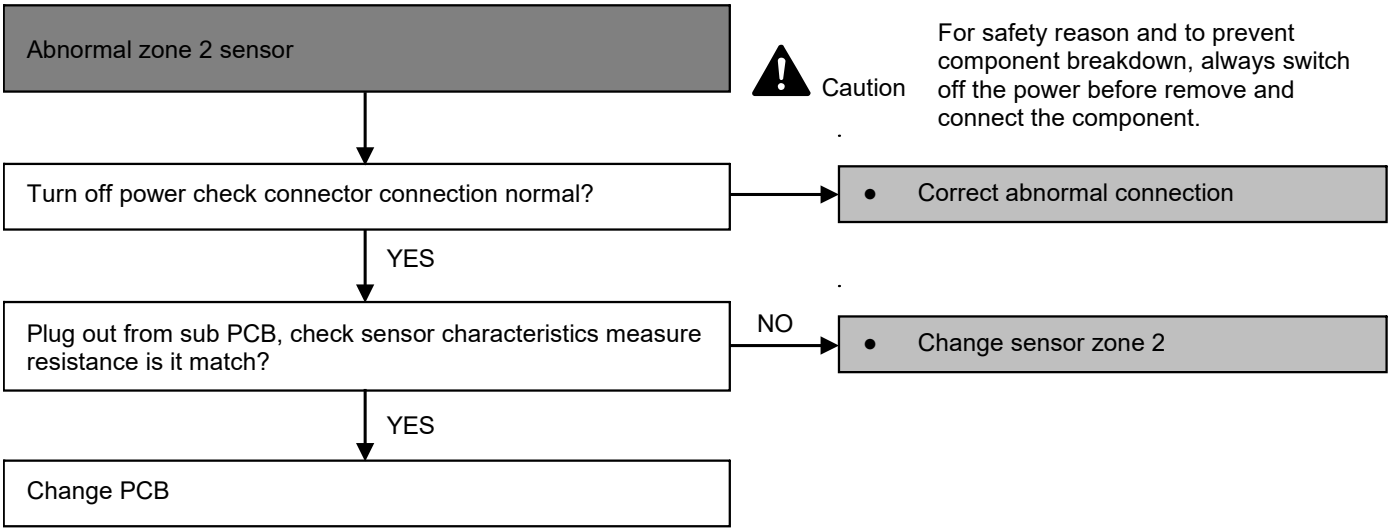
18.5.12 Abnormal Zone 2 Sensor (H44)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



18.5.13 Water Flow Switch Abnormality (H62)

Malfunction Decision Conditions:

During operation of cooling and heating, the water flow detected by the indoor water flow switch is used to determine water flow error.

Malfunction Caused:

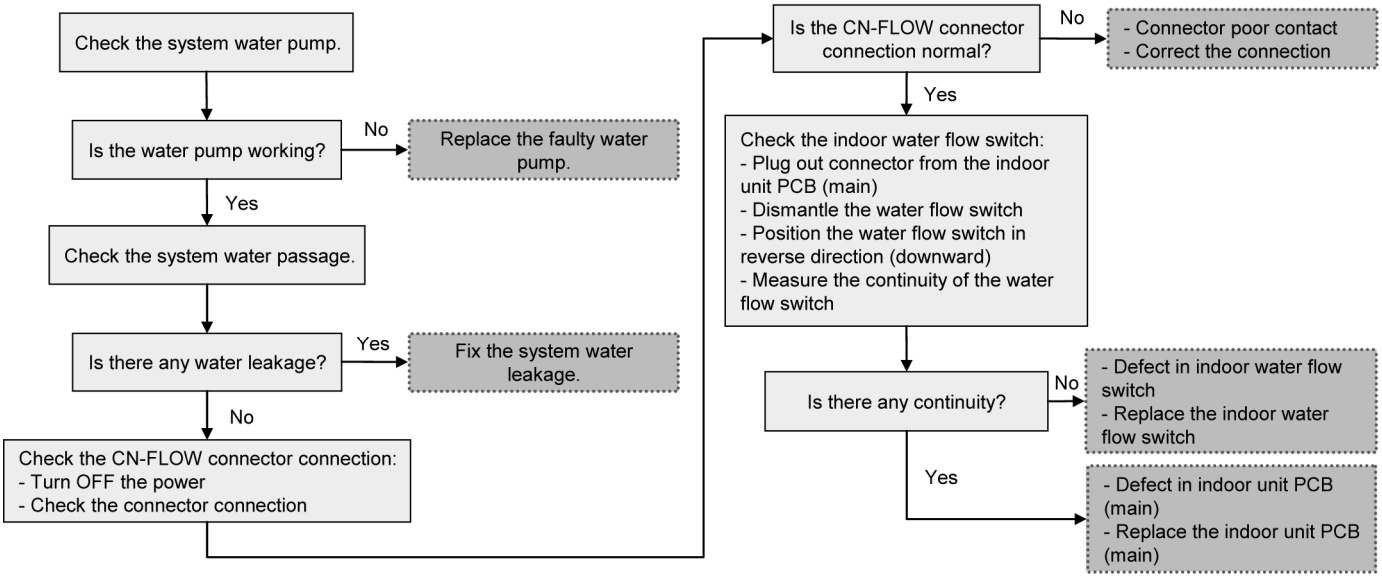
1 Faulty water pump.
2 Water leak in system.
3 Faulty connector connection.
4 Faulty water flow switch.
5 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 10 seconds (but no judgment for 9 minutes after compressor startup/restart).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.14 Outdoor High Pressure Abnormality (H64)

Malfunction Decision Conditions:

During operation of cooling and heating, when the outdoor high pressure sensor output signal is 0 Vdc or 5 Vdc.

Malfunction Caused:

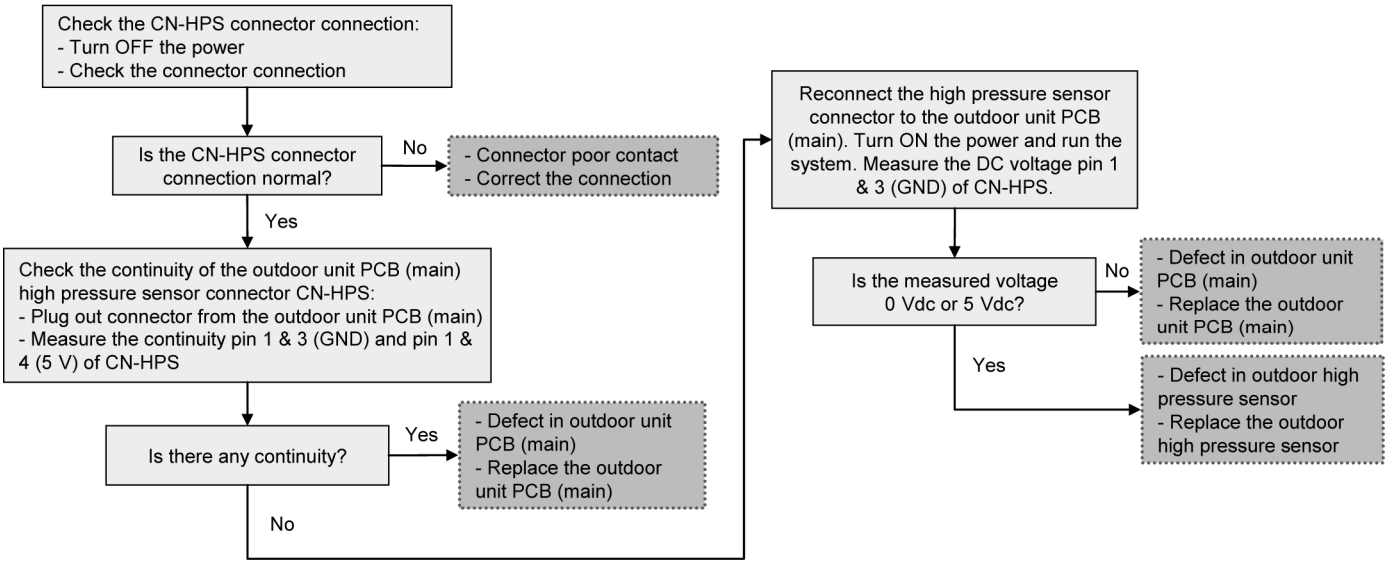
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.15 Deice Circulation Error (H65)

Malfunction Decision Conditions:

During startup and operation of deice (mode 2), the water flow (> 5l/min) detected by the water flow switch is used to determine deice circulation error.

Malfunction Caused:

1 Water flow in air-to-water heatpump indoor unit circuitry.
2 Faulty indoor unit water flow switch.
3 Faulty indoor unit water pump.
4 Faulty indoor unit PCB.

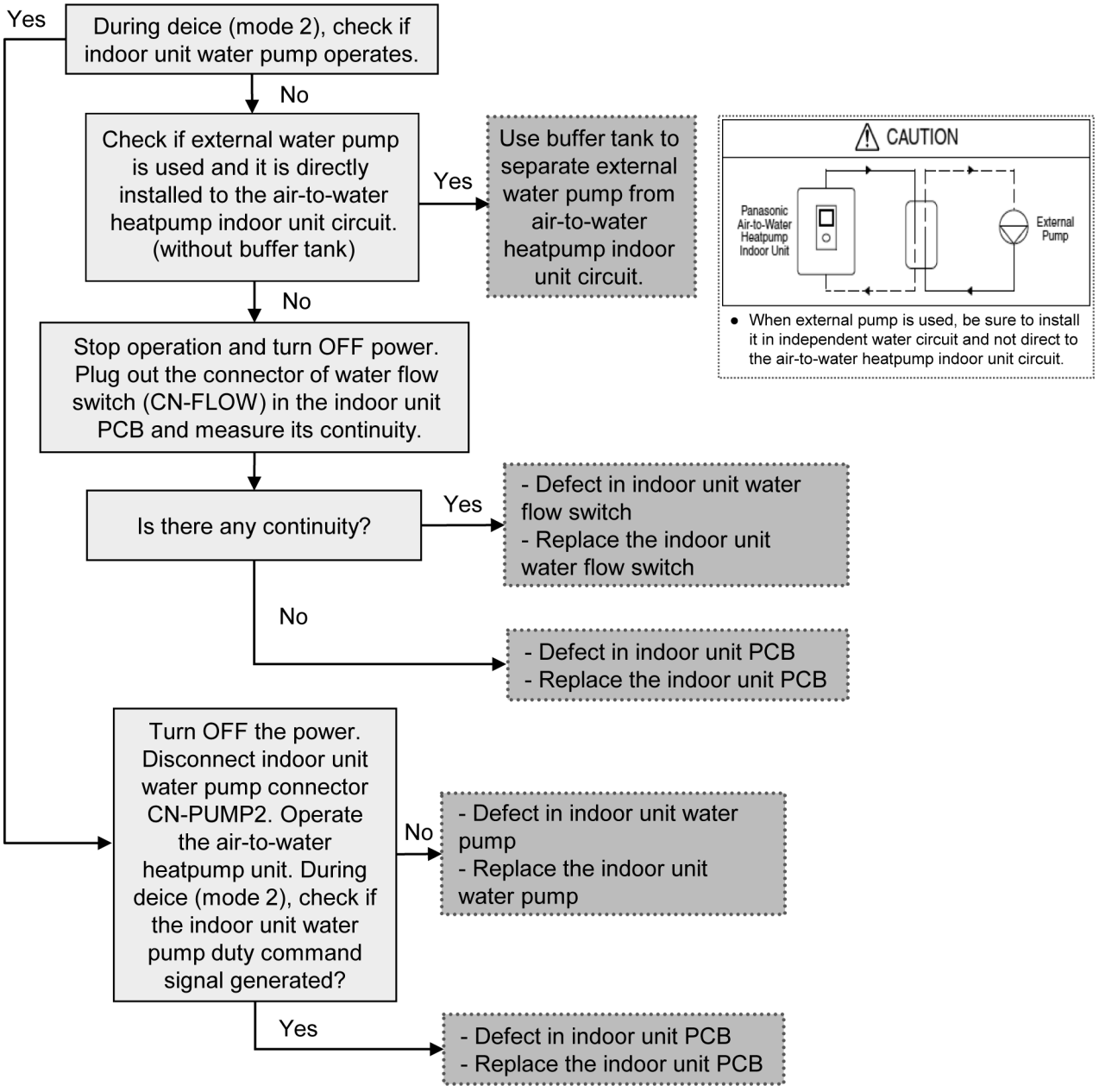
Abnormality Judgment:

Continue for 10 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



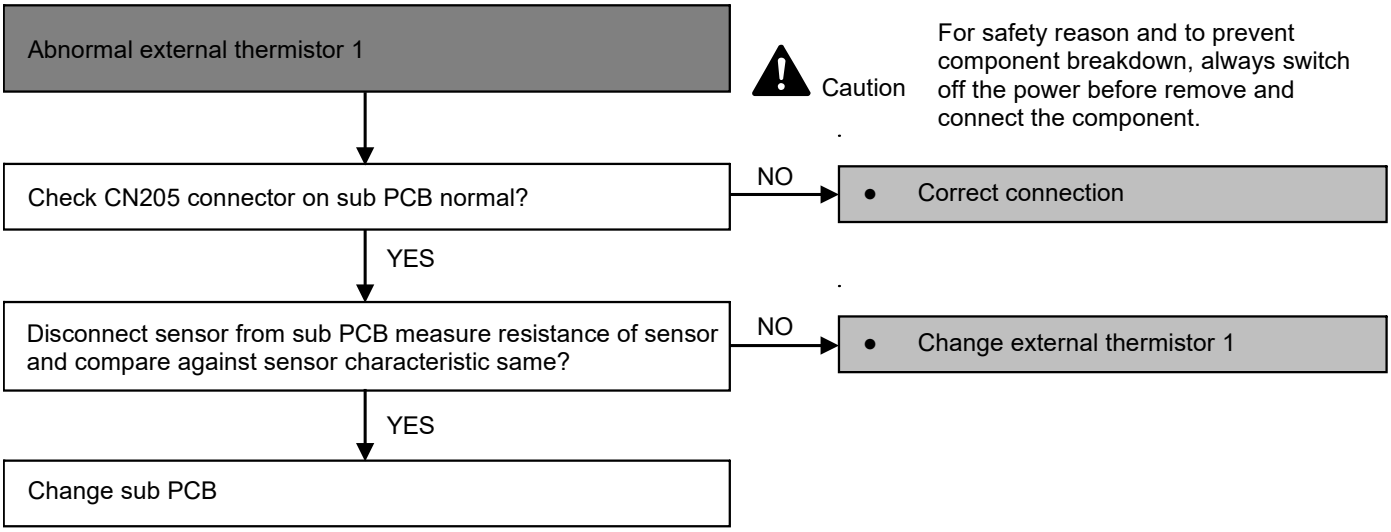
18.5.16 Abnoraml External Thermistor 1 (H67)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 1 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



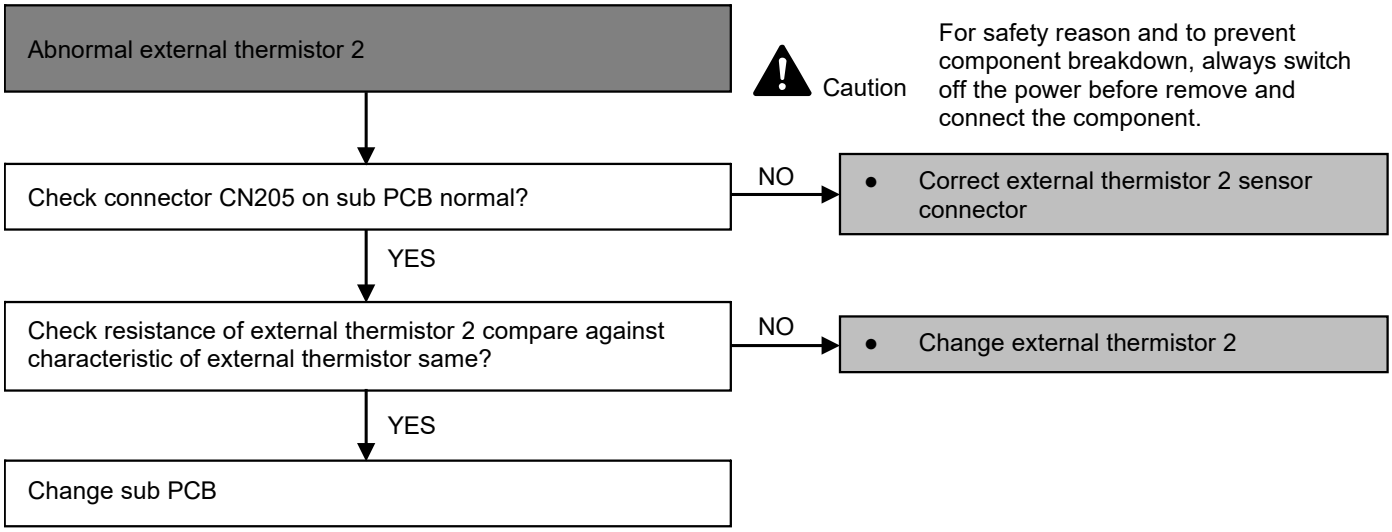
18.5.17 Abnoraml External Thermistor 2 (H68)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 2 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



18.5.18 Indoor Backup Heater OLP Abnormality (H70)

Malfunction Decision Conditions:

During operation of indoor backup heater, when no power supplies to indoor backup heater or OLP open circuit.

Malfunction Caused:

1 Faulty power supply connector connection.
2 Faulty connector connection.
3 Faulty indoor backup heater overload protector (OLP).
4 Faulty indoor unit PCB (main).

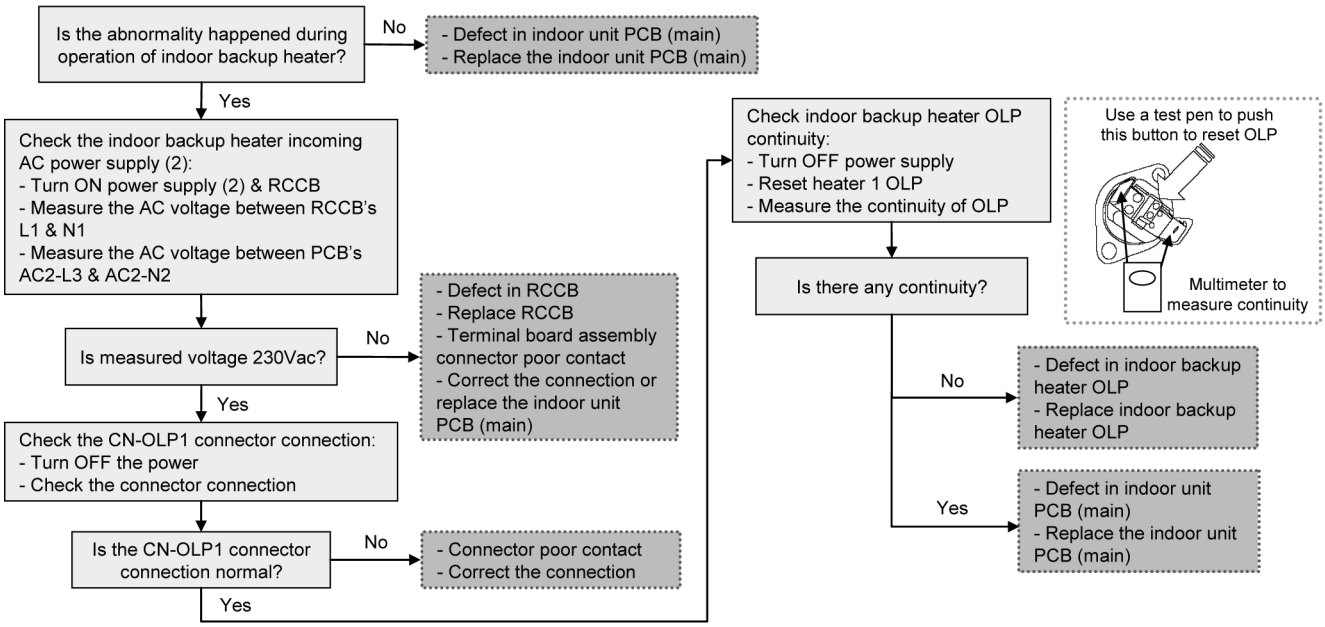
Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.19 Tank Temperature Sensor Abnormality (H72)

Malfunction Decision Conditions:

When tank connection is set to ON, the temperatures detected by the tank temperature sensor are used to determine sensor error.

Malfunction Caused:

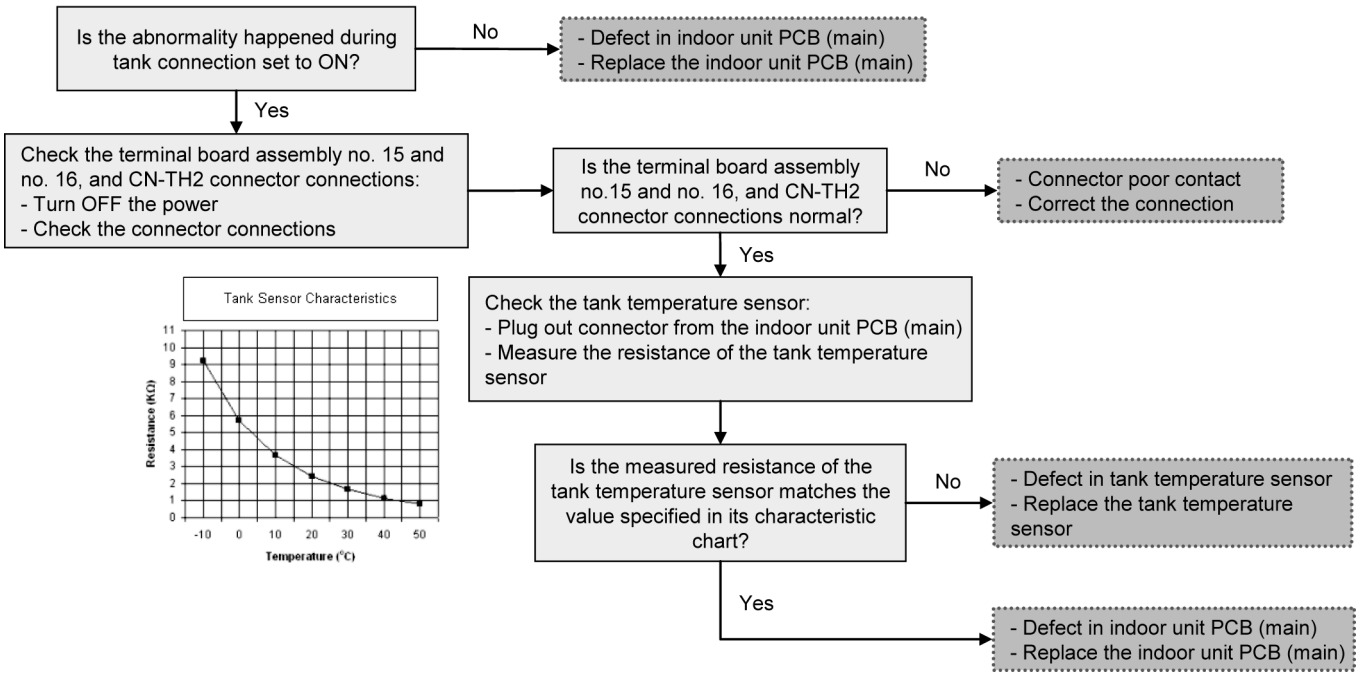
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.20 PCB Communication Error (H74)

Malfunction Decision Conditions:

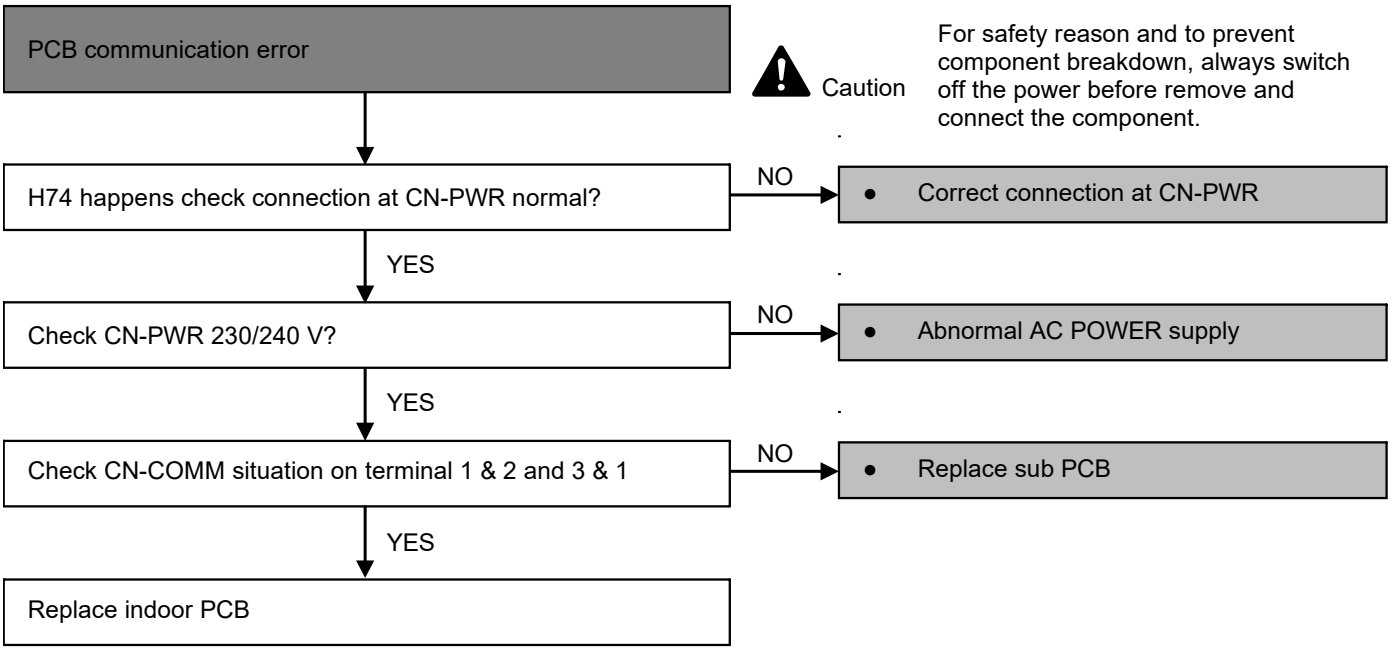
When External PCB connection is select “YES” and no communication with External PCB micon for 10 seconds and above.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty indoor PCB.
3 Faulty indoor sub PCB.

Abnormality Judgment:

After 1 minute operation started.



18.5.21 Indoor-Control Panel Communication Abnormality (H76)

Malfunction Decision Conditions:

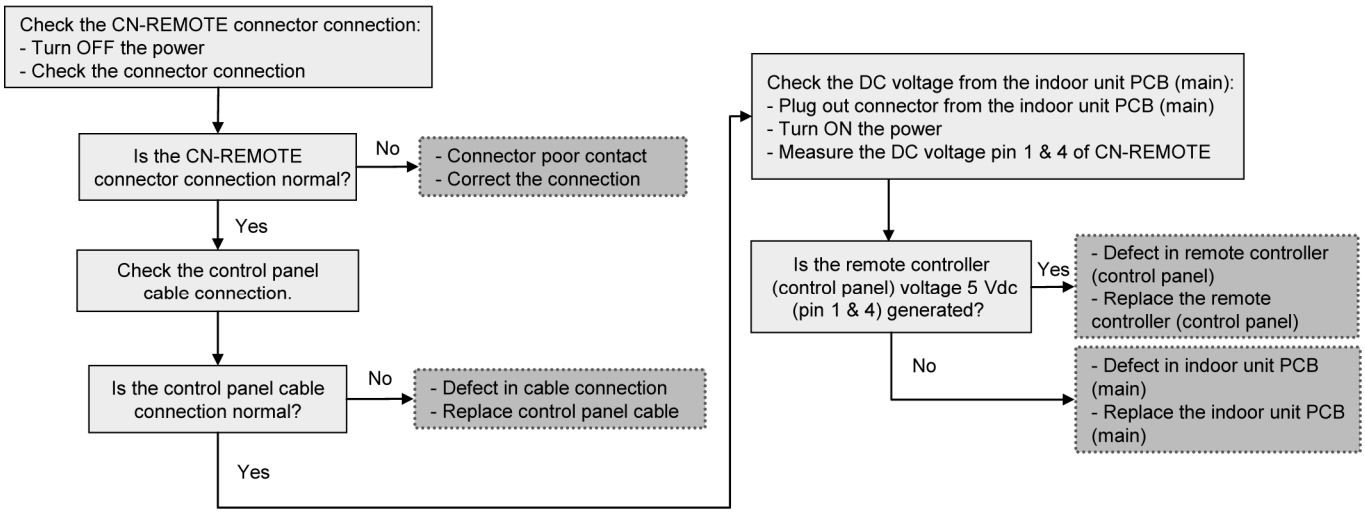
During standby and operation of cooling and heating, indoor-control panel error occur.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty control panel.
3 Faulty indoor unit PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.22 Indoor/Outdoor Abnormal Communication (H90)

Malfunction Decision Conditions:

During operation of cooling and heating, the data received from outdoor unit in indoor unit signal transmission is checked whether it is normal.

Malfunction Caused:

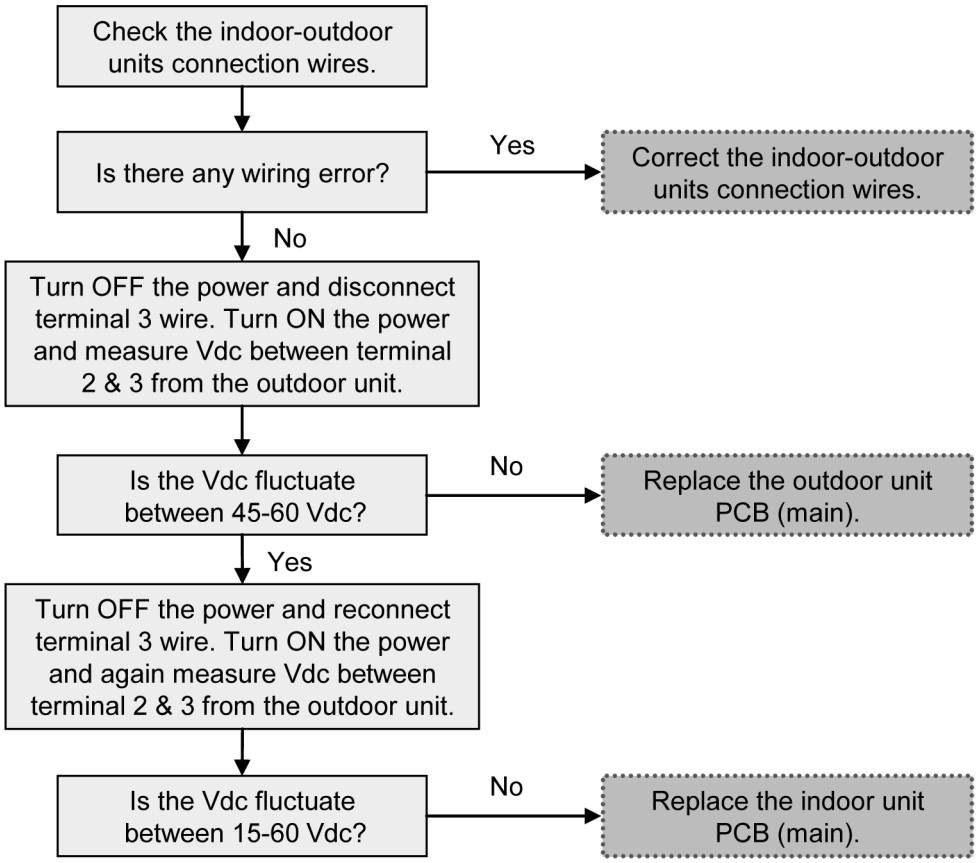
1 Faulty outdoor unit PCB (main).
2 Faulty indoor unit PCB (main).
3 Indoor-outdoor signal transmission error due to wrong wiring.
4 Indoor-outdoor signal transmission error due to breaking of wire in the connection wires between the indoor and outdoor units.
5 Indoor-outdoor signal transmission error due to disturbed power supply waveform.

Abnormality Judgment:

Continue for 1 minute after operation.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.23 Tank Booster Heater OLP Abnormality (H91)

Malfunction Decision Conditions:

During operation of tank booster heater, and tank booster heater OLP open circuit.

Malfunction Caused:

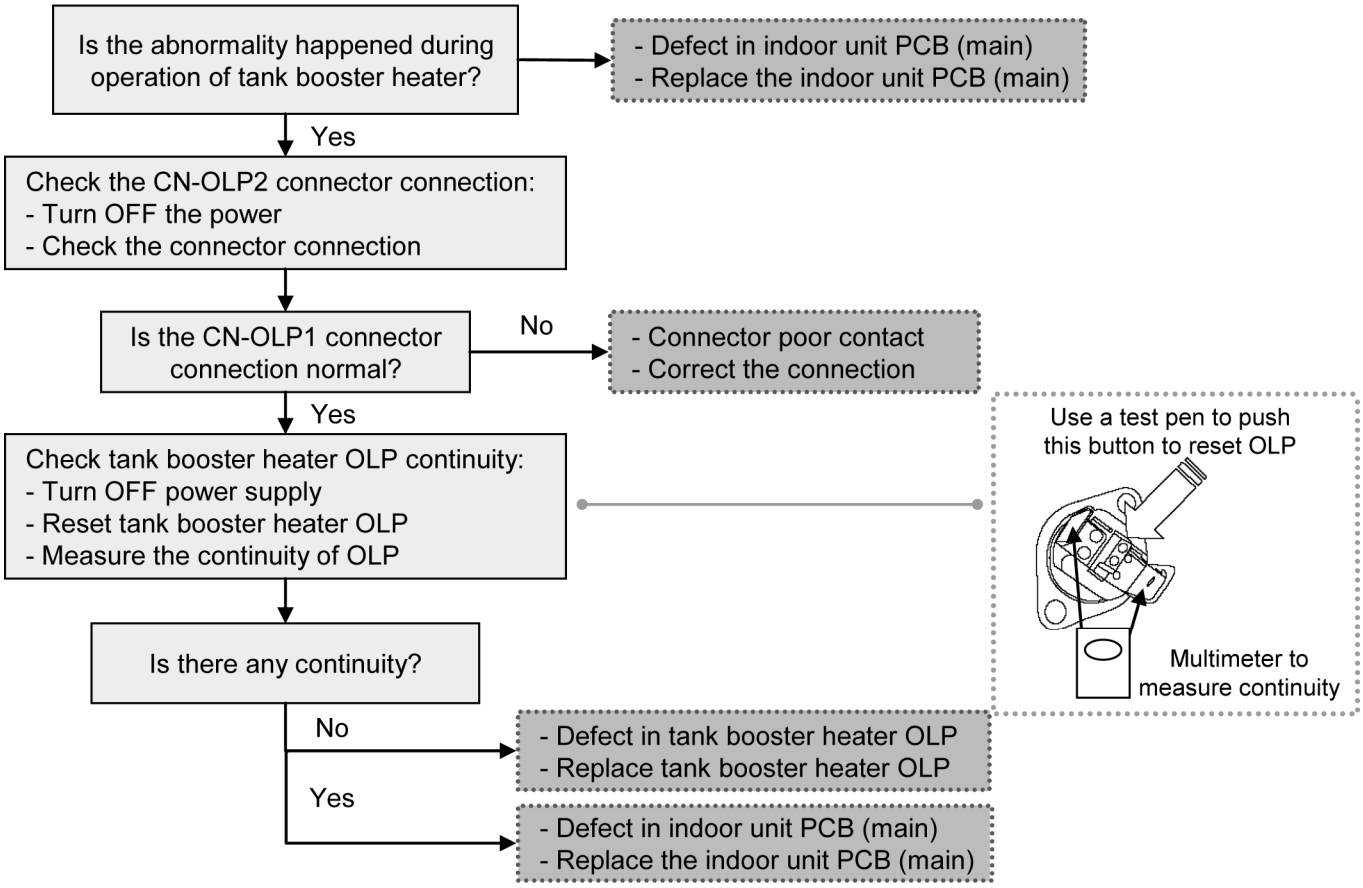
1 Faulty connector connection.
2 Faulty tank booster heater overload protector (OLP).
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.24 Unspecified Voltage between Indoor and Outdoor (H95)

Malfunction Decision Conditions:

The supply power is detected for its requirement by the indoor/outdoor transmission.

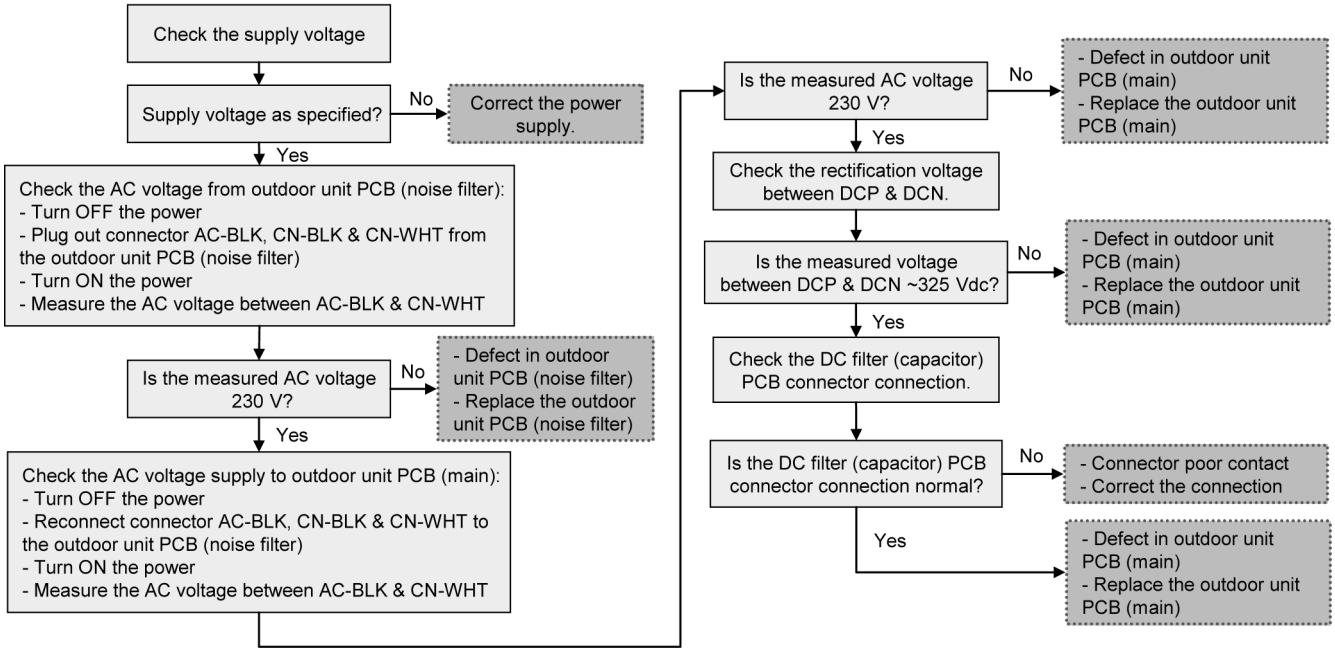
Malfunction Caused:

1 Insufficient power supply.
2 Faulty outdoor unit PCB (noise filter/main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.25 Outdoor High Pressure Protection (H98 / F95)

Malfunction Decision Conditions:

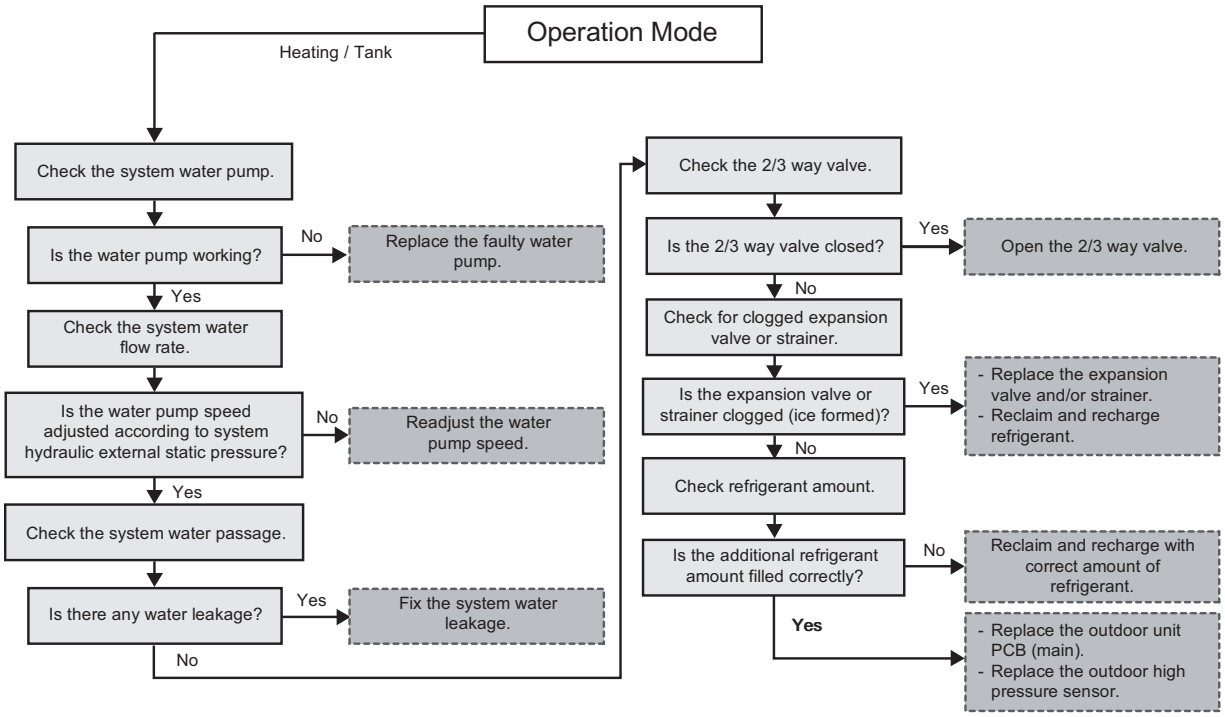
During operation of cooling / heating, when pressure 4.2 MPa and above is detected by outdoor high pressure sensor.

Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 Dust accumulation in the outdoor unit heat exchanger.
5 Air short circuit at outdoor.
6 Faulty outdoor unit fan motor.
7 2/3 way closed.
8 Clogged expansion valve or strainer.
9 Excessive refrigerant.
10 Faulty outdoor high pressure sensor.
11 Faulty outdoor unit PCB (main).

Troubleshooting:

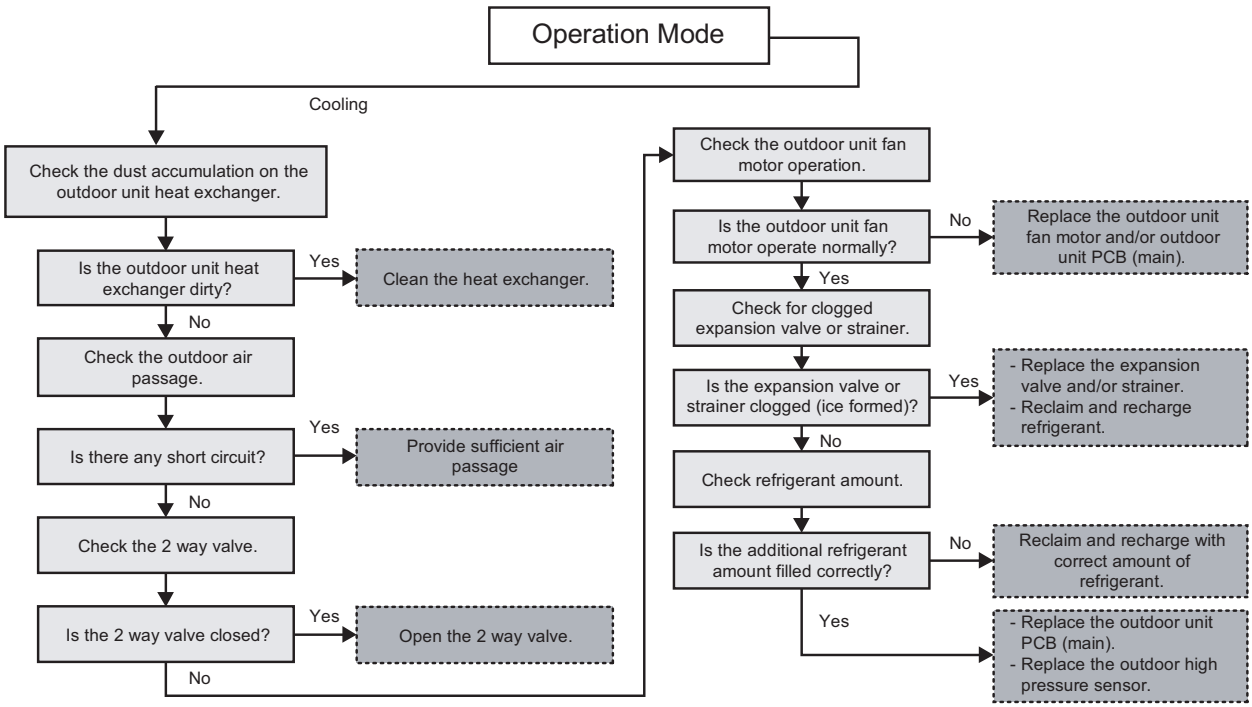
Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.26 Indoor Freeze-up Protection (H99)

Malfunction Decision Conditions:

During anti-freezing control in cooling operation, when the indoor refrigerant liquid temperature < 0°C.

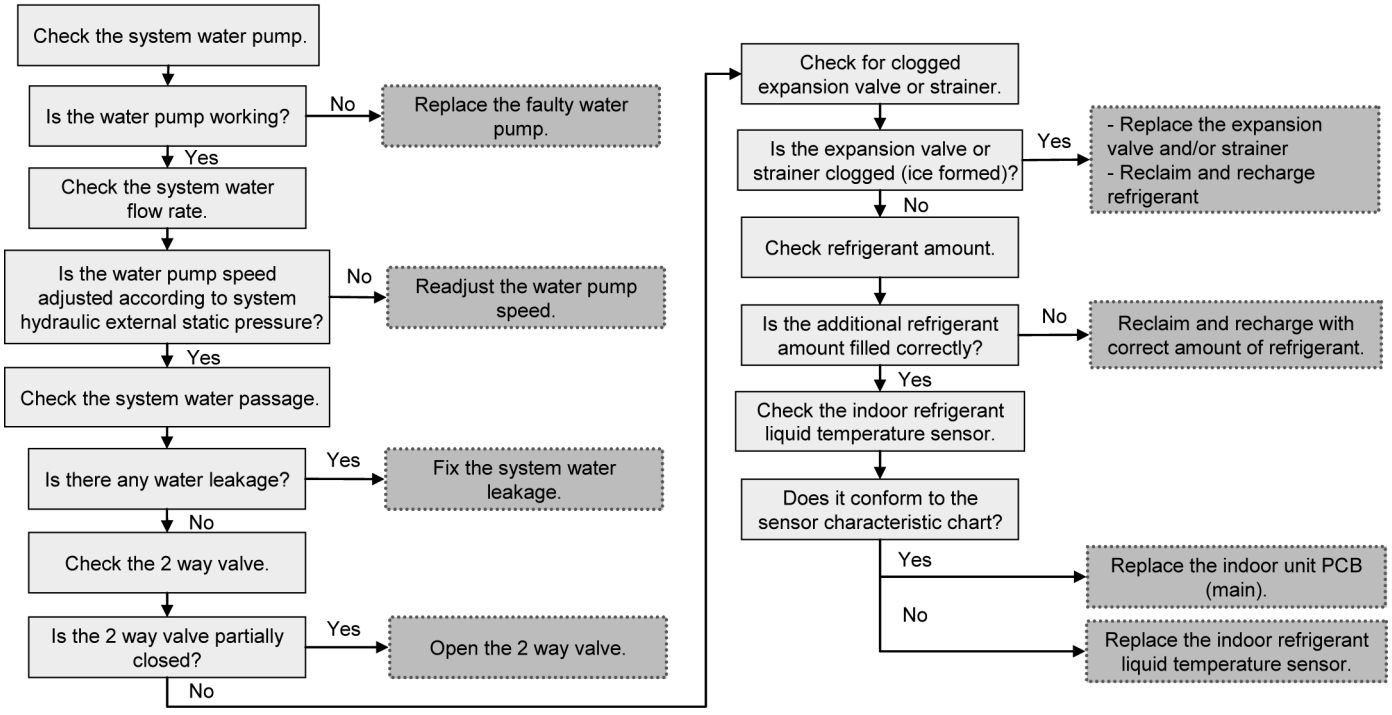
Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 2 way valve partially closed.
5 Clogged expansion valve or strainer.
6 Refrigerant shortage (refrigerant leakage).
7 Faulty indoor refrigerant liquid temperature sensor.
8 Faulty indoor unit PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.27 Outdoor High Pressure Switch Activate (F12)

Malfunction Decision Conditions:

During operation of cooling and heating, when pressure 4.3 MPa and above is detected by outdoor high pressure switch.

Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 Faulty water pump.
4 Insufficient water flow rate in system.
5 Water leak in system.
6 2/3 way valve closed.
7 Clogged expansion valve or strainer.
8 Excessive refrigerant.
9 Faulty outdoor high pressure sensor and switch.
10 Faulty outdoor unit PCB.

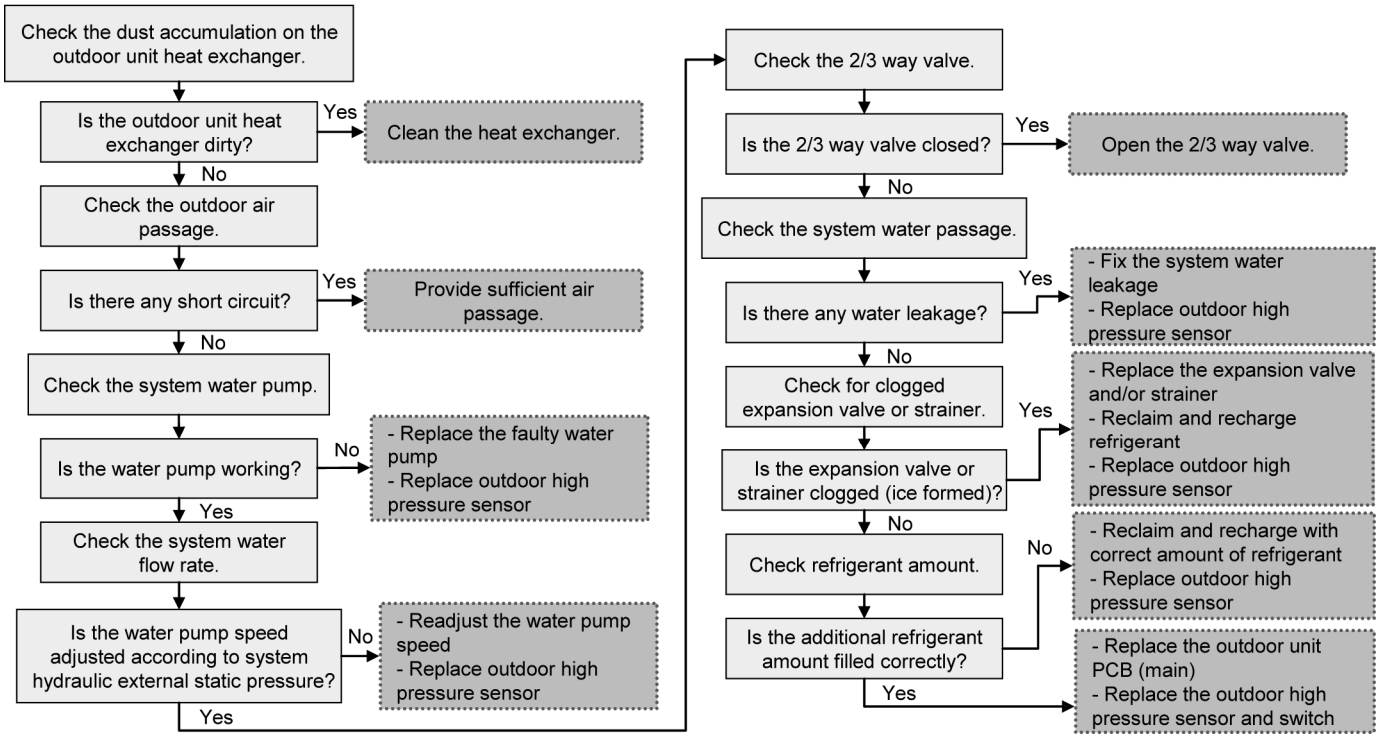
Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.28 Compressor Rotation Failure (F14)

Malfunction Decision Conditions:

A compressor rotation failure is detected by checking the compressor running condition through the position detection circuit.

Malfunction Caused:

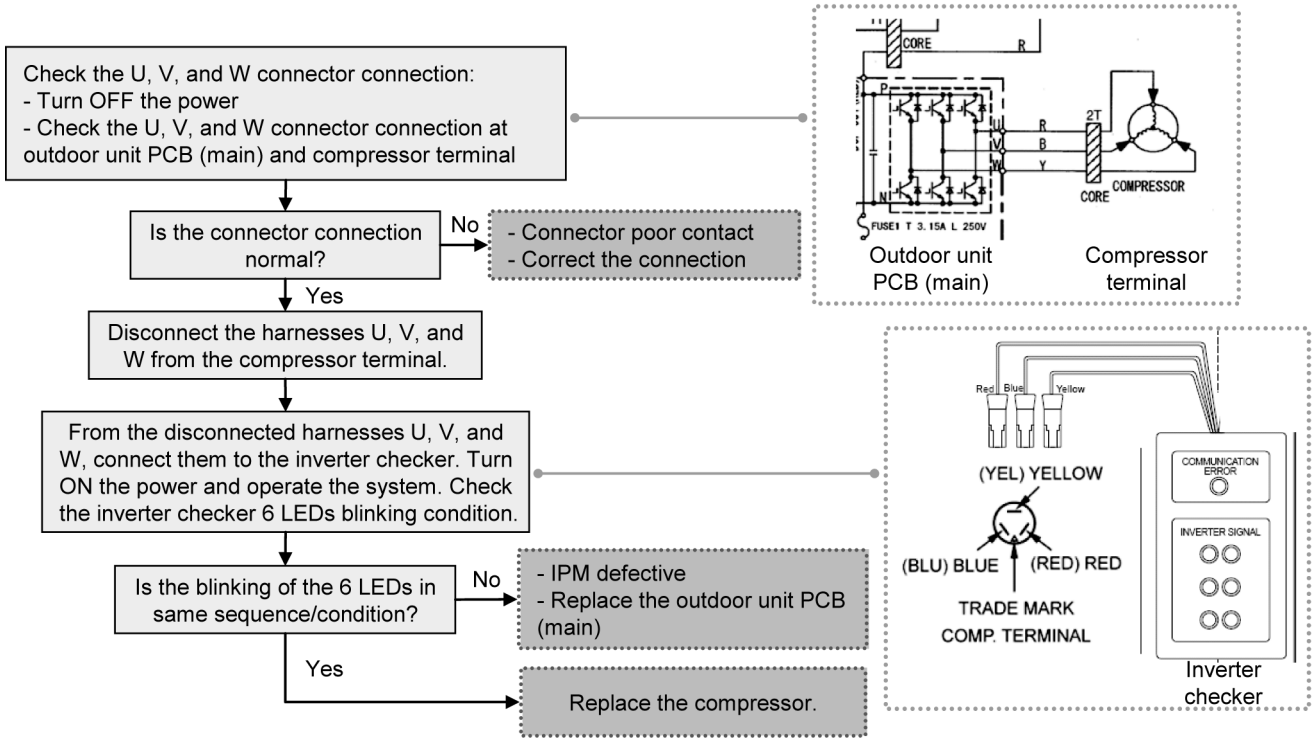
1 Compressor terminal disconnect.
2 Faulty outdoor unit PCB (main).
3 Faulty compressor.

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.29 Outdoor Fan Motor (DC Motor) Mechanism Locked (F15)

Malfunction Decision Conditions:

The rotation speed detected by the Hall IC of the fan motor during fan motor operation is used to determine abnormal fan motor (feedback of rotation > 2550 rpm or < 20 rpm).

Malfunction Caused:

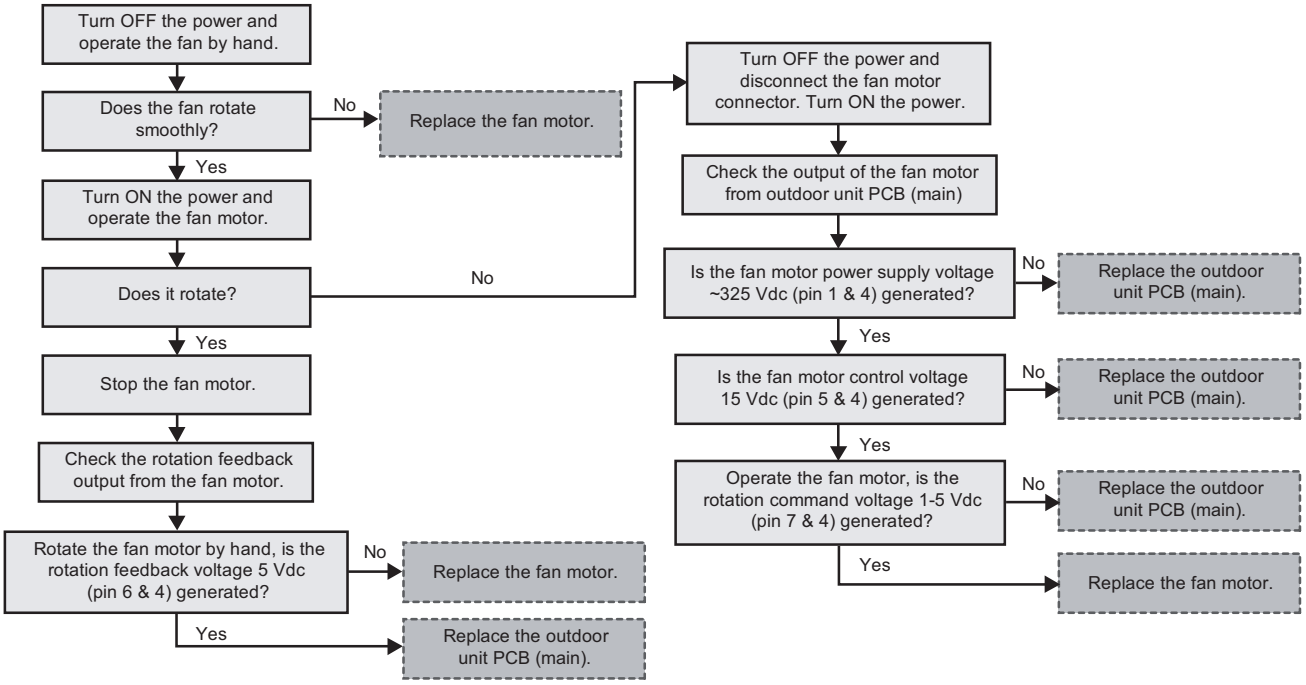
1 Operation stop due to short circuit inside the fan motor winding.
2 Operation stop due to breaking of wire inside the fan motor.
3 Operation stop due to breaking of fan motor lead wires.
4 Operation stop due to fan motor Hall IC malfunction.
5 Operation error due to faulty outdoor unit PCB.

Abnormality Judgment:

Continue 2 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.30 Input Over Current Detection (F16)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor current above 28.0A (Heating) and 20.0A (Cooling) is detected by the current transformer (CT) in the outdoor unit PCB.

Malfunction Caused:

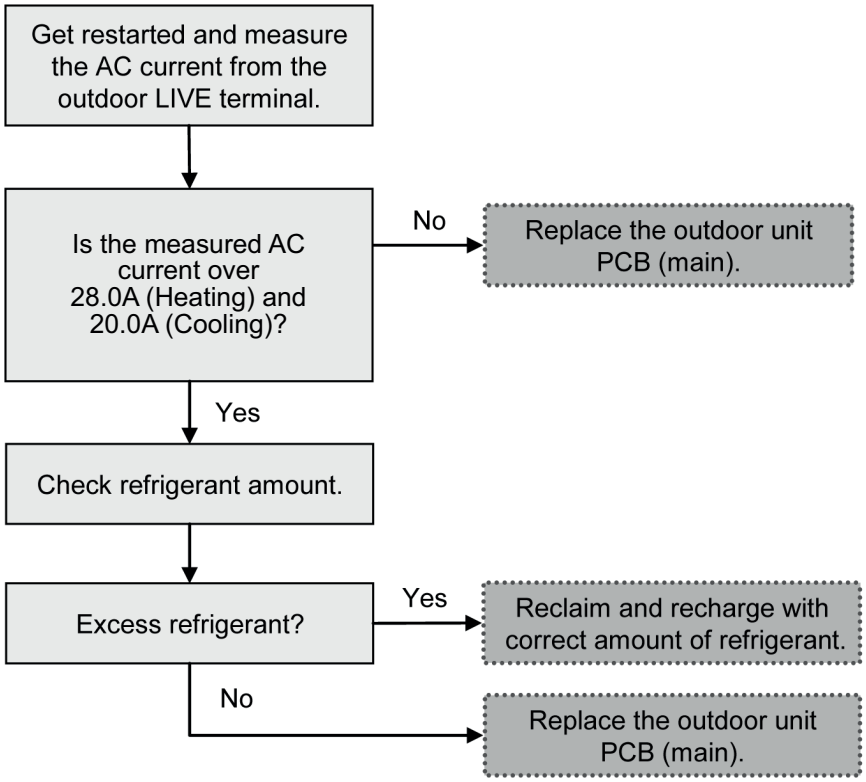
1 Excessive refrigerant.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.31 Compressor Overheating (F20)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature above 117°C is detected by the outdoor discharge pipe temperature sensor.

Malfunction Caused:

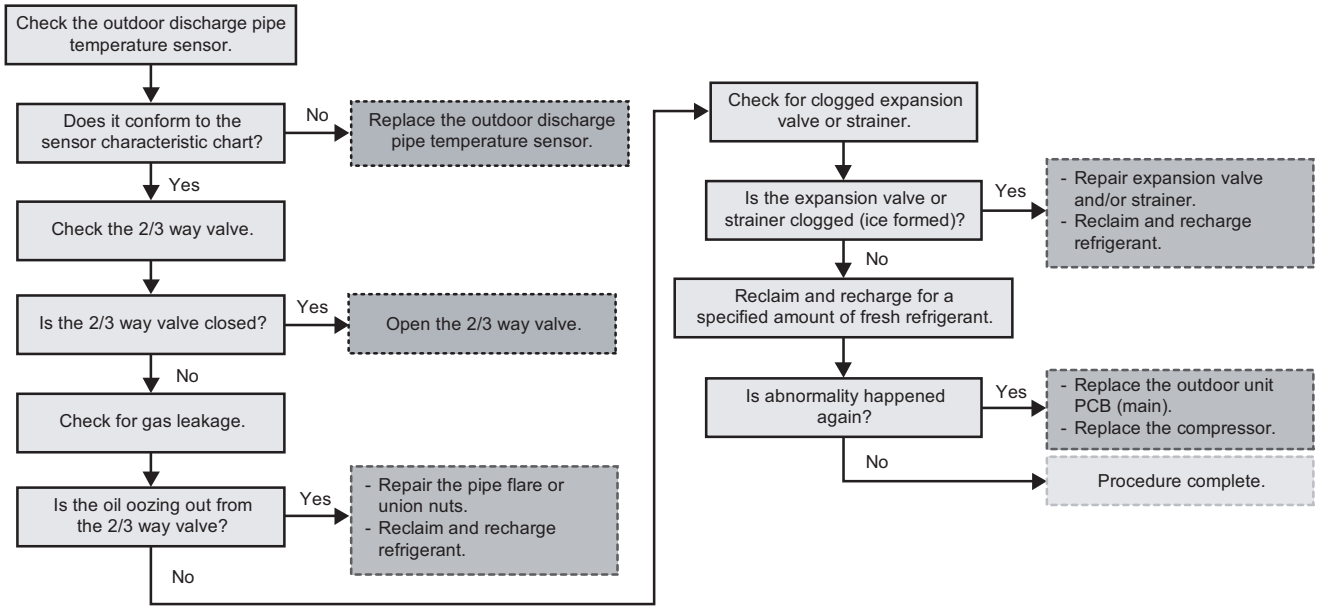
1 Faulty outdoor discharge pipe temperature sensor.
2 2/3 way valve closed.
3 Refrigerant shortage (refrigerant leakage).
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Faulty compressor.

Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.32 IPM Overheating (F22)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature 95°C is detected by the outdoor IPM temperature sensor.

Malfunction Caused:

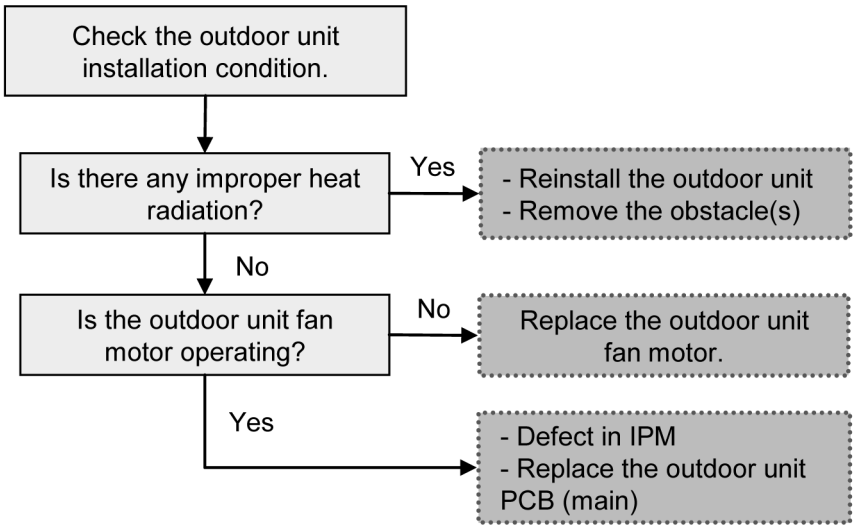
1 Faulty outdoor unit fan motor.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.33 Output Over Current Detection (F23)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor DC current is above 53 A is detected by the IPM DC Peak sensing circuitry in the outdoor unit PCB (main).

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty compressor.

Abnormality Judgment:

Continue for 7 times.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

Check the compressor winding resistance:
- Turn OFF the power and disconnect the harnesses U, V, and W
- Measure the winding resistance between U-V, V-W, and W-U

→ Are the compressor's winding resistance (U-V, V-W, or U-W) shorted?

Yes → - Compressor defective due to winding shorted
- Replace the compressor

No → - Outdoor electronic circuit defect due to power transistor shorted
- Replace the outdoor unit PCB (main)

- Checking the power transistor
- Never touch any live parts for at least 10 minutes after turning off the circuit breaker.
- If unavoidably necessary to touch a live part, make sure the power transistor's supply voltage is below 50 V using the tester.
- For the UVW, make measurements at the Faston terminal on the board or the relay connector.

Tester's negative terminal	Power transistor (+)	UVW	Power transistor (−)	UVW
Tester's positive terminal	UVW	Power transistor (+)	UVW	Power transistor (−)
Normal resistance	Several kohms to several Mohms			
Abnormal resistance	0 or ∞			

18.5.34 Refrigeration Cycle Abnormality (F24)

Malfunction Decision Conditions:

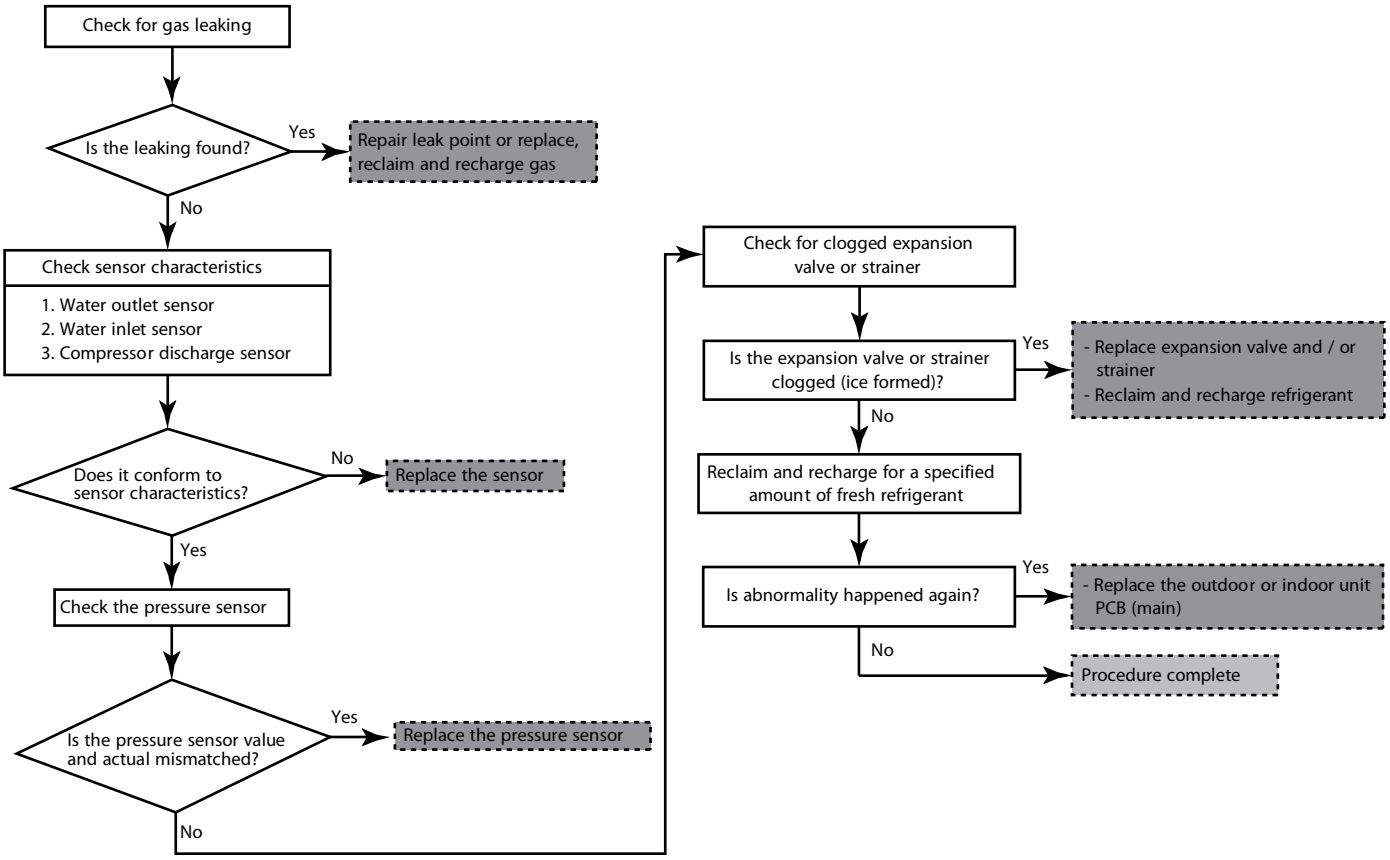
1 During operation running (heating / cooling) for more than 5 minutes expect deice, pumpdown and test mode.
2 During heating / cooling, water outlet and water inlet difference is less than 1°C.
3 During heating / cooling, high pressure < 1.90MPa (275 Psi) for more than 10 minutes or < 1.6MPa (232 Psi) for more than 4 minutes.
4 During heating / cooling, discharge temperature - saturation temperature of high pressure ≥ 60°C.

Malfunction Caused:

1 Refrigerant shortage (refrigerant leakage).
2 Faulty indoor water inlet, indoor water outlet, compressor discharge temp sensor or high pressure sensor.
3 2/3 way valve closed.
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Poor compression of compressor.

Abnormality Judgment:

Continue 2 times in 30 minutes.



18.5.35 Four Way Valve Abnormality (F25)

Malfunction Decision Conditions:

1 During heating operation, when the indoor pipe temperature of thermostat ON indoor unit < 0°C.
2 During cooling operation, when the indoor pipe temperature of thermostat ON indoor unit > 45°C.

Malfunction Caused:

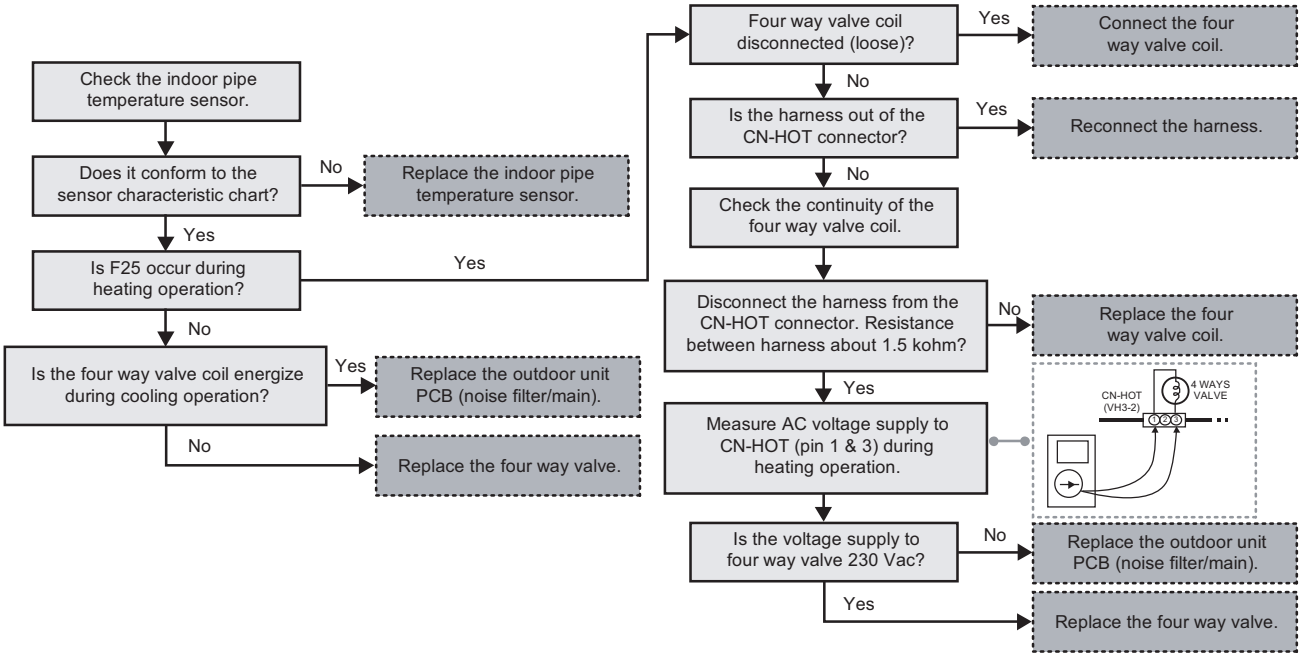
1 Faulty sensor.
2 Faulty connector connection.
3 Faulty outdoor unit PCB (noise filter/main).
4 Faulty four way valve.

Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.36 Outdoor High Pressure Switch Abnormal (F27)

Malfunction Decision Conditions:

During compressor stop, and outdoor high pressure switch is remain opened.

Malfunction Caused:

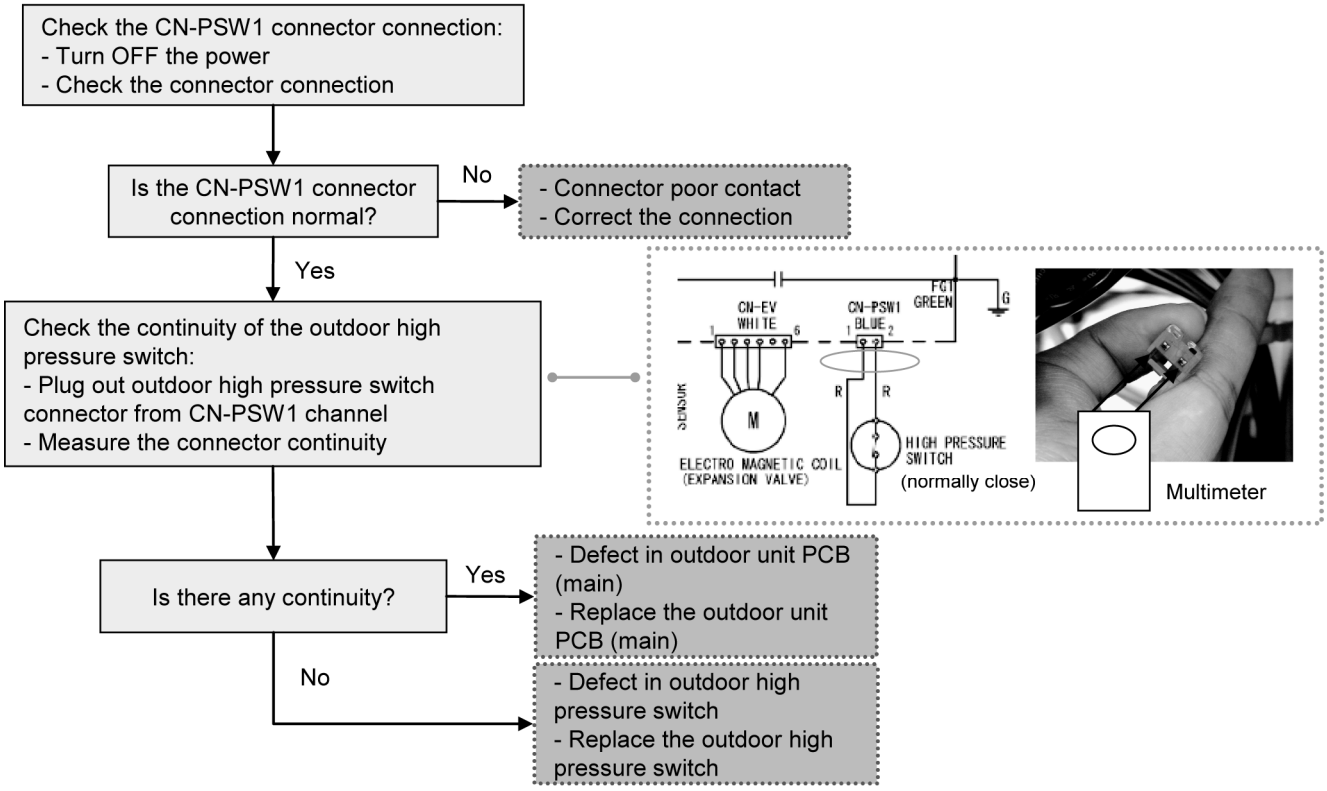
1 Faulty connector connection.
2 Faulty switch.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 1 minute.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.37 Indoor Water Outlet Temperature Sensor 2 Abnormality (F30)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor 2 are used to determine sensor error.

Malfunction Caused:

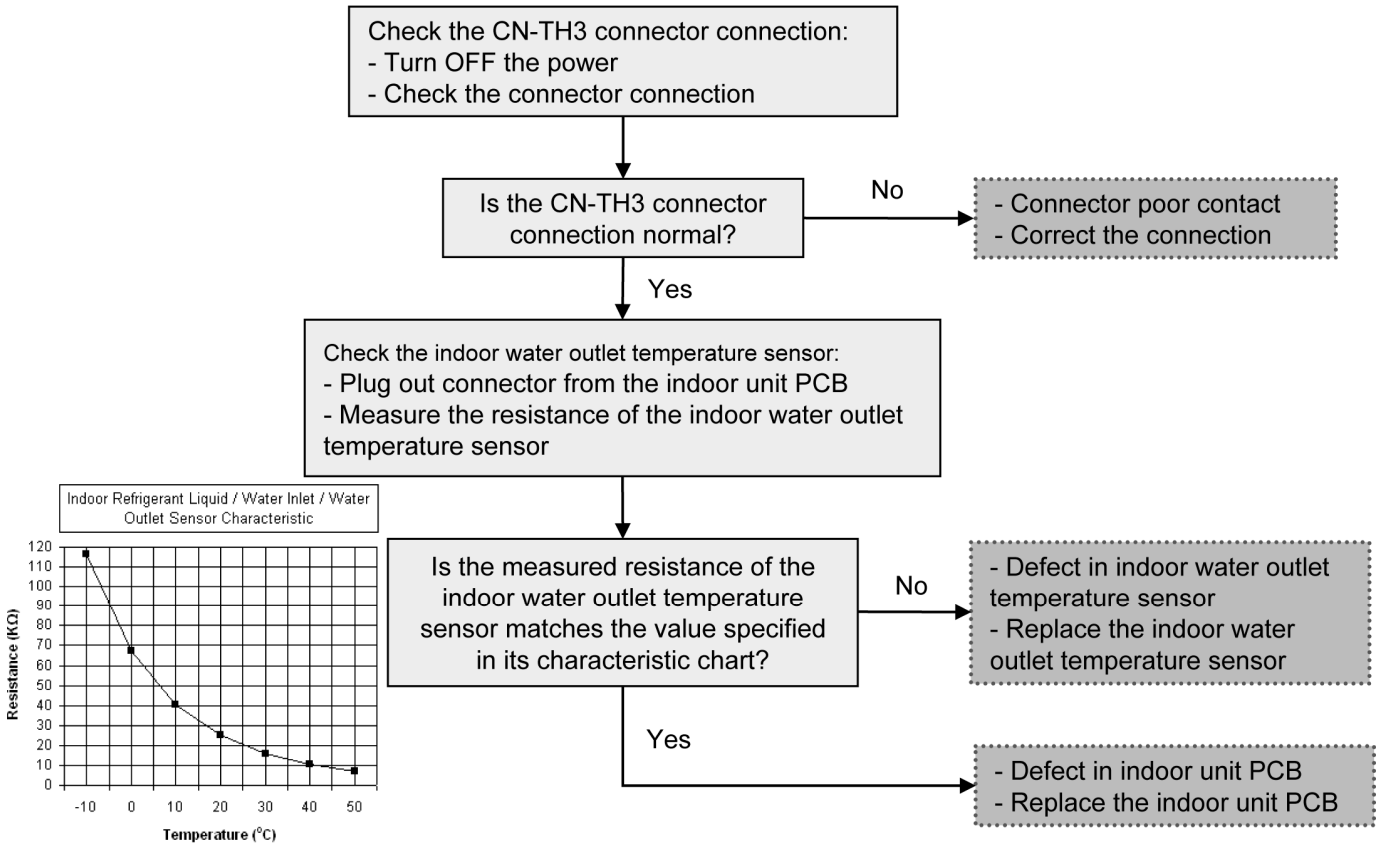
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB.

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.38 Outdoor Air Temperature Sensor Abnormality (F36)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor air temperature sensor are used to determine sensor error.

Malfunction Caused:

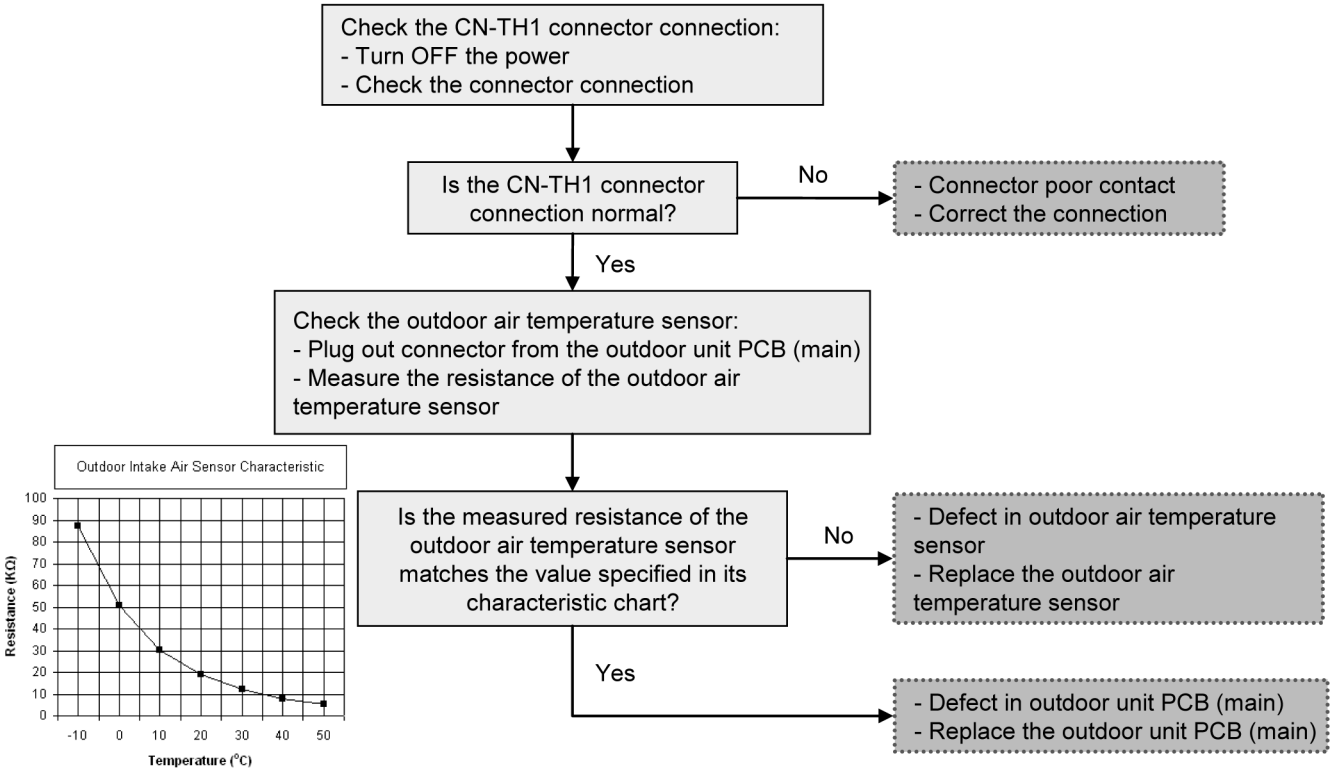
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.39 Indoor Water Inlet Temperature Sensor Abnormality (F37)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water inlet temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

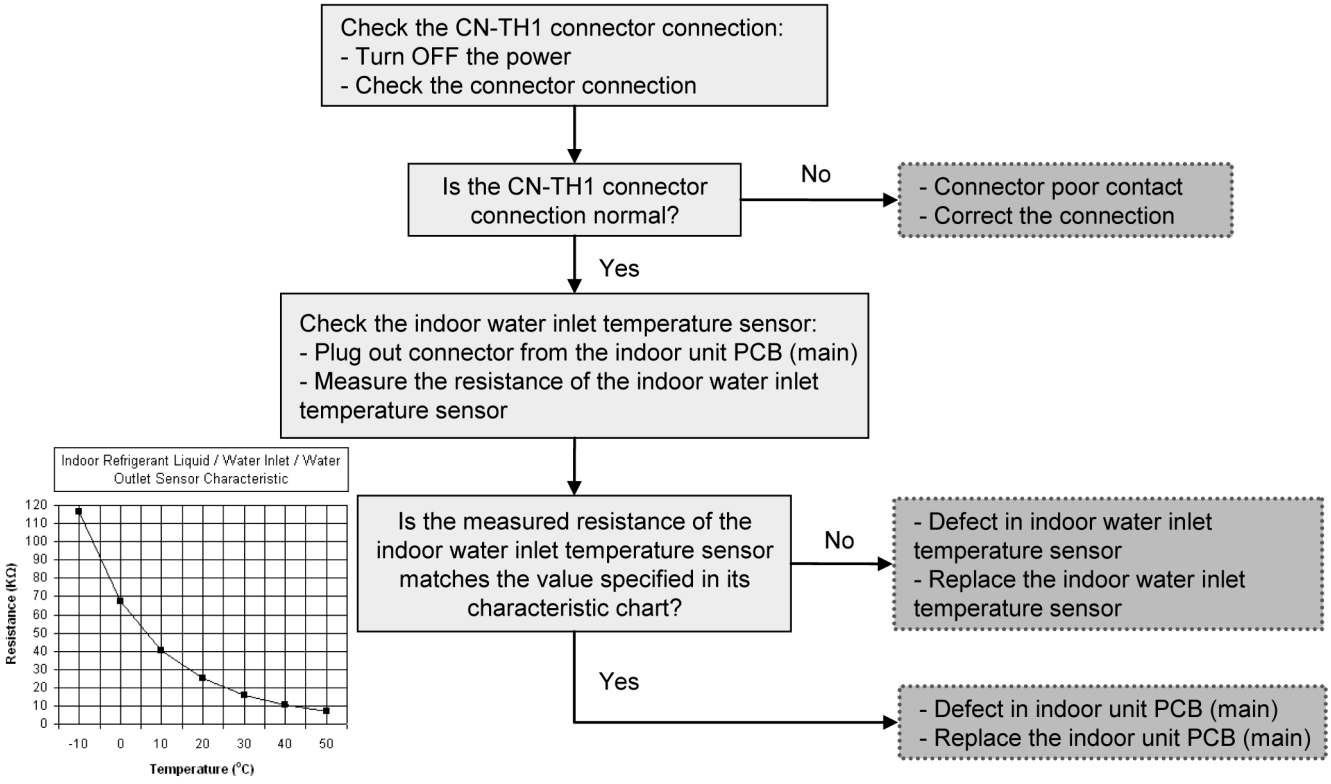
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.40 Outdoor Discharge Pipe Temperature Sensor Abnormality (F40)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor discharge pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

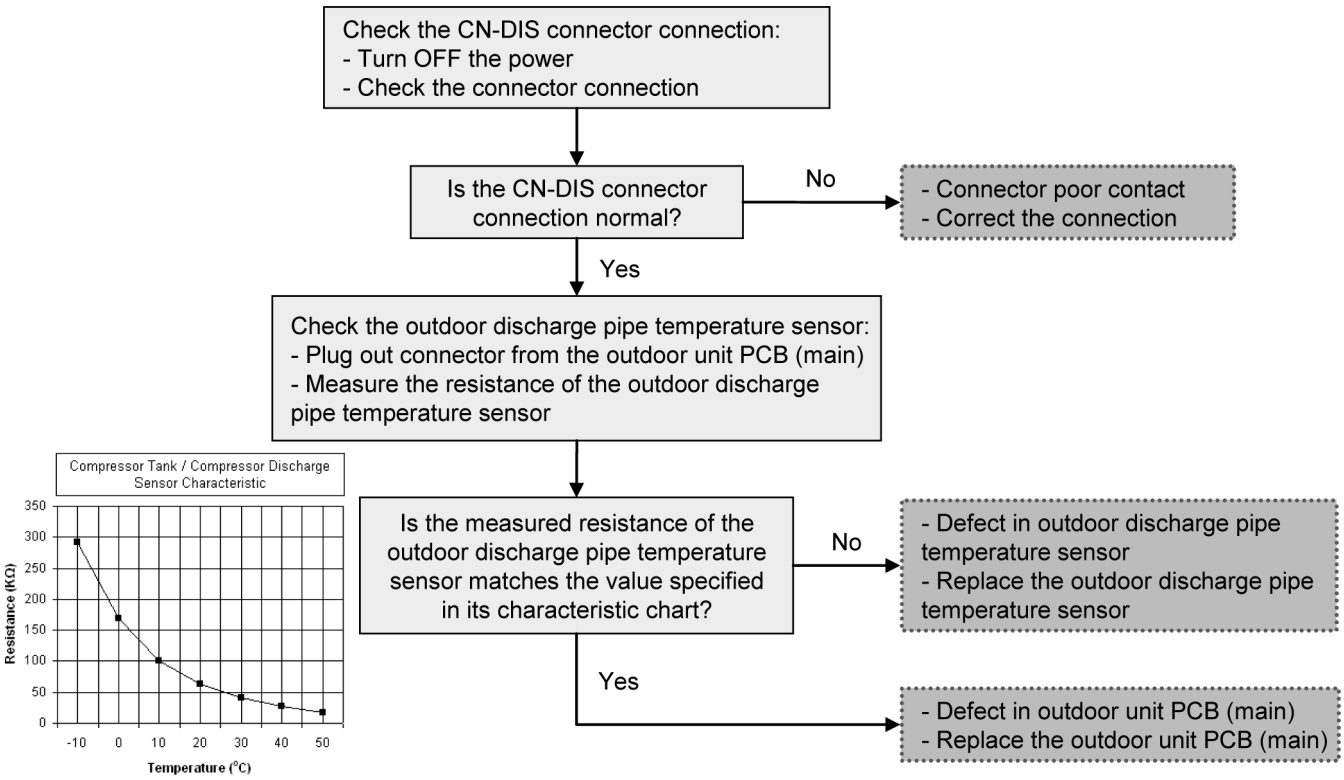
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.41 Power Factor Correction (PFC) Abnormality (F41)

Malfunction Decision Conditions:

During operation of cooling and heating, when the PFC protection circuitry in the outdoor unit PCB (main) senses abnormal high DC voltage level.

Malfunction Caused:

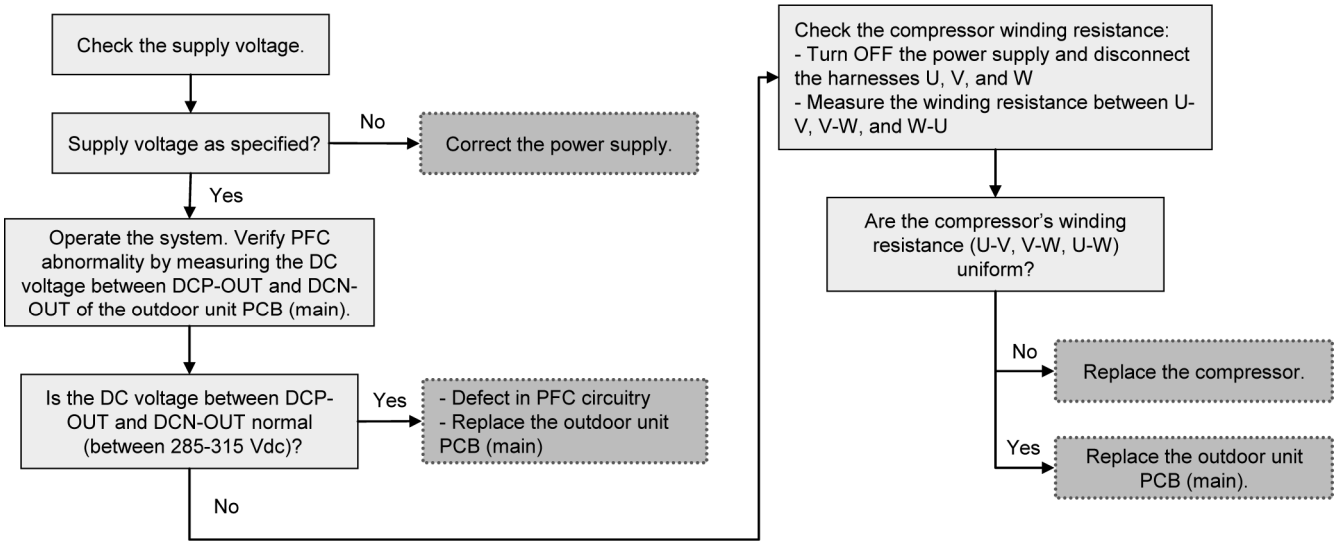
1 Power supply surge.
2 Compressor windings not uniform.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 4 times in 10 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.42 Outdoor Pipe Temperature Sensor Abnormality (F42)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

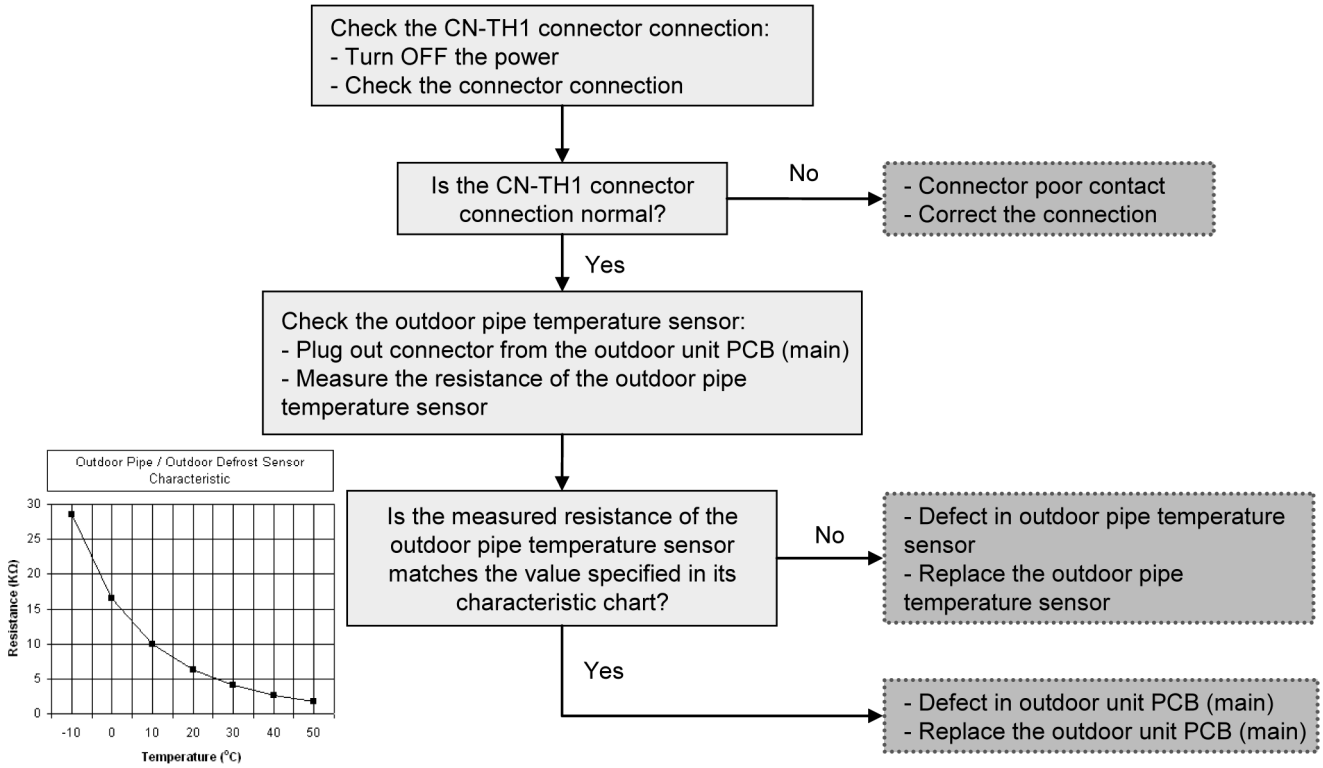
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.43 Outdoor Defrost Temperature Sensor Abnormality (F43)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor defrost temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

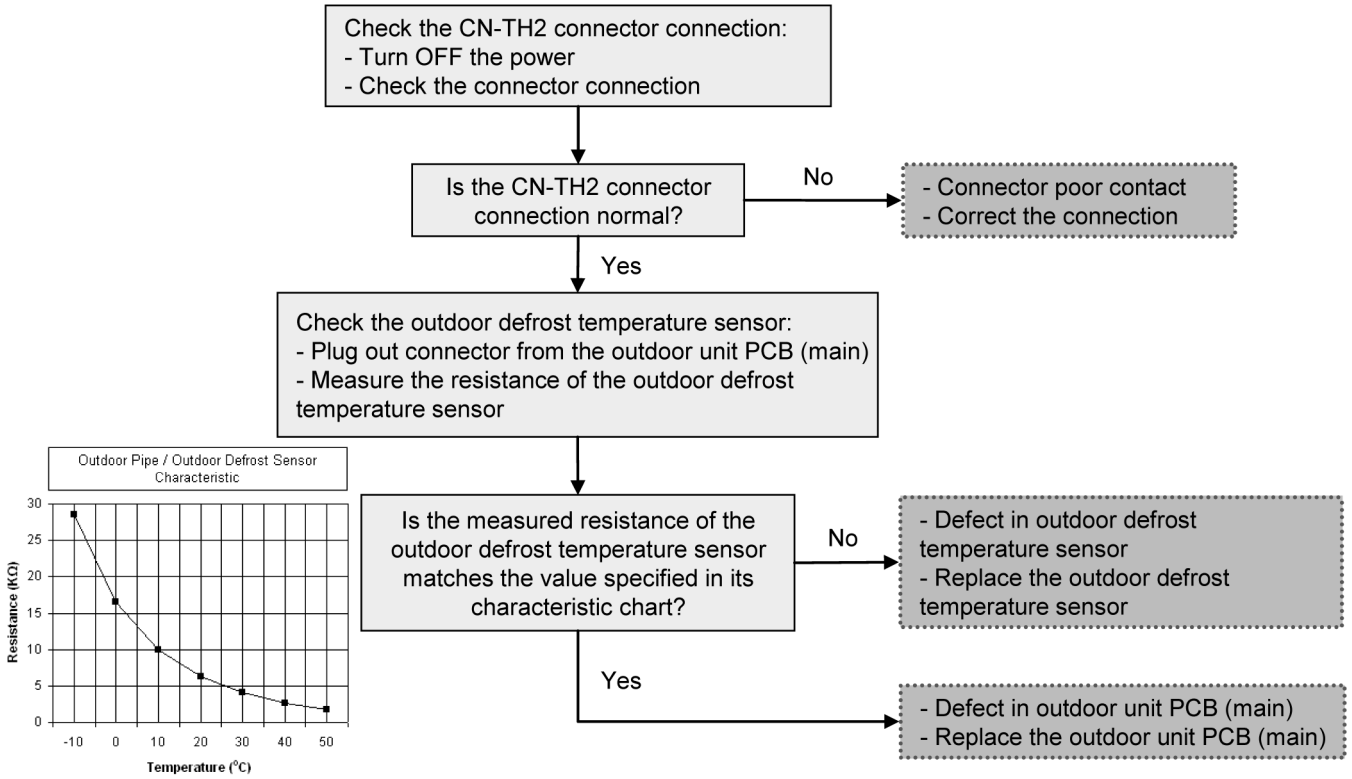
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



18.5.44 Indoor Water Outlet Temperature Sensor Abnormality (F45)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor are used to determine sensor errors.

Malfunction Caused:

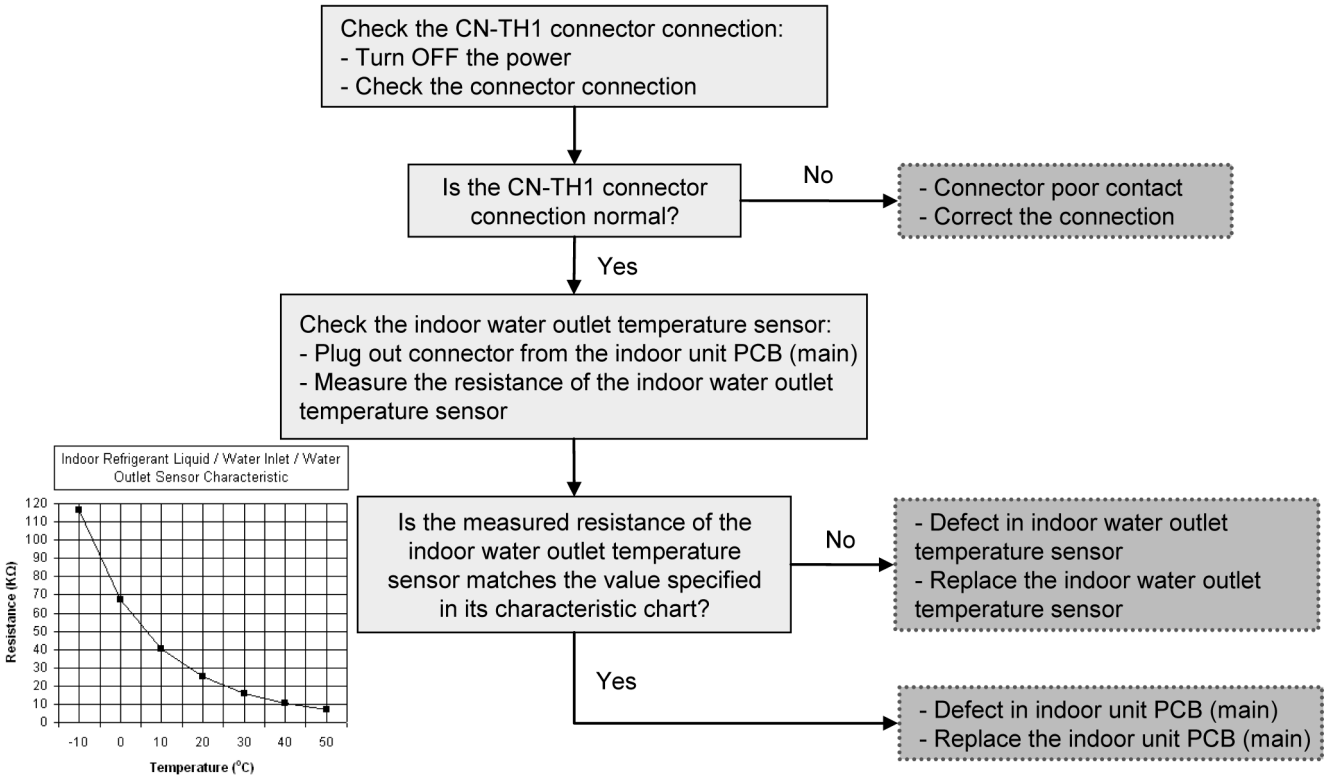
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



18.5.45 Outdoor Current Transformer Open Circuit (F46)

Malfunction Decision Conditions:

A current transformer (CT) open circuit is detected by checking the compressor running frequency (⩾ rated frequency) and CT detected input current (< 1.6 A) for continuously 20 seconds.

Malfunction Caused:

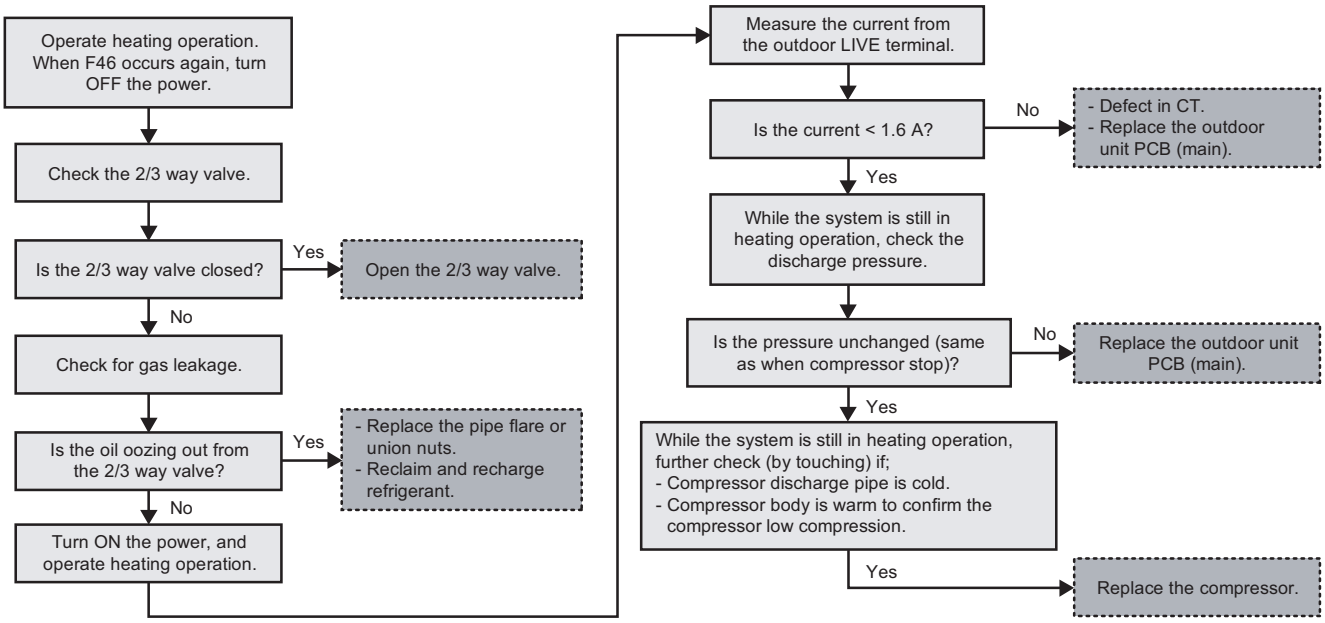
1 CT defective.
2 Faulty outdoor unit PCB (main).
3 Compressor defective (low compression).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19. Disassembly and Assembly Instructions

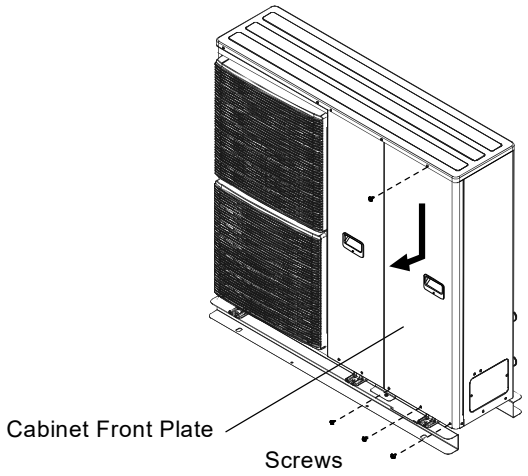
⚠ WARNING

High Voltage are generated in the electrical parts area by the capacitor. Ensure that the capacitor has discharged sufficiently before proceeding with repair work. Failure to heed this caution may result in electric shocks.

⚠ WARNING

Be sure to switch off all the power supply before performing each of the below action.

19.1 To Remove The Cabinet Front Plate

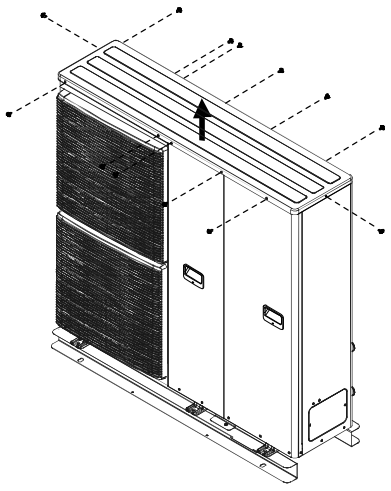


1. Remove 4 mounting screws.

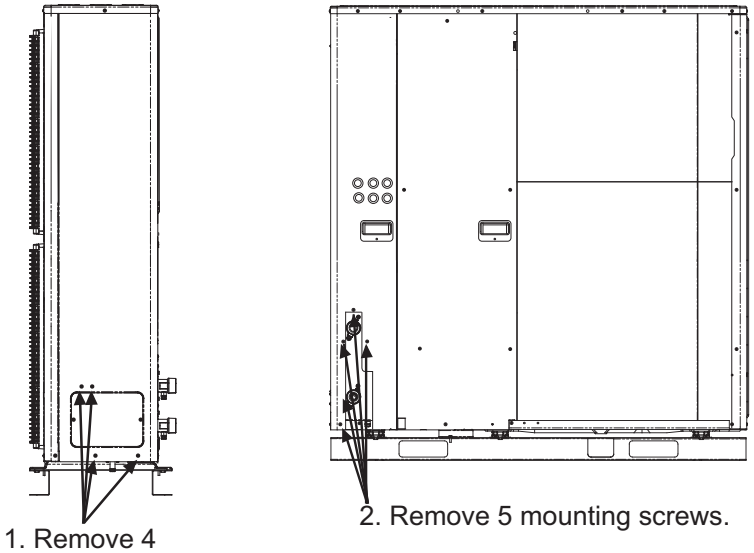
2. Slide the cabinet front plate downward to release the pawls. Then, pull it toward front to remove it.

19.2 To Remove The Cabinet Top Plate

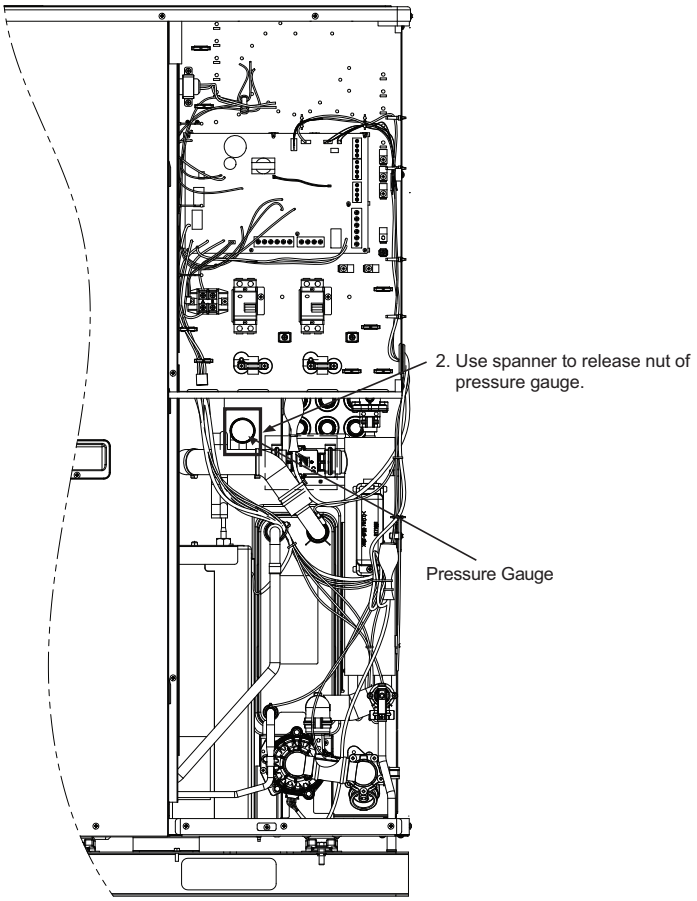
1. Remove the 13 mounting screws
2. Lift the cabinet top plate upward to remove it.



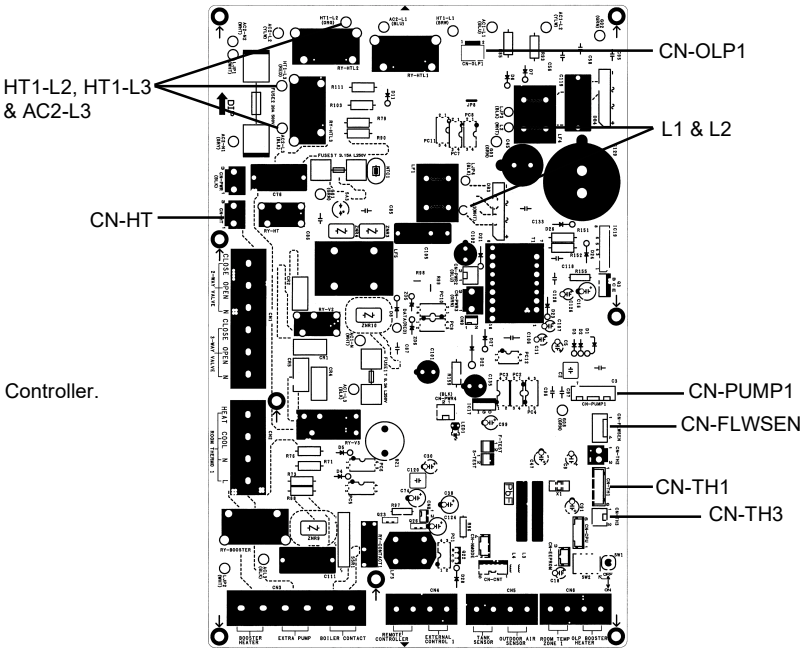
19.3 To Remove The Cabinet Rear Plate



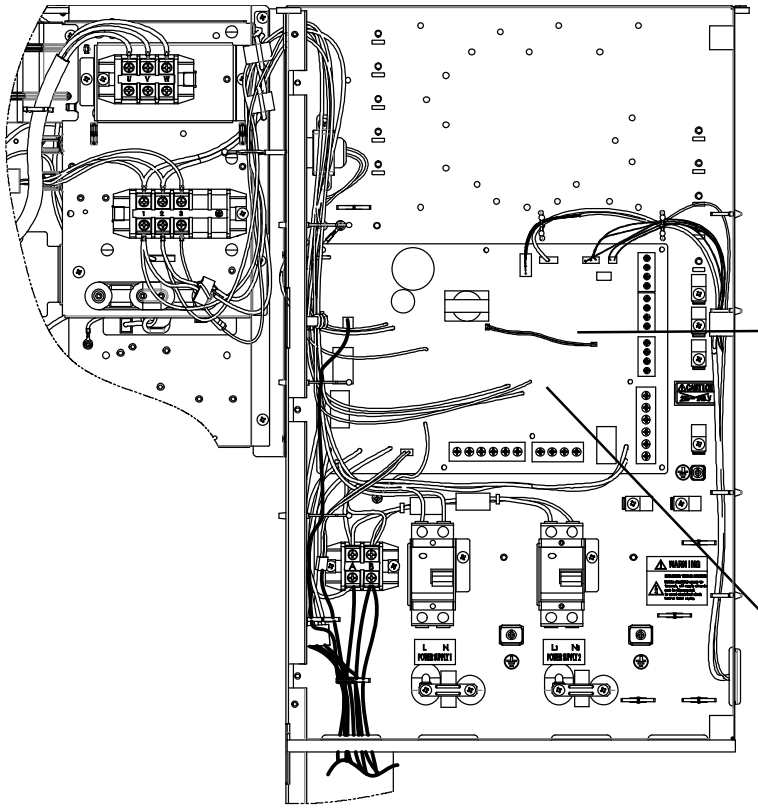
19.4 To Remove Pressure Gauge



19.5 To Remove Water System Electronic Controller Board



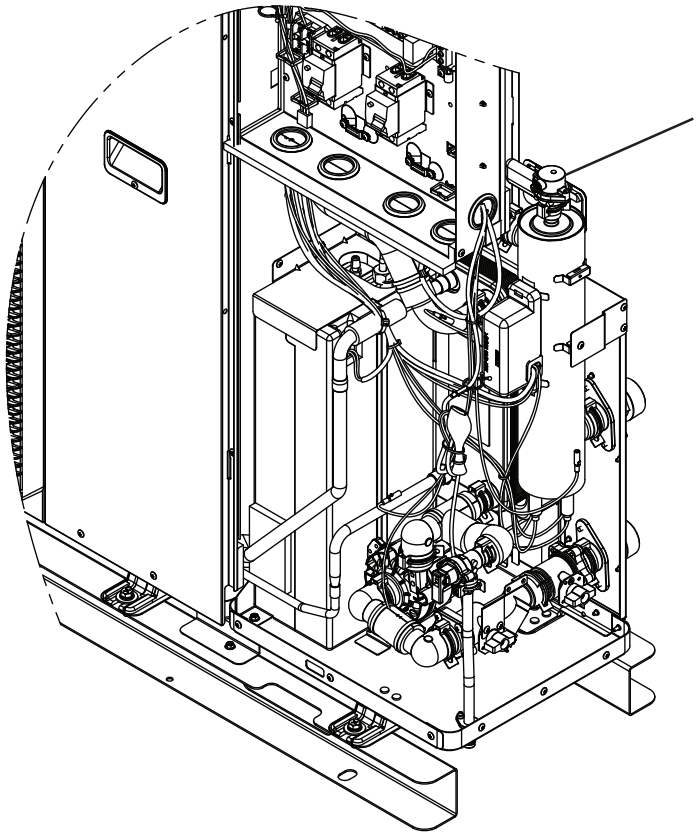
1. Remove all the connectors from the Electronic Controller.



Electronic Controller Board

2. Disconnect all lead wires of
G01 (Green), G02 (Green), G03 (Green)
AC2-N2 (White), AC2-L2 (Yellow)
Data (Red), AC1-N (White)
AC1-L3 (Black) and ACL3 (Black)

19.6 To Remove Air Purge Valve



1. Turn the Air Purge Valve anti-clockwise, then release the retaining ring and pull out the air purge valve.

⚠ During reinstall Air Purge Valve.

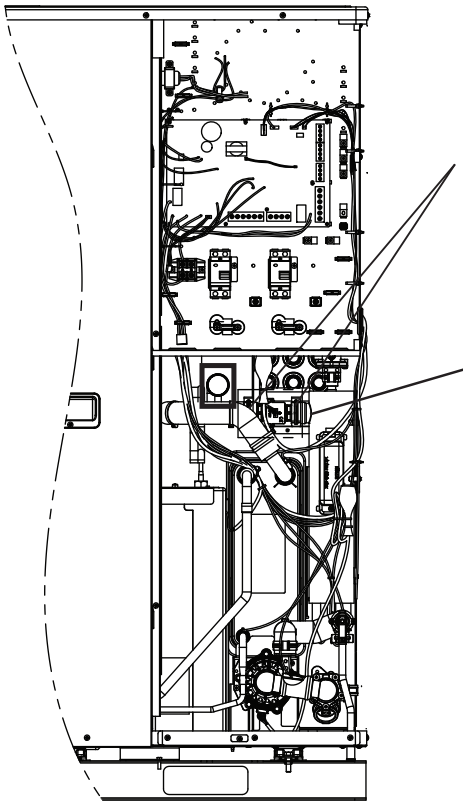
Make sure plug is fully close in clockwise direction.



Packing

Apply some water based grease on packing surface before fix with shut off valve.

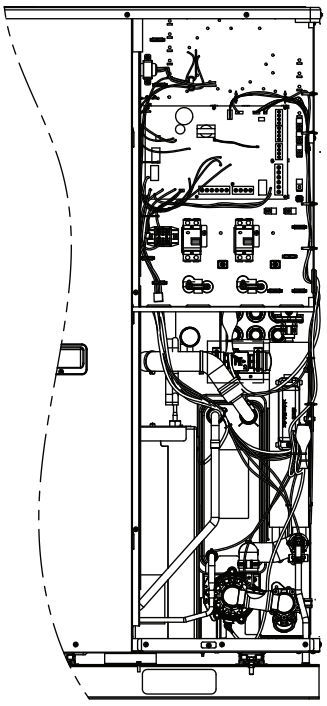
19.7 To Remove Flow Sensor



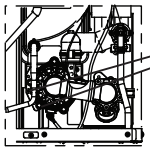
1. Disconnect the connector of CN-FLWSEN from the Water side Electronic Controller Board
2. Remove 2 retaining ring
3. Lift up slowly to remove the flow sensor.

⚠ When reinstall the water pipe, use grease or water at the joining

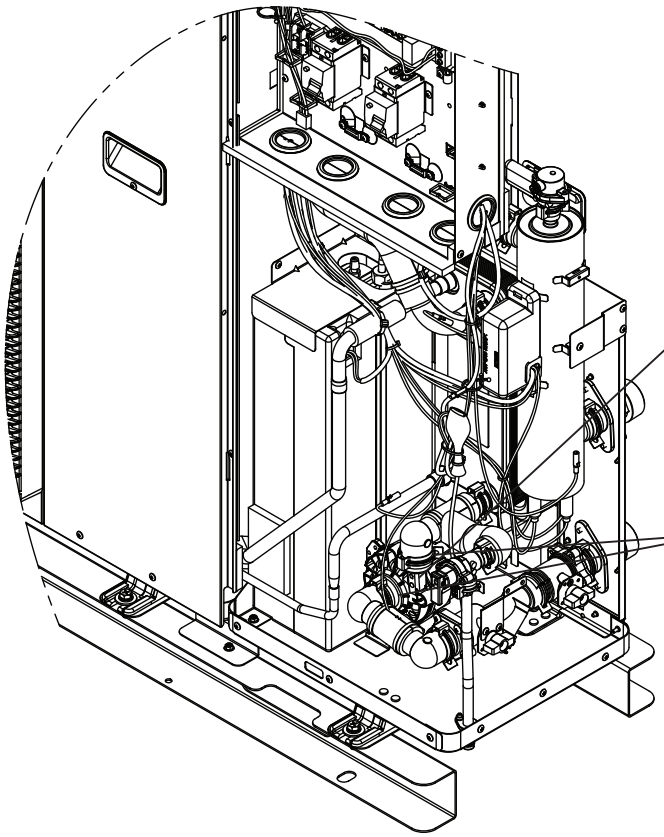
19.8 To Remove Water Pump



1. Disconnect CN-PUMP1 connector from Electronic Controller Board.

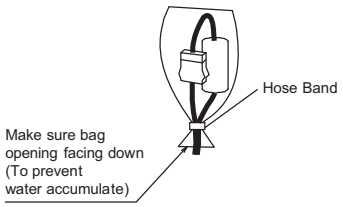


2. Remove 2 screw from Pump Bracket.



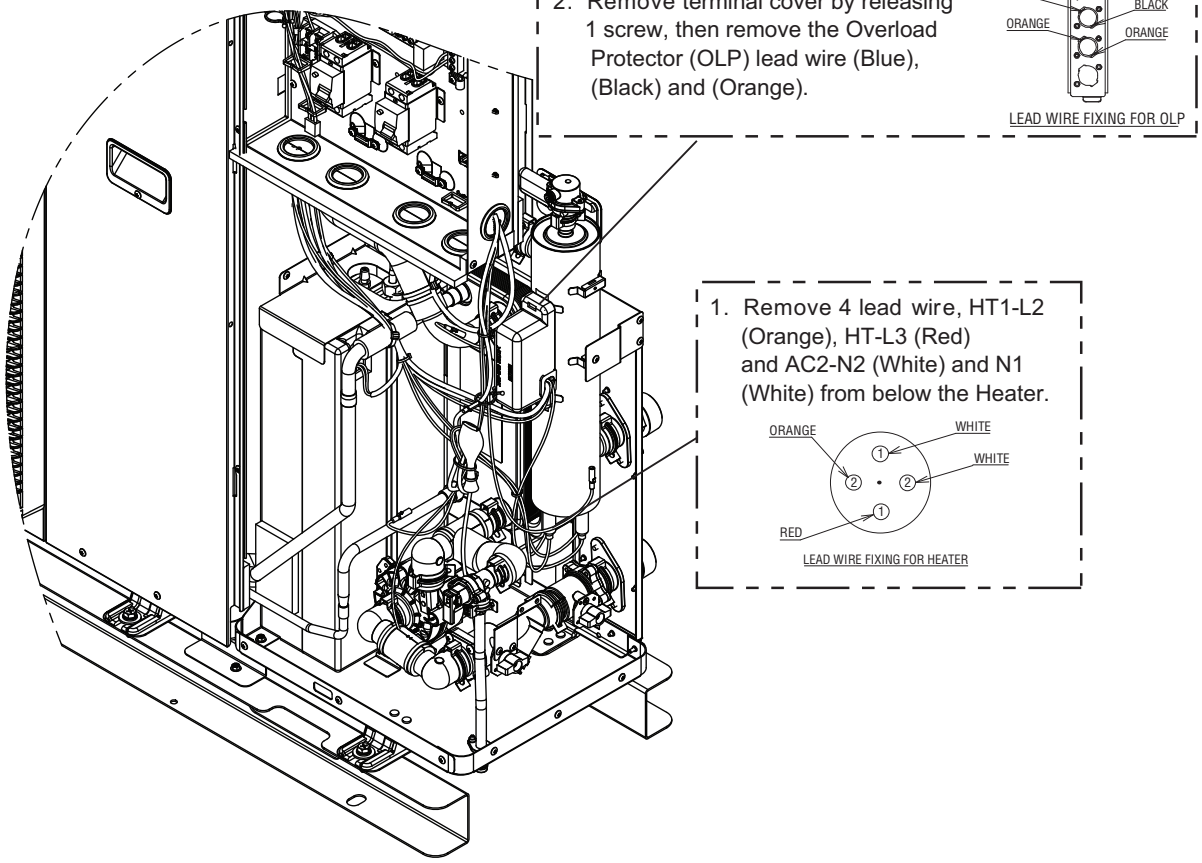
3. Remove 2 retaining ring, then slowly pull out the water pump.

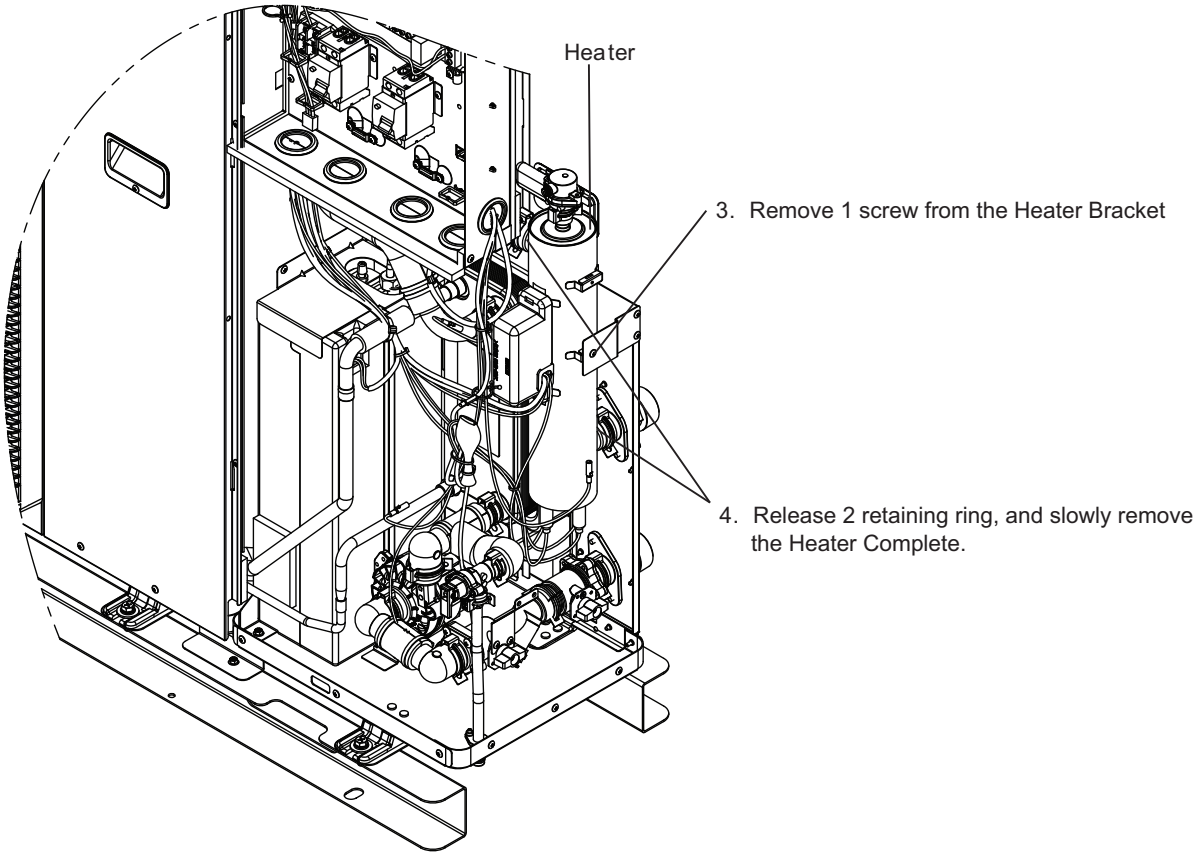
⚠ If pump is replace, please follow below instruction.



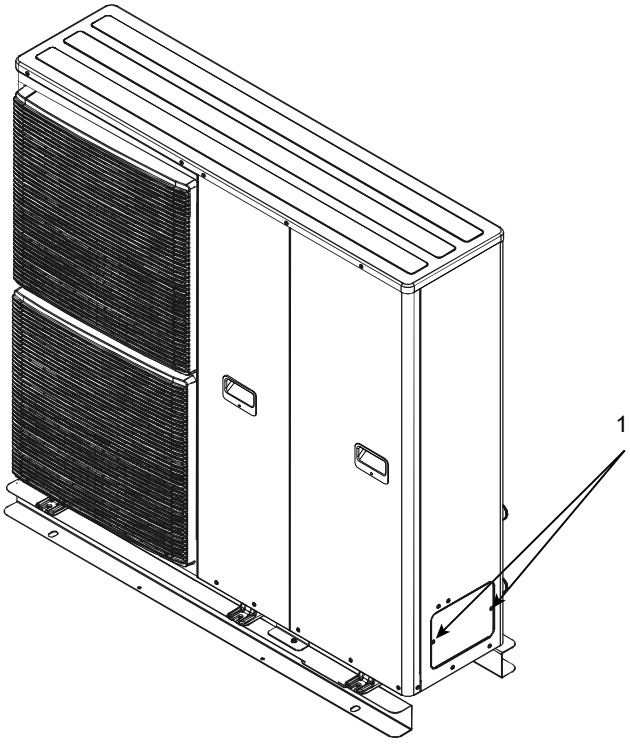
1. Please fix bag around the connector as shown, and seal it with a band.

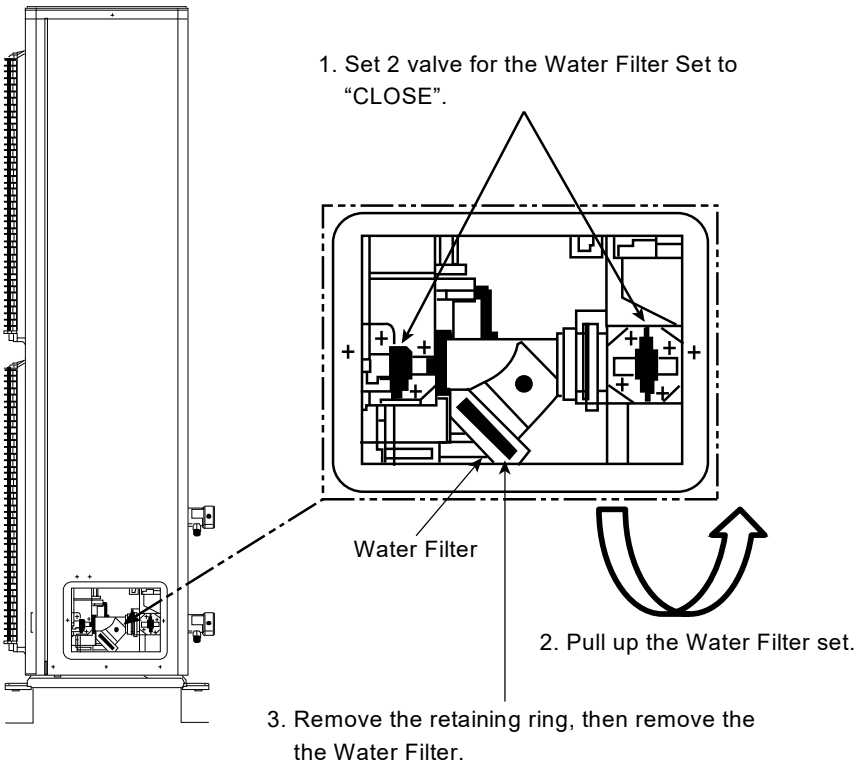
19.9 To Remove Heater





19.10 To Remove Filter





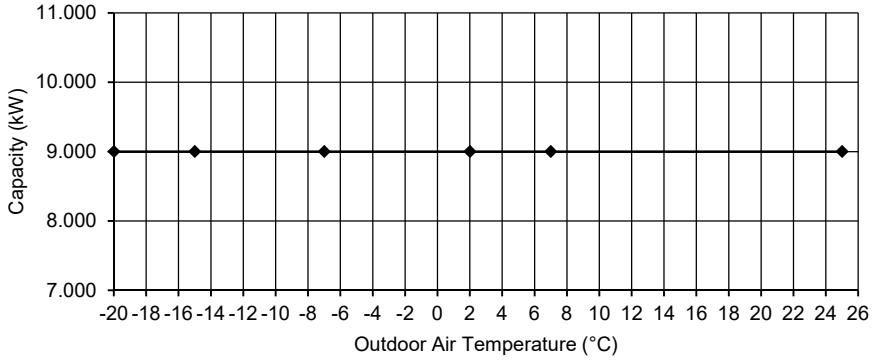
20. Technical Data

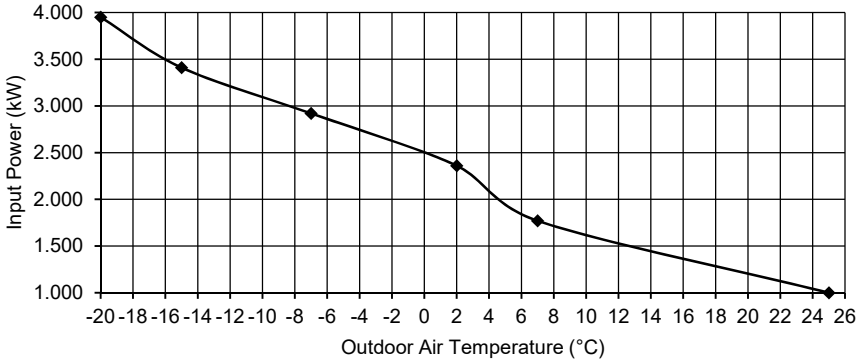
20.1 Operation Characteristics

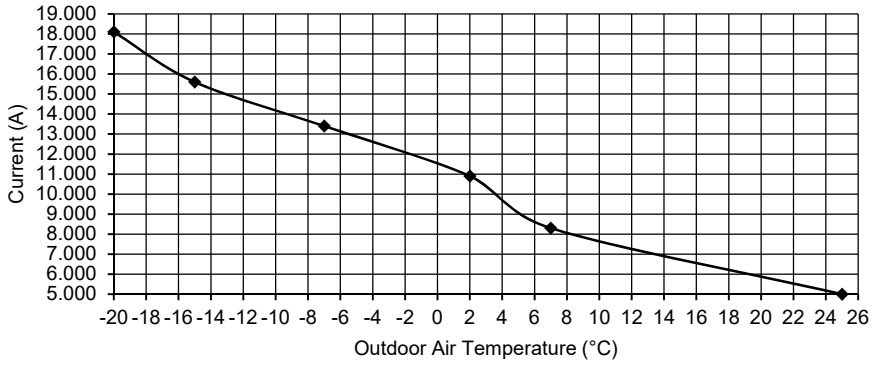
20.1.1 WH-MXC09J3E5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C







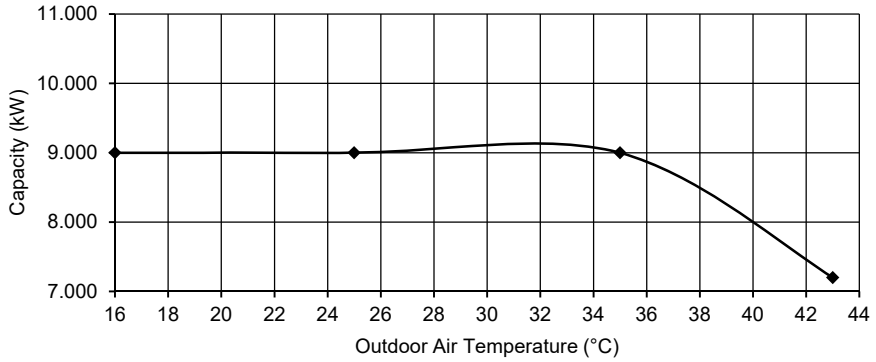
Cooling Characteristics at Different Outdoor Air Temperature

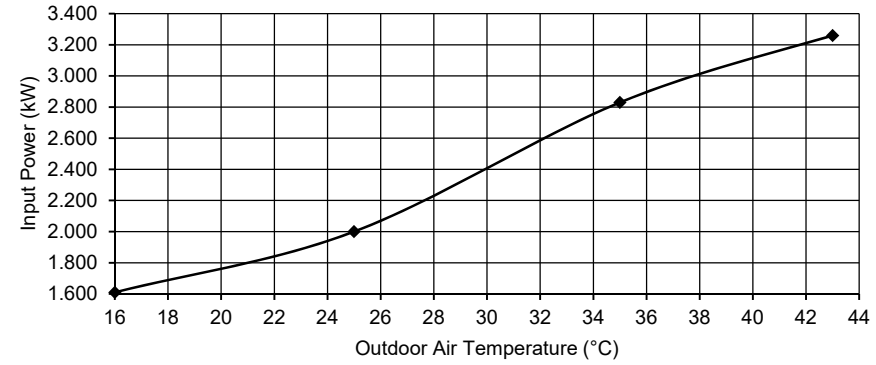
Condition

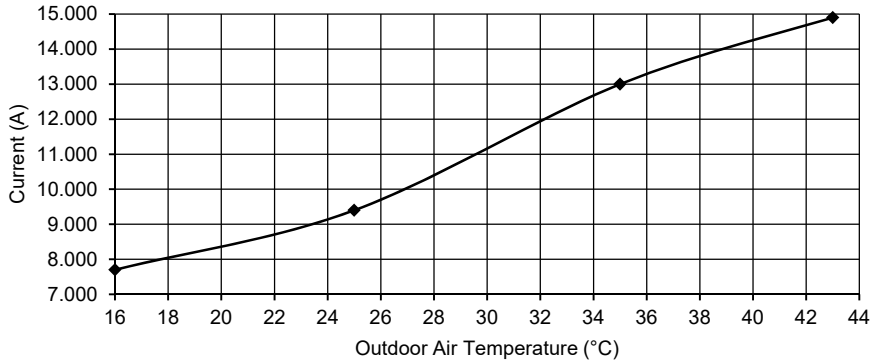
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C







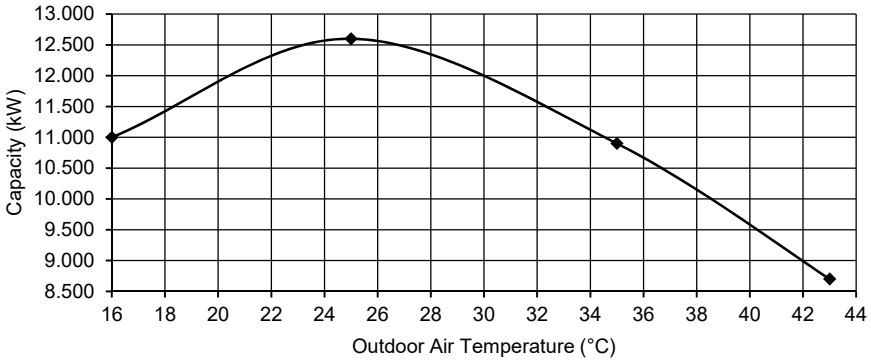
Cooling Characteristics at Different Outdoor Air Temperature

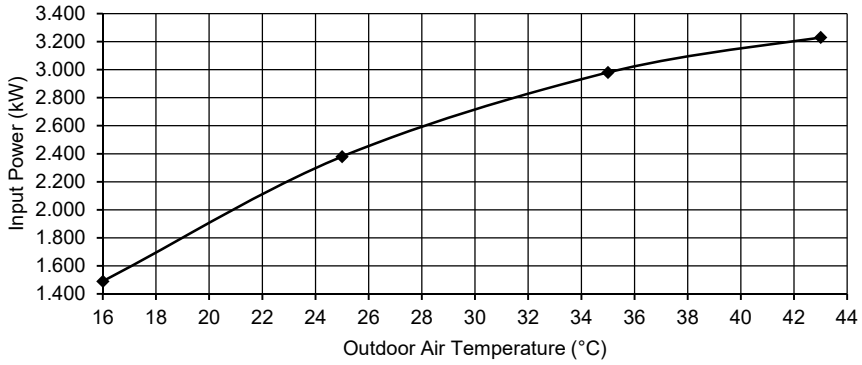
Condition

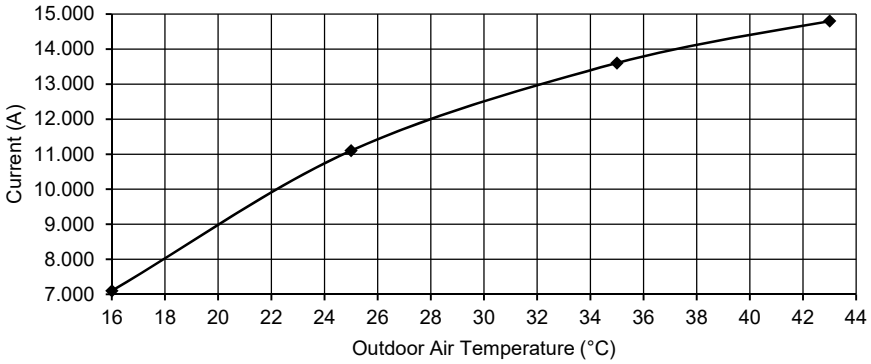
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C







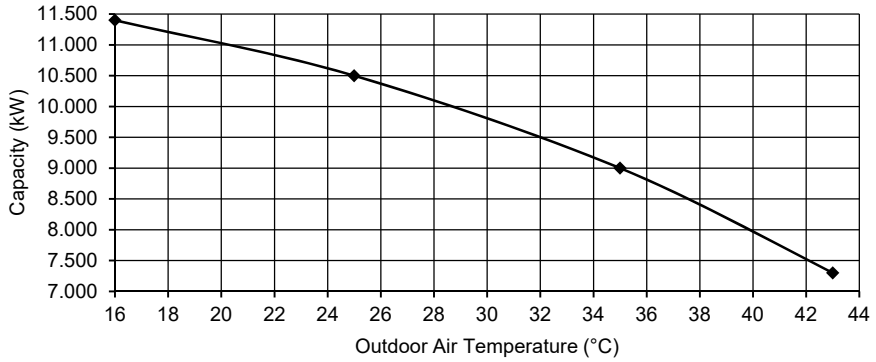
Cooling Characteristics at Different Outdoor Air Temperature

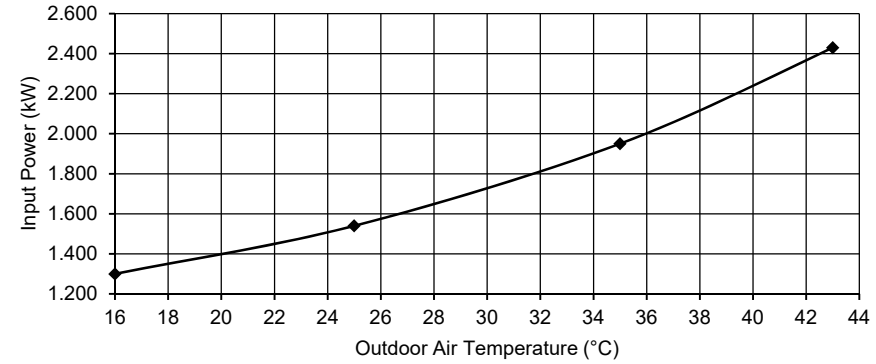
Condition

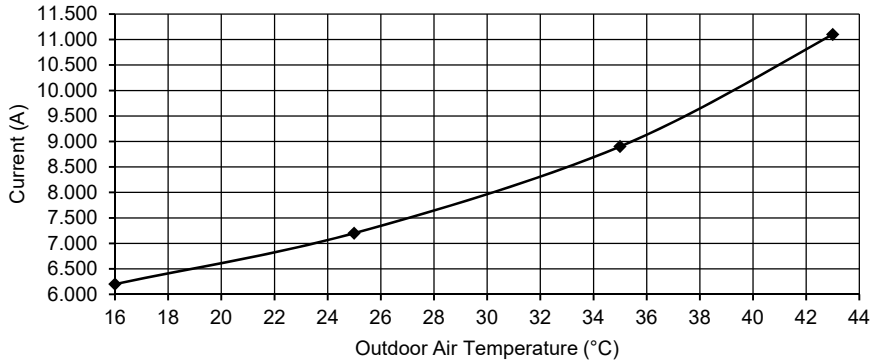
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C



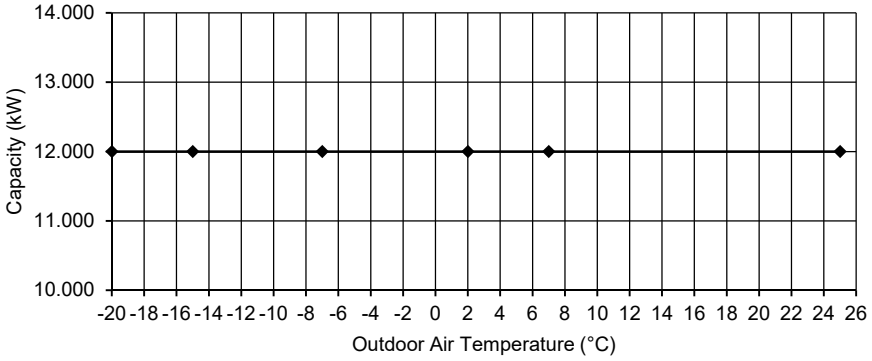


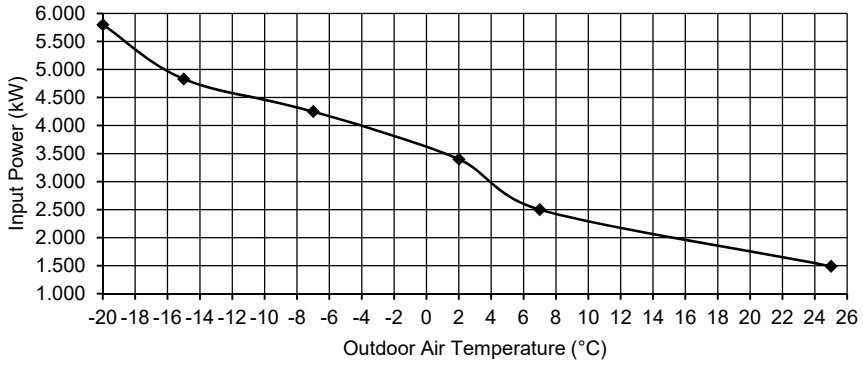


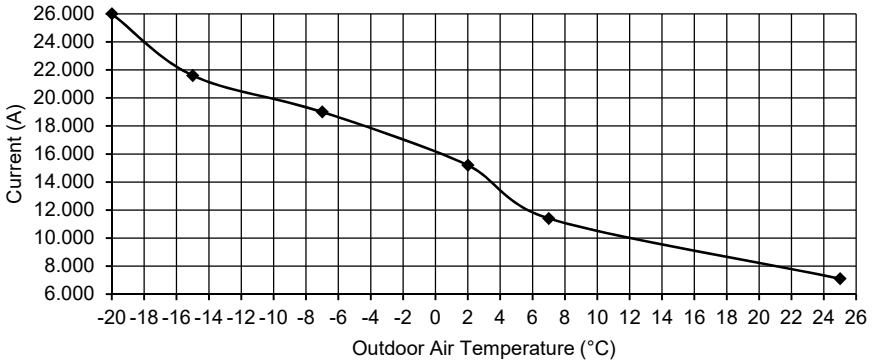
20.1.2 WH-MXC12J6E5

Heating Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C







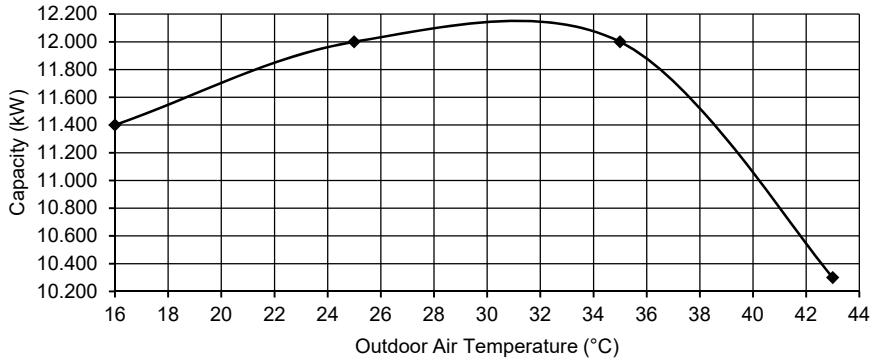
Cooling Characteristics at Different Outdoor Air Temperature

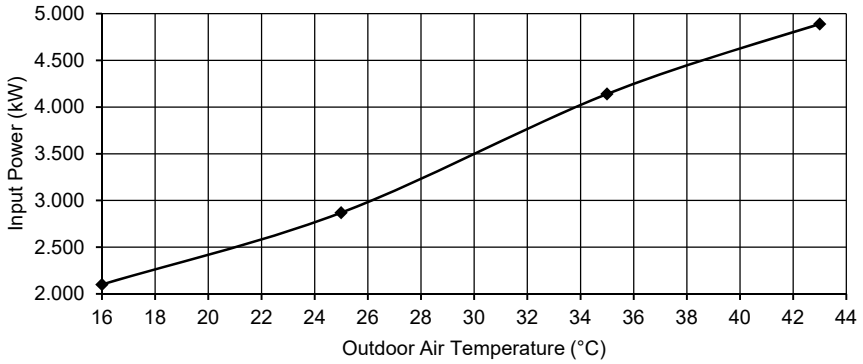
Condition

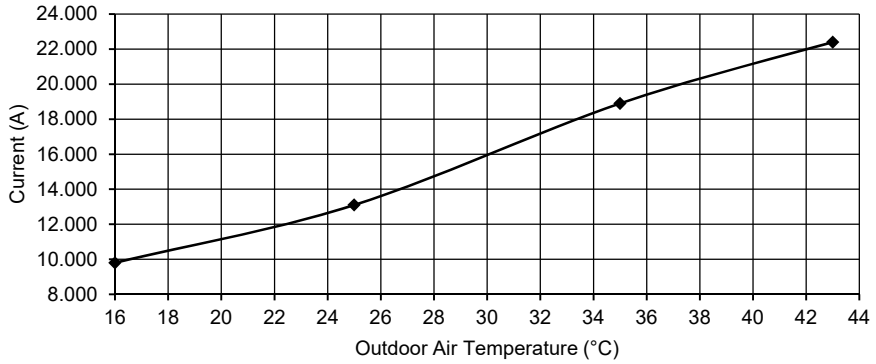
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C







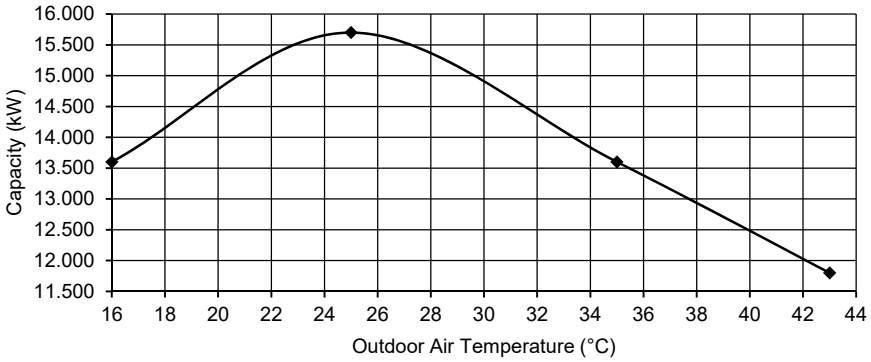
Cooling Characteristics at Different Outdoor Air Temperature

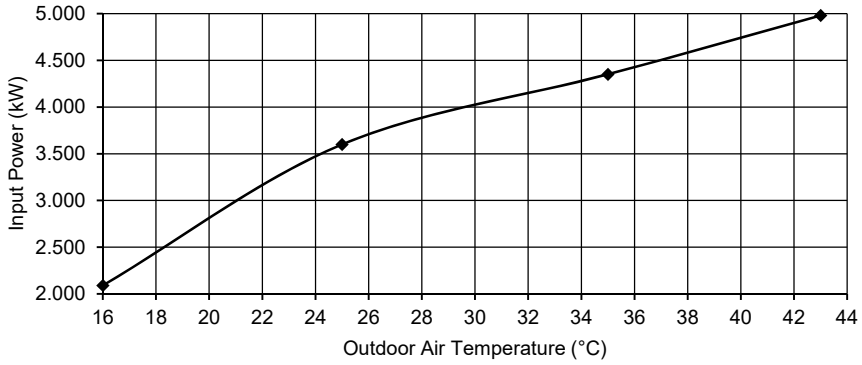
Condition

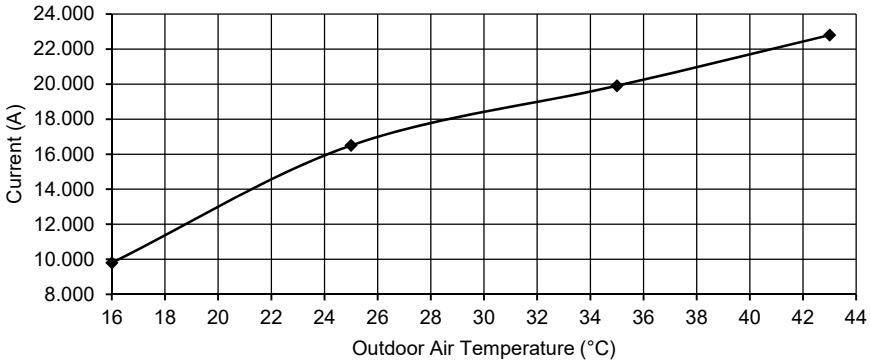
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C







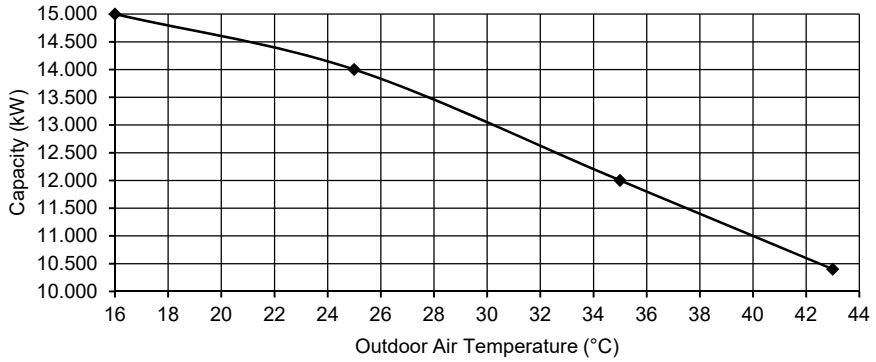
Cooling Characteristics at Different Outdoor Air Temperature

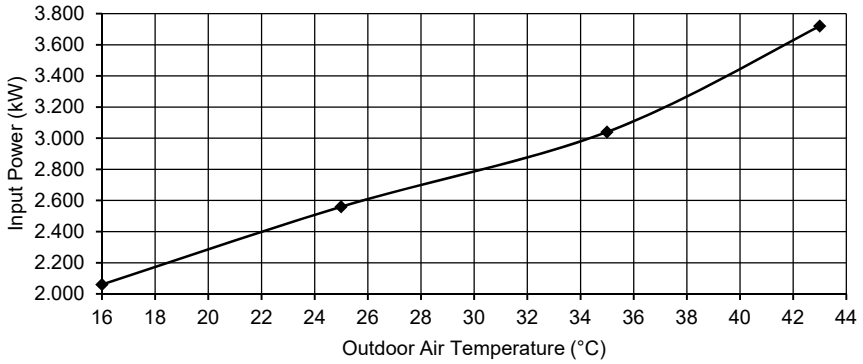
Condition

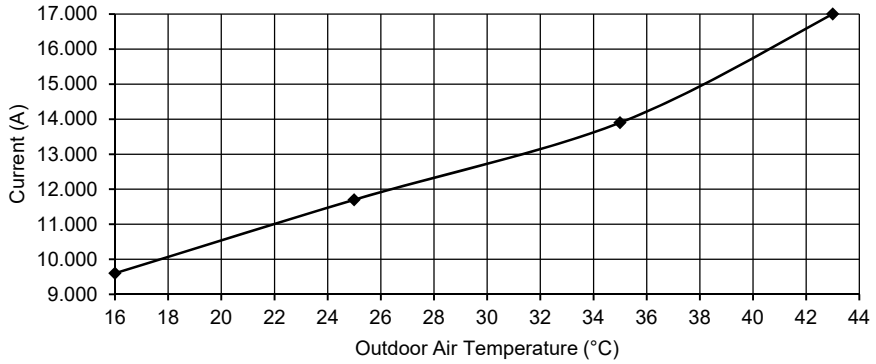
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C







20.2 Heating Capacity Table

20.2.1 WH-MXC09J3E5

Water Out	25ºC			35ºC		
Outdoor Air	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)
-20	9000	3440	15.7	9000	3950	18.1
-15	9000	2980	13.6	9000	3410	15.6
-7	10500	2720	12.4	9000	2920	13.4
2	10800	2140	9.9	9000	2360	10.9
7	9000	1380	6.5	9000	1770	8.3
25	9000	770	3.8	9000	1000	5.0

Water Out	45ºC			55ºC			60ºC			
Outdoor Air	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)	
-20	9000	4650	21.3	7900	5580	25.5	-	-	-	*
-15	9000	4040	18.5	9000	4830	22.1	8700	5370	24.6	
-7	9000	3540	16.2	9000	4240	19.4	9000	4620	21.1	
2	9000	2910	13.5	9000	3550	16.4	9000	4050	18.7	
7	9000	2370	11.1	9000	2920	13.7	9000	3290	15.4	
25	10000	1670	7.9	10000	2280	10.7	11000	2860	13.4	

Note:
* Between outdoor ambient -15°C and -20°C, the water outlet temperature gradually decreases from 60°C to 55°C.
* Only when ΔT is set to 15°C, the set temperature above 60°C will take effect.

20.2.2 WH-MXC12J6E5

Water Out	25ºC			35ºC		
Outdoor Air	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)
-20	12000	5020	22.5	12000	5800	26.0
-15	12000	4140	18.6	12000	4830	21.6
-7	13500	4300	19.3	12000	4250	19.0
2	14500	3230	14.5	12000	3400	15.2
7	12000	2000	9.2	12000	2500	11.4
25	12000	1200	5.7	12000	1490	7.1

Water Out	45ºC			55ºC			60ºC			
Outdoor Air	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)	
-20	11000	5950	26.7	10000	6500	29.1	-	-	-	*
-15	11000	5200	23.3	10500	6000	26.9	8900	6300	28.2	
-7	12000	5020	22.5	12000	6000	26.9	11000	6300	28.2	
2	12000	4200	18.8	12000	4950	22.2	12000	5770	25.9	
7	12000	3240	14.8	12000	3940	18.0	12000	4520	20.7	
25	12000	2100	10.0	12000	2750	13.1	12000	3110	14.9	

Note:
* Between outdoor ambient -15°C and -20°C, the water outlet temperature gradually decreases from 60°C to 55°C.
* Only when ΔT is set to 15°C, the set temperature above 60°C will take effect.

20.3 Cooling Capacity Table

20.3.1 WH-MXC09J3E5

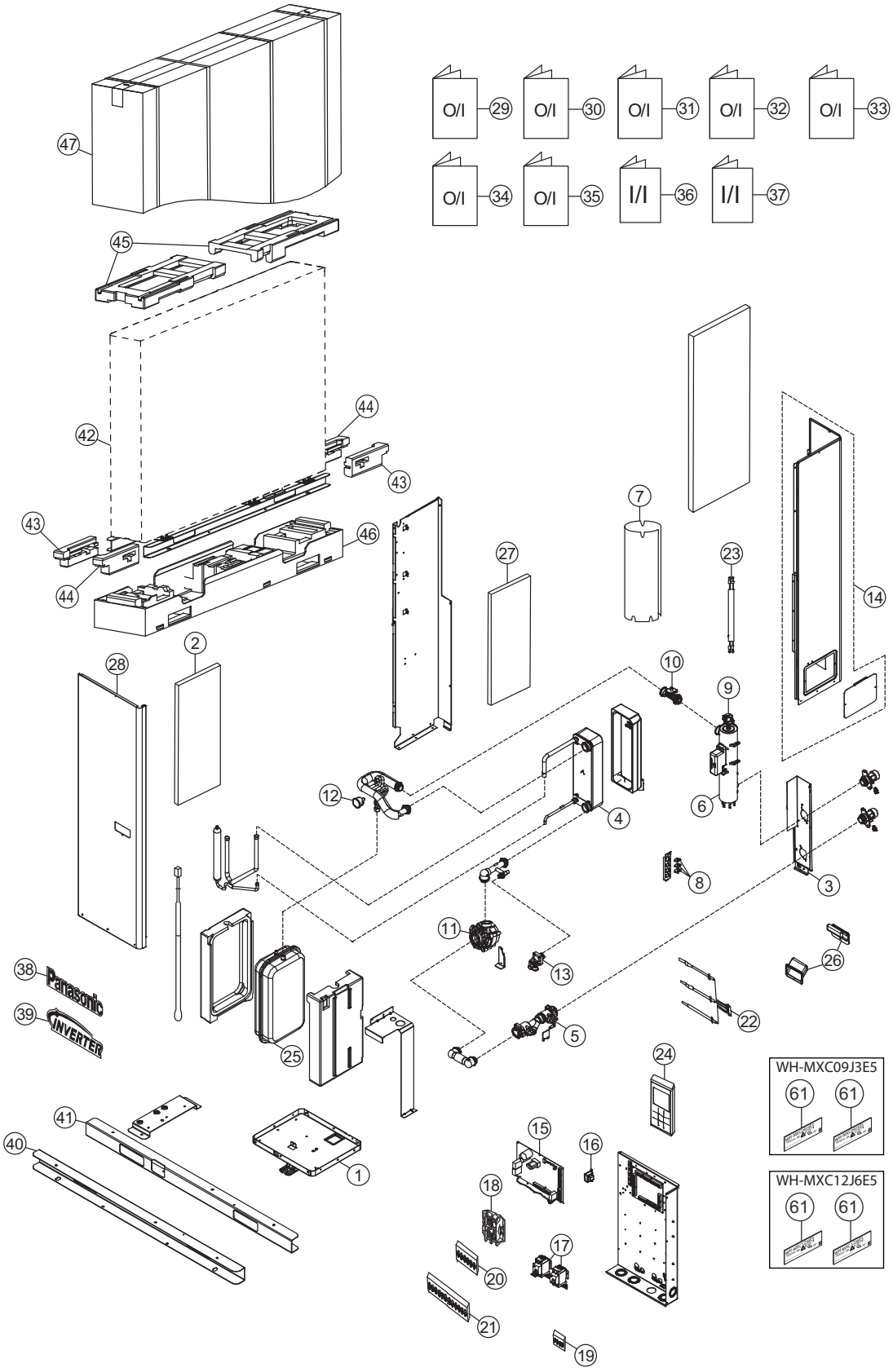
OD	Water Out 7ºC			Water Out 14ºC			Water Out 18ºC		
	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)
16	9000	1610	7.7	11000	1490	7.1	11400	1300	6.2
25	9000	2000	9.4	12600	2380	11.1	10500	1540	7.2
35	9000	2830	13.0	10900	2980	13.6	9000	1950	8.9
43	7200	3260	14.9	8700	3230	14.8	7300	2430	11.1

20.3.2 WH-MXC12J6E5

OD	Water Out 7ºC			Water Out 14ºC			Water Out 18ºC		
	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)	Capacity(W)	Input Power(W)	Current(A)
16	11400	2100	9.8	13600	2090	9.8	15000	2060	9.6
25	12000	2870	13.1	15700	3600	16.5	14000	2560	11.7
35	12000	4140	18.9	13600	4350	19.9	12000	3040	13.9
43	10300	4890	22.4	11800	4980	22.8	10400	3720	17.0

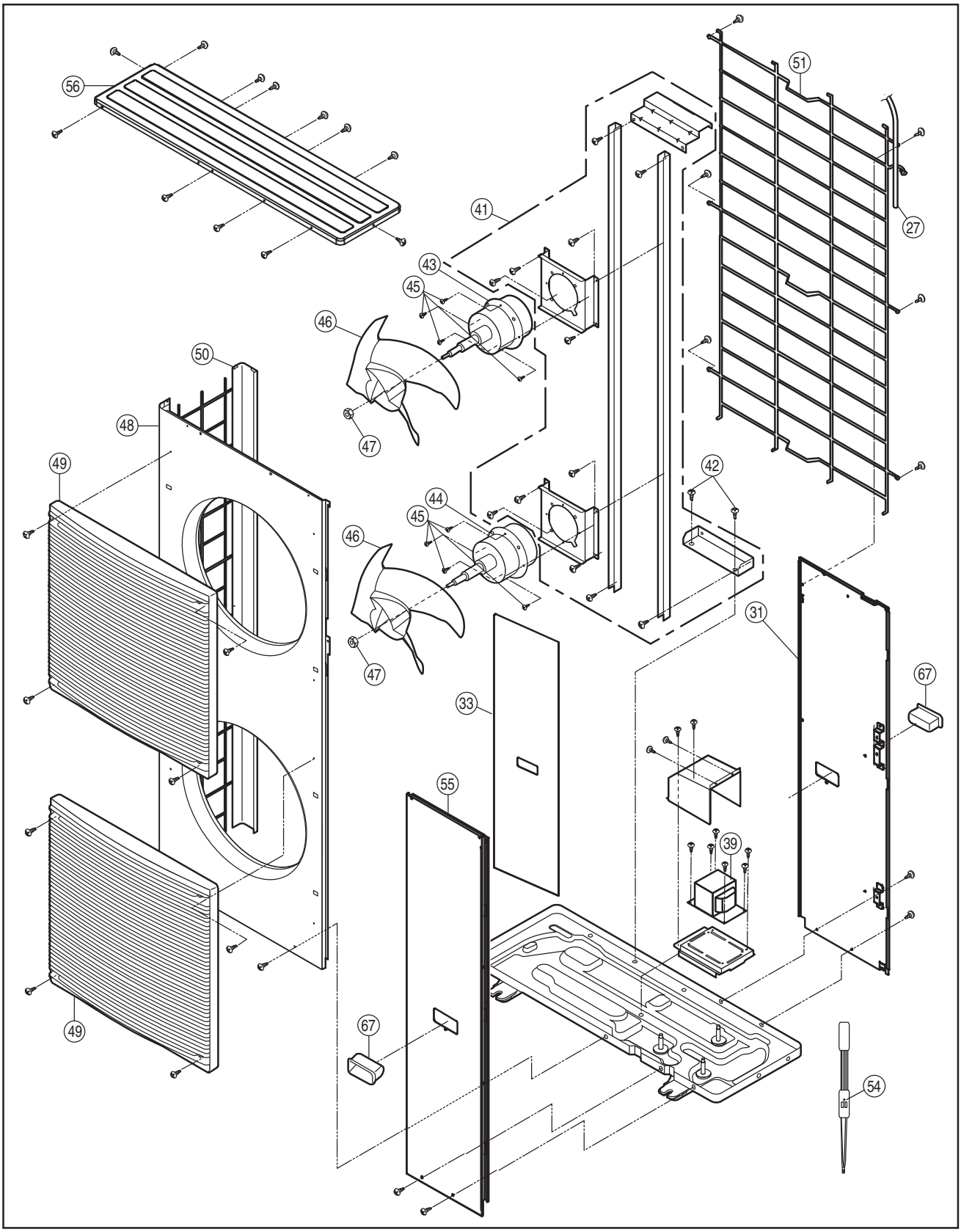
21. Exploded View and Replacement Parts List

21.1 Water System

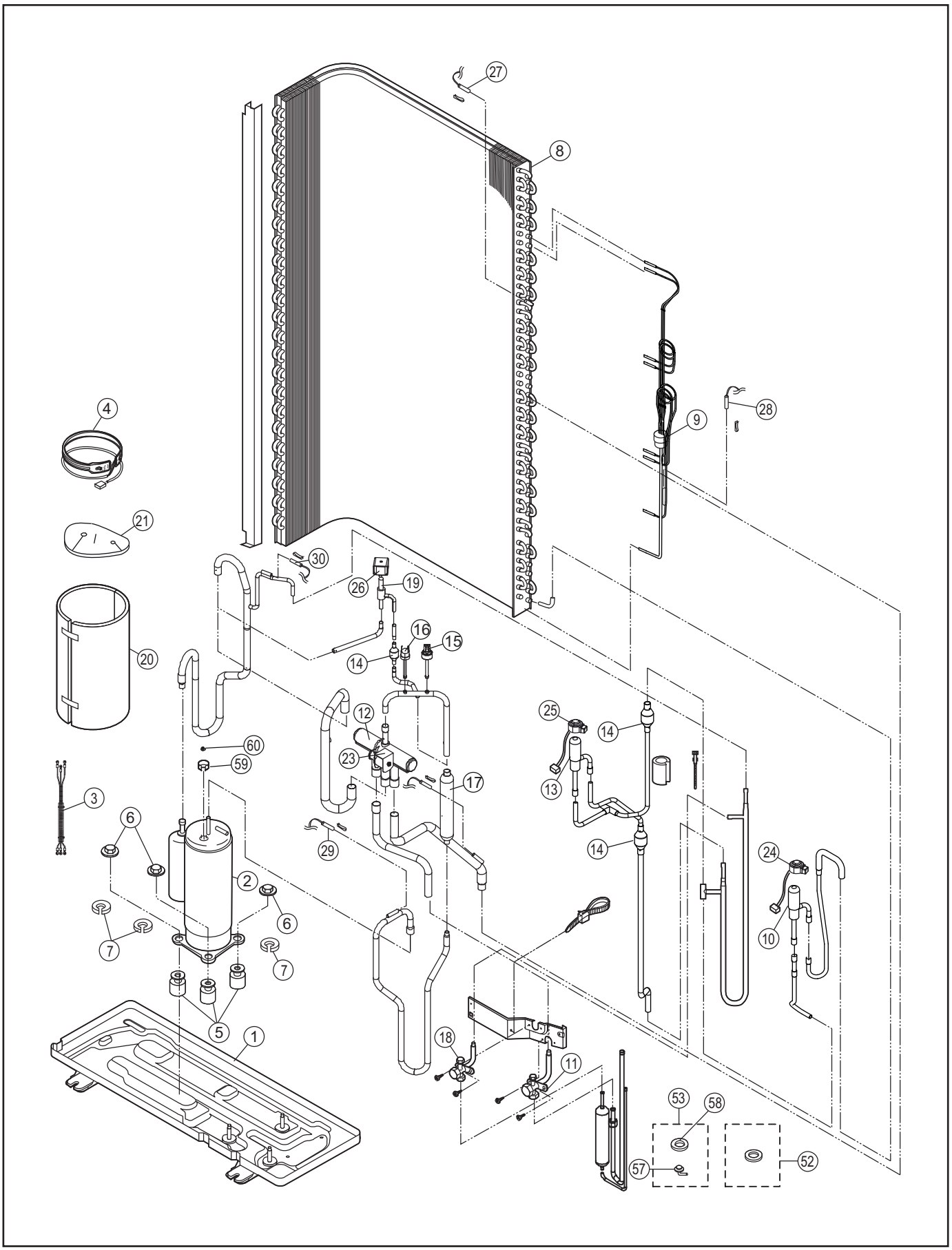


Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

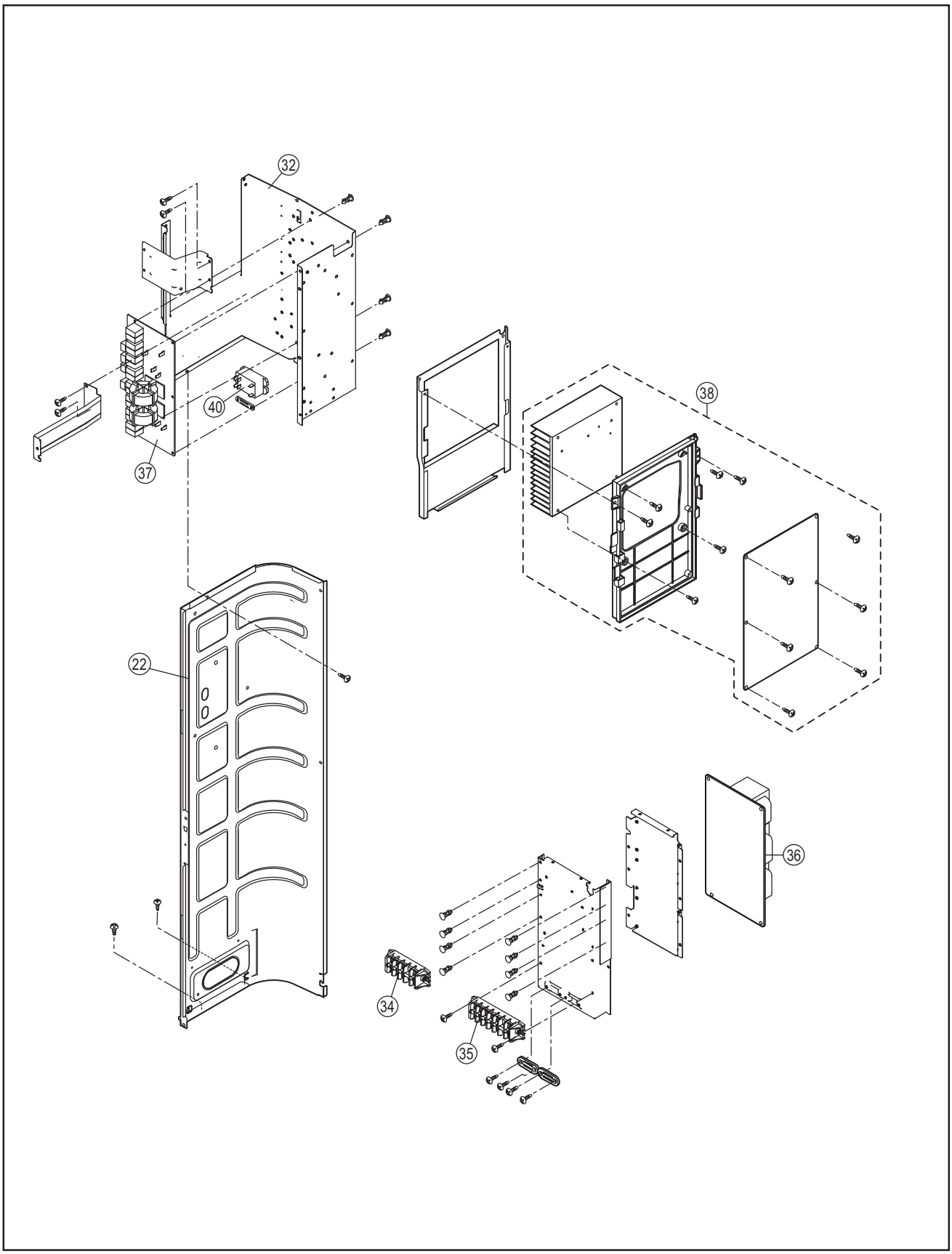
21.2 Refrigerant System



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

<Water System>

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MXC09J3E5	WH-MXC12J6E5	REMARK
	1	BASE PAN ASS'Y	1	ACXD52K02760	←	
	2	SOUND PROOF MATERIAL	1	CWG302594	←	
	3	HOLDER COUPLING	1	ACXH35-01510A	←	
	4	HOT WATER COIL-COMPLETE	1	ACXB90C01450	←	
	5	FILTER COMPLETE	1	ACXB51C00100	←	
⚠	6	HEATER ASS'Y	1	ACXA34K00490	ACXA34K00500	
	7	SOUND PROOF MATERIAL	1	ACXG30-10192	ACXG30-07663	
⚠	8	THERMOSTAT	2/3	CWA151074	←	
	9	AIR PURGE VALVE	1	ACXB62-00130	←	
	10	FLOW SWITCH	1	ACXB62-00930	←	
⚠	11	PUMP	1	ACXB53-00390	←	
	12	PRESSURE GAUGE	1	CWB070003	←	
	13	PRESSURE RELIEF VALVE	1	ACXB62-00740	←	
	14	CABINET SIDE PLATE	1	ACXE04-12210A	←	
⚠	15	ELECTRONIC CONTROLLER (MAIN)	1	ACXA73C70280	ACXA73C70290	
⚠	16	REACTOR	1	G0C103Z00003	←	
⚠	17	CIRCUIT BREAKER	2	CWA181005	←	
⚠	18	TERMINAL BOARD ASS'Y(A,B)	1	CWA28K1238	←	
⚠	19	TERMINAL BOARD ASS'Y(1,2,3,4,5,6)	1	CWA28K1341	←	
⚠	20	TERMINAL BOARD ASS'Y(7,8,9,10,11,12)	1	CWA28K1340	←	
⚠	21	TERMINAL BOARD ASS'Y(13,14,15,16,17,18)	1	CWA28K1343	←	
⚠	22	SENSOR-COMPLETE(WATER IN,OUT, REF TEMP)	1	ACXA50C16670	←	
⚠	23	SENSOR-COMPLETE(HEX WATER OUTLET)	1	ACXA50C13140	←	
⚠	24	REMOTE CONTROL COMPLETE	1	ACXA75C15751	←	
⚠	25	RECEIVER	1	CWB141039	←	
	26	HANDLE	4	ACXE16-00100	←	
	27	SOUND PROOF MATERIAL	1	CWG302599	←	
	28	CABINET FRONT PLATE	1	ACXE06-03550A	←	
	29	OPERATING INSTRUCTION	1	ACXF55-29310	←	
	30	OPERATING INSTRUCTION	1	ACXF55-29320	←	
	31	OPERATING INSTRUCTION	1	ACXF55-29330	←	
	32	OPERATING INSTRUCTION	1	ACXF55-29340	←	
	33	OPERATING INSTRUCTION	1	ACXF55-29350	←	
	34	OPERATING INSTRUCTION	1	ACXF55-29360	←	
	35	OPERATING INSTRUCTION	1	ACXF55-29370	←	
	36	INSTALLATION INSTRUCTION	1	ACXF60-41770	←	
	37	INSTALLATION INSTRUCTION	1	ACXF60-41780	←	
	38	PANASONIC BADGE	1	CWE373439	←	
	39	INVERTER BADGE	1	CWE373441	←	
	40	CONNECTING BAR	1	CWE261313A	←	
	41	CONNECTING BAR	1	CWE261214A	←	
	42	BAG	1	CWG861452	←	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MXC09J3E5	WH-MXC12J6E5	REMARK
	43	SHOCK ABSORBER	2	CWG713660	←	
	44	SHOCK ABSORBER	2	CWG713661	←	
	45	SHOCK ABSORBER	2	CWG713662	←	
	46	BASE BOARD-COMPLETE	1	ACXG62C02080	←	
	47	C.C.CASE	1	ACXG50-57120	←	
	61	MODEL LABEL	2	ACXF85-62530	ACXF85-62520	

Note:

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- “O” marked parts are recommended to be kept in stock.

<Refrigerant System>

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MXC09J3E5	WH-MXC12J6E5	REMARK
	1	BASE PAN ASS'Y	1	CWD52K1276	←	
⚠	2	COMPRESSOR	1	9JD420XAA22	←	
⚠	3	LEADWIRE COMPRESSOR	1	ACXA60C62280	←	
⚠	4	CRANKCASE HEATER	1	ACXA34-00110	←	
	5	BUSHING - COMPRESSOR MOUNT	3	CWH50055	←	
	6	NUT-COMPRESSOR MOUNT	3	CWH561049	←	
	7	PACKING	3	CWB811017	←	
	8	CONDENSER COMPLETE	1	ACXB32C23050	←	
	9	TUBE ASSY (CAP.TUBE)	1	ACXT07K09170	←	
	10	EXPANSION VALVE	1	CWB051029	←	
	11	3-WAYS VALVE (GAS)	1	CWB011699	←	
	12	4-WAYS VALVE	1	CWB001046	←	
	13	EXPANSION VALVE	1	CWB051091	←	
	14	STRAINER	3	CWB111032	←	
⚠	15	HIGH PRESSURE SENSOR	1	ACXA50-05990	←	
⚠	16	PRESSURE SWITCH	1	ACXA10-00650	←	
	17	DISCHARGE MUFFLER	1	ACXB12-01450	←	
	18	3-WAYS VALVE(LIQUID)	1	CWB011700	←	
	19	2-WAYS VALVE(LIQUID)	1	CWB021637	←	
	20	SOUND PROOF MATERIAL	1	ACXG30-11370	←	
	21	SOUND PROOF MATERIAL	1	ACXG30-11380	←	
	22	SOUND PROOF BOARD	1	CWH151085	←	
⚠	23	V-COIL COMPLETE(4 WAY VALVE)	1	CWA43C2169J	←	
⚠	24	V-COIL COMPLETE(MAIN EXPANSION VALVE)	1	ACXA43C06900	←	
⚠	25	V-COIL COMPLETE(BYPASS EXPANSION VALVE)	1	CWA43C2335	←	
⚠	26	V-COIL COMPLETE(2-WAY VALVE)	1	CWA43C2607	←	
⚠	27	SENSOR-COMP.(OUTDOOR AIR,PIPE TEMP)	1	CWA50C2730	←	
⚠	28	SENSOR-COMP.(DEFROST TEMP)	1	CWA50C2577	←	
⚠	29	SENSOR-COMP.(DISCHARGE TEMP)	1	CWA50C2576	←	
⚠	30	HIGH PRESSURE SENSOR	1	CWA50C3165	←	
	31	CABINET REAR PLATE-COMPLETE	1	ACXE02-02020A	←	
	32	CONTROL BOARD CASING	1	CWH10K1228	←	
	33	SOUND PROOF MATERIAL	1	CWG302598	←	
⚠	34	TERMINAL BOARD ASS'Y(U,V,W,A)	1	CWA28K1334	←	
⚠	35	TERMINAL BOARD ASS'Y(1,2,3)	1	CWA28K1213	←	
⚠	36	ELECTRONIC CO.(CAPACITOR)	1	ACXA73-36390	←	
⚠	37	ELECTRONIC CO.(NOISE FILTER)	1	ACXA73-36910	←	
⚠	38	ELECTRONIC CONTROLLER (MAIN)	1	ACXA73C70410R	ACXA73C70420R	
⚠	39	REACTOR	1	G0C592J00006	←	
⚠	40	ELECTRO MAGNETIC SWITCH	1	K6C1AYY00003	←	
	41	FAN MOTOR BRACKET	1	ACXD54K02340	←	
	42	SCREW-FAN MOTOR BRACKET	2	CWH551040J	←	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-MXC09J3E5	WH-MXC12J6E5	REMARK
⚠	43	FAN MOTOR(UPPER)	1	L6CBYYYL0362	←	
⚠	44	FAN MOTOR(LOWER)	1	L6CBYYYL0363	←	
	45	SCREW-FAN MOTOR MOUNT	8	CWH551323	←	
	46	PROPELLER FAN ASSY	2	CWH00K1006	←	
	47	NUT	2	CWH561092	←	
	48	CABINET FRONT PLATE	1	CWE061098A	←	
	49	DISCHARGE GRILLE	2	CWE201154	←	
	50	CABINET SIDE PLATE ASS'Y	1	CWE04K1023A	←	
	51	WIRE NET	1	CWD041069	←	
	52	ACCESSORY CO.(RUBBER CAP)	1	CWH82C1839	←	
	53	ACCESSORY CO.(DRAIN ELBOW)	1	CWG87C900	←	
⚠	54	HEATER	1	CWA341071	←	
	55	CABINET FRONT PLATE	1	CWE061274A	←	
	56	CABINET TOP PLATE	1	CWE03C1105	←	
	57	FLEXIBLE PIPE (L-TUBE)	1	CWH5850080	←	
	58	PACKING-L.TUBE	1	CWB81012	←	
	59	TERMINAL COVER	1	CWH171051	←	
	60	GASKET FOR TERMINAL COVER	1	CWB811177	←	

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